Physical Chemistry for Combustion
Chapter 2: Chemical Kinetics

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outlines

1 Introduction

2 Basic definitions in chemical kinetics

3 Mathematical description of reaction rate

4 The kinetics of complex reactions

5 Kinetics modeling in combustion

6 Chemical kinetics studies in combustion

7 Numerical analyses of chemical kinetics

General description

Chemical kinetics

Describes the rate at which a chemical transformation is taking place under given thermodynamic conditions

Example : $H_2+\frac{1}{2}O_2=H_2O$

$\tau_{1/2}=10^{25}$ year at ambiant T and P

=> Both thermodynamic and chemical kinetics have to be considered

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Basic definitions in chemical kinetics (1)

Reaction stoichiometry (1)

Global reaction for methane combustion in oxygen

$$CH_4 + 2O_2 = CO_2 + 2H_2O$$

To establish this equation

- **On the left hand side, write the reactants**
- **On the right hand side, write the products**
- **Equilibrate the reaction by applying the law of mass conservation**

Global reaction for methane combustion in air

$$CH_4 + 2(O_2 + 3.76N_2) = CO_2 + 2H_2O + 2 \times 3.76N_2$$

N_2 is considered as inert

X_{N2}/X_{O2} is fixed at 3.76

Basic definitions in chemical kinetics (2)

Reaction stoichiometry (2)

Global reaction for ammonia combustion in oxygen

Basic definitions in chemical kinetics (2)

Reaction stoichiometry (2)

Global reaction for ammonia combustion in oxygen

$$NH_3 + \frac{3}{4}O_2 = \frac{1}{2}N_2 + \frac{3}{2}H_2O$$

By convention, the products in the global reaction equation correspond to the most stable form of the chemical elements which compose the reactants. The actual composition at equilibrium is not taken into account for describing the reaction stoichiometry.

For ammonia combustion

- **N_2 is the most stable form for N**
- **H_2O is the most stable form for H and O**

Basic definitions in chemical kinetics (3)

Reaction stoichiometry (3)

To characterize the respective quantities of fuel and oxidant, we use the equivalence ratio (Φ)

$$\Phi = \frac{\left(\frac{X_F}{X_{Ox}}\right)_{mix}}{\left(\frac{X_F}{X_{Ox}}\right)_{sto}}$$

To calculate the equivalence ratio for Case 1 : no dilution

$$X_F + X_{Ox} = 1$$

$$X_F = \frac{\Phi}{\Phi + R_s}$$

$$X_{Ox} = \frac{R_s}{\Phi + R_s}$$

$$R_s = \left(\frac{X_{Ox}}{X_F}\right)_{sto}$$

Basic definitions in chemical kinetics (4)

Reaction stoichiometry (4)

To calculate the equivalence ratio for Case 2 a : with a fixed X_{Dil} / X_{Ox}

$$X_F = \frac{\frac{\Phi}{\Phi+R_s}}{\frac{\Phi}{\Phi+R_s}+(R_d+1)\frac{R_s}{\Phi+R_s}}$$

$$X_{Ox} = \frac{\frac{R_s}{\Phi+R_s}}{\frac{\Phi}{\Phi+R_s}+(R_d+1)\frac{R_s}{\Phi+R_s}}$$

$$X_{Dil} = \frac{R_d\frac{R_s}{\Phi+R_s}}{\frac{\Phi}{\Phi+R_s}+(R_d+1)\frac{R_s}{\Phi+R_s}}$$

$$R_d = \frac{X_{Dil}}{X_{Ox}}$$

Basic definitions in chemical kinetics (5)

Reaction stoichiometry (5)

To calculate the equivalence ratio for Case 2 b : with a fixed X_{Dil}

$$X_F = (1 - X_{Dil}) \frac{\Phi}{\Phi + R_s}$$

$$X_{Ox} = (1 - X_{Dil}) \frac{R_s}{\Phi + R_s}$$

Basic definitions in chemical kinetics (6)

Reaction stoichiometry (6)

Under certain conditions (combustion in engine), the air excess (ε) is preferred

$$\varepsilon = \frac{1}{\Phi}$$

Basic definitions in chemical kinetics (7)

Reaction rate (1)

Gas phase reaction

$$\nu_A A + \nu_B B = \nu_C C + \nu_D D$$

Total reaction rate corresponds to the number of molecule consummed/produced during the period dt

$$v_A = -\frac{dN_A}{dt} = -\frac{dn_A}{dt}$$

$$v_C = \frac{dN_C}{dt} = \frac{dn_C}{dt}$$

Basic definitions in chemical kinetics (8)

Reaction rate (2)

Specifc reaction rate corresponds to the number of molecule consummed/produced during the period dt per volume unit

$$v_A = -\frac{1}{V}\frac{dN_A}{dt} = -\frac{1}{V}\frac{dn_A}{dt} = -\frac{d[A]}{dt}$$

$$v_C = \frac{1}{V}\frac{dN_C}{dt} = \frac{1}{V}\frac{dn_C}{dt} = \frac{d[C]}{dt}$$

Basic definitions in chemical kinetics (9)

Reaction rate (3)

For a unity stoichiometric coefficient and V=1

	$\nu_A A$	+	$\nu_B B$	=	$\nu_C C$	+	$\nu_D D$
t=0	a		b		0		0
t	a - $\nu_A x$		b - $\nu_B x$		$\nu_C x$		$\nu_D x$

$$-\frac{d[A]}{\nu_A} = -\frac{d(a-\nu_A x)}{\nu_A} = dx = -\frac{d(b-\nu_B x)}{\nu_B}$$

$$v = \frac{dx}{dt} = -\frac{d[A]}{\nu_A dt} = -\frac{d[B]}{\nu_B dt} = \frac{d[C]}{\nu_C dt} = \frac{d[D]}{\nu_D dt}$$

Basic definitions in chemical kinetics (10)

Reaction rate (4)

Reaction rate as a function of the advancement degree (σ)

$$\sigma = \sigma_A = \frac{\nu_A x}{a} = \sigma_B = \frac{\nu_B x}{b}$$

$$v_\sigma = \frac{d\sigma}{dt}$$

v_σ has unit of 1/s

Basic definitions in chemical kinetics (11)

Reaction order (1)

Rate of reaction can be expressed as

$$v = k[A]^{\alpha}[B]^{\beta}$$

k is the rate constant

α and β are the partial orders of reaction with respect to A and B

$\alpha + \beta = \gamma$ corresponds to the global order of the reaction

Partial and global order can be non-integer

$$v = k[A]^{1/2}[B]$$

Basic definitions in chemical kinetics (11)

Reaction order (2)

Order of reaction can be zero

$$v = k$$

Order of reaction can be indefinite

$$H_2 + Br_2 = 2HBr$$

$$v = \frac{k[H_2][Br_2]^{3/2}}{[Br_2]+k'[HBr]}$$

Partial orders for Br_2 and HBr are indefinite

=> Global reaction order is indefinite

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Mathematical description of reaction rate (1)

Zero-order reaction (1)

Possible if chemical kinetics is controlled by a physical process such as a vaporization process or an heterogeneous catalytic process

Reaction rate

$$v = k = -\frac{d[A]}{dt}$$

Integration

$$\int -d[A] = \int k dt + cst$$

$$-[A] = kt + cst$$

At t=0, $[A] = [A]_0$

$$[A]_0 - [A] = kt$$

Mathematical description of reaction rate (2)

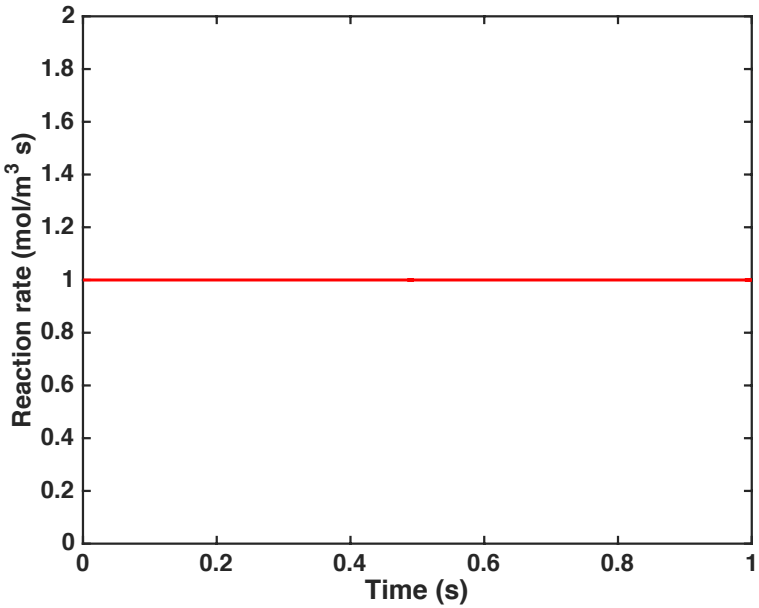
Zero-order reaction (2)

Half reaction time

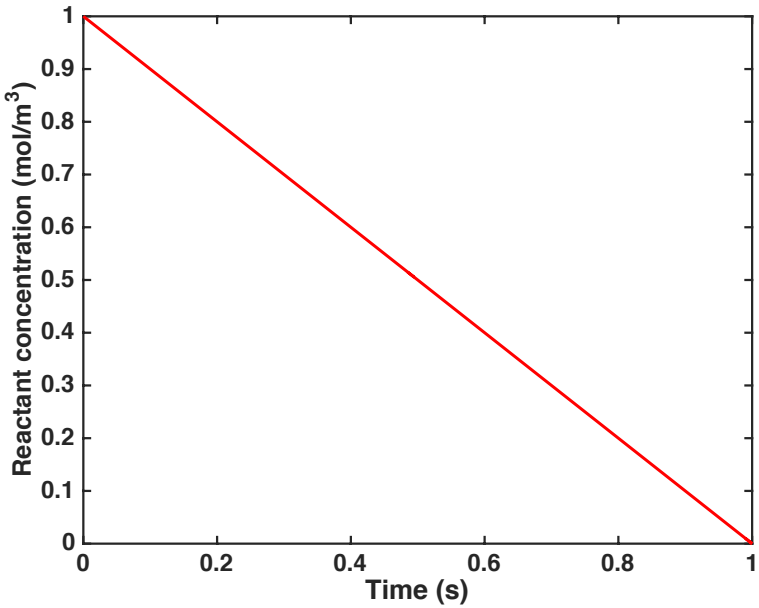
at $t = t_{1/2}$, $[A] = [A]_0/2$

$$[A]_0 - \frac{[A]_0}{2} = kt_{1/2}$$

=> $$t_{1/2} = \frac{[A]_0}{2k}$$



a) Reaction rate



b) Reactant concentration

Mathematical description of reaction rate (3)

First-order reaction (1)

Table of advancement of a first-order reaction

	A	=	P
t=0	a		0
t	a - x		x

Reaction rate

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k(a-x)$$

Integration

$$\int k dt = \int \frac{dx}{a-x} + cst$$

$$kt = \ln \frac{1}{a-x} + cst$$

Mathematical description of reaction rate (4)

First-order reaction (2)

At t=0, $x = 0$ so that

$$cst = \ln a$$

We get

$$kt = \ln \frac{a}{a-x}$$

$$[A] = [A]_0 \exp(-kt)$$

Half reaction time, at $t = t_{1/2}$, $[A] = [A]_0/2$

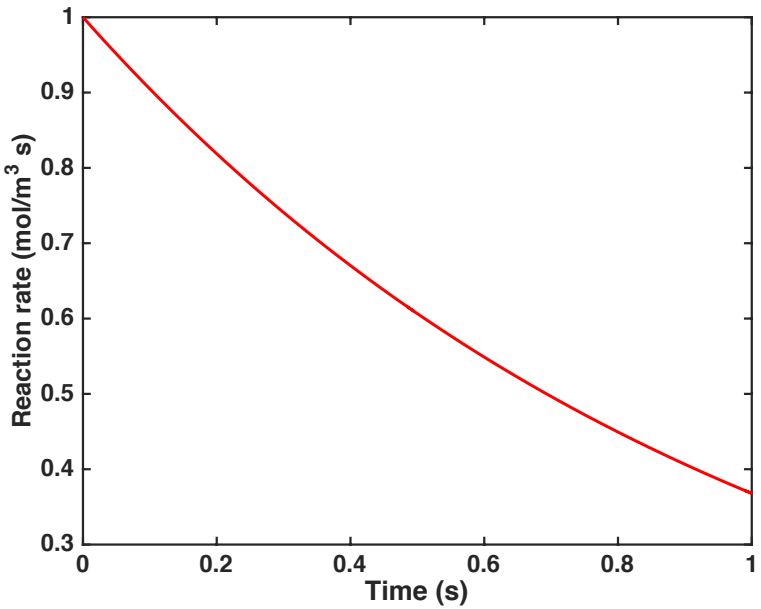
$$kt_{1/2} = \ln \frac{[A]_0}{[A]_0 - [A]_0/2}$$

=> $t_{1/2} = \frac{\ln 2}{k}$

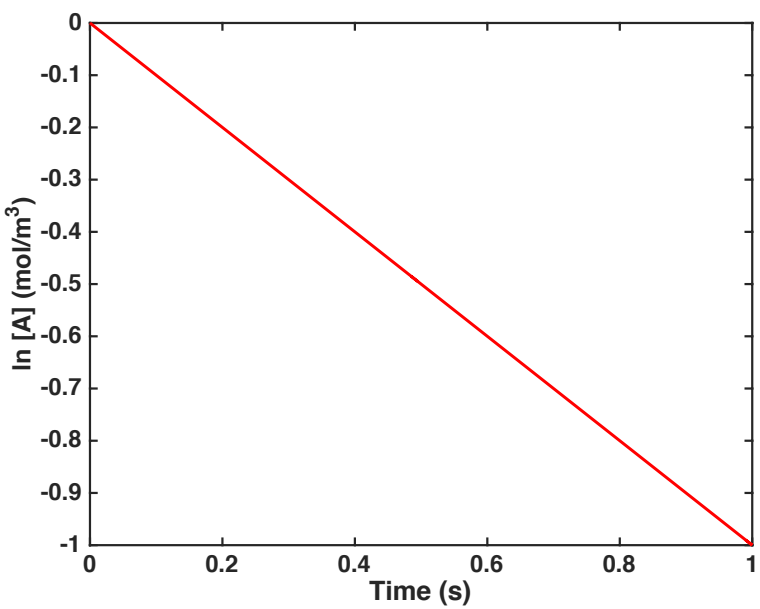
Mathematical description of reaction rate (5)

First-order reaction (3)

Evolution of reaction rate and reactant concentration



a) Reaction rate



b) Reactant concentration

Mathematical description of reaction rate (6)

Second-order reaction with 1 reactant (1)

Table of advancement of a second-order reaction with 1 reactant

	A	=	P
t=0	a		0
t	a - x		x

Reaction rate

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k(a-x)^2$$

Integration

$$\int kdt = \int \frac{dx}{(a-x)^2} + cst$$

$$kt = \frac{1}{a-x} + cst$$

Mathematical description of reaction rate (7)

Second-order reaction with 1 reactant (2)

At t=0, $x = 0$ so that

$$cst = -\frac{1}{a}$$

We get

$$kt = \frac{1}{[A]} - \frac{1}{[A]_0}$$

Half reaction time, at $t = t_{1/2}$, $[A] = [A]_0/2$

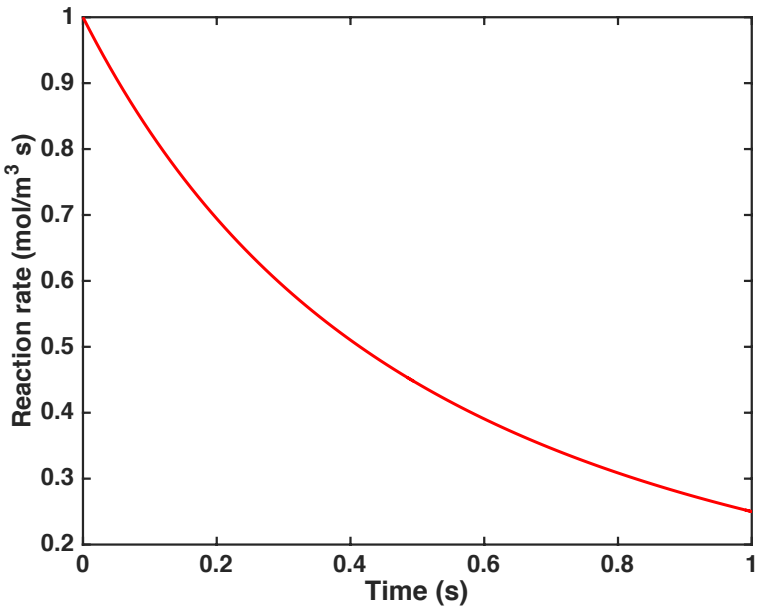
$$kt_{1/2} = \frac{1}{[A]_0/2} - \frac{1}{[A]_0}$$

=> $$t_{1/2} = \frac{1}{k[A]_0}$$

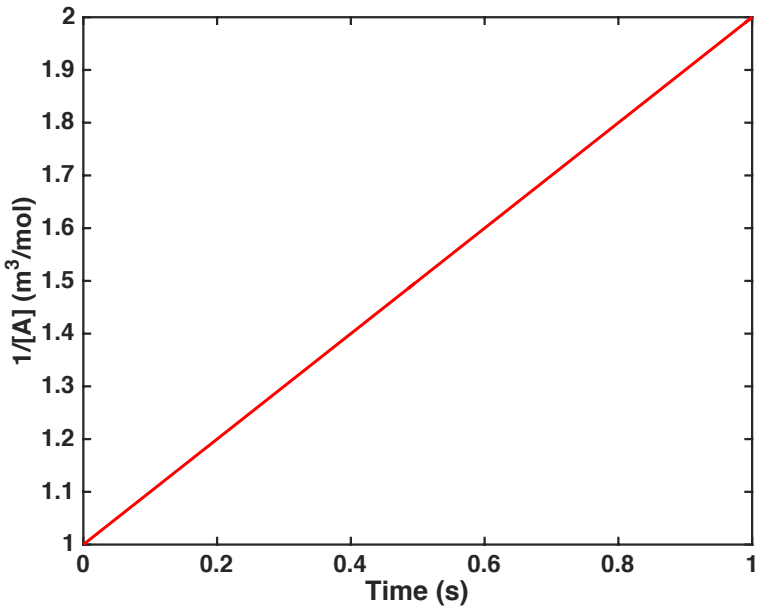
Mathematical description of reaction rate (8)

Second-order reaction with 1 reactant (3)

Evolution of reaction rate and reactant concentration



a) Reaction rate



b) Reactant concentration

Mathematical description of reaction rate (9)

Second-order reaction with 2 reactants (1)

Table of advancement of a second-order reaction with 2 reactants

	A	+	B	=	P
t=0	a		b		0
t	a - x		b - x		x

Reaction rate

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k(a-x)(b-x)$$

Integration

$$\int k dt = \int \frac{dx}{(a-x)(b-x)} + cst$$

Mathematical description of reaction rate (10)

Second-order reaction with 2 reactants (2)

Let us consider that

$$\frac{1}{(a-x)(b-x)} = \frac{A}{(a-x)} + \frac{B}{(b-x)}$$

We can deduce

$$\frac{1}{(b-x)} = A + B\frac{(a-x)}{(b-x)}$$

and

$$\frac{1}{(a-x)} = A\frac{(b-x)}{(a-x)} + B$$

For $x = a$, we get

$$A = \frac{1}{(b-a)}$$

Mathematical description of reaction rate (11)

Second-order reaction with 2 reactants (3)

For $x = b$, we get

$$B = \frac{1}{(a-b)} = -A$$

Re-writing the rate law

$$kt = \frac{1}{b-a}\left(\int \frac{dx}{(a-x)} - \int \frac{dx}{(b-x)}\right) + cst$$

which leads to

$$kt = \frac{1}{b-a} \ln \frac{(b-x)}{(a-x)} + cst$$

At t=0, $x = 0$

$$cst = \frac{1}{b-a} \ln \frac{a}{b}$$

Mathematical description of reaction rate (12)

Second-order reaction with 2 reactants (4)

We obtain

$$kt = \frac{1}{b-a} \ln \frac{a}{b} \frac{(b-x)}{(a-x)} = \frac{1}{[B]_0 - [A]_0} \ln \frac{[A]_0}{[B]_0} \frac{[B]}{[A]}$$

and finally

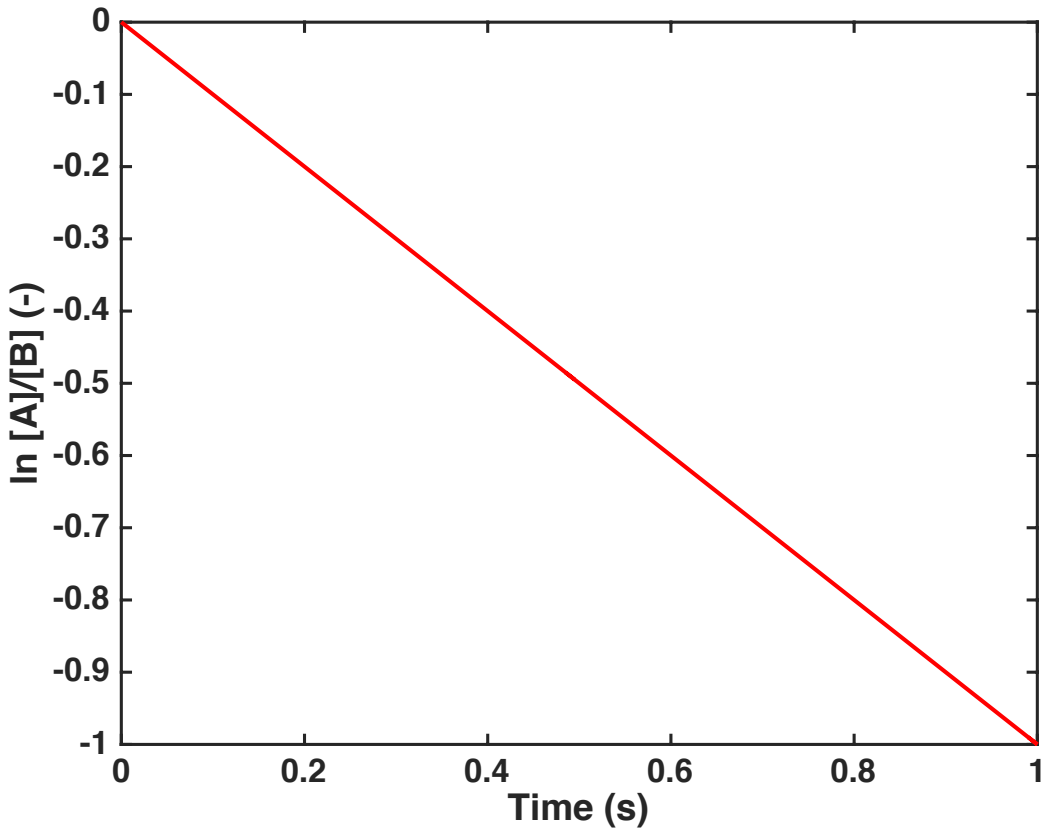
$$\ln \frac{[A]}{[B]} = ([A]_0 - [B]_0)\, kt + \ln \frac{[A]_0}{[B]_0}$$

Since there are 2 reactants, it is not possible to clearly define a half-reaction time

Mathematical description of reaction rate (13)

Second-order reaction with 2 reactants (5)

Evolution of the reactant concentration



Mathematical description of reaction rate (14)

Third-order reaction with 1 reactant (1)

Table of advancement of a third-order reaction with 1 reactants

	A	=	P
t=0	a		0
t	a - x		x

Reaction rate

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k(a-x)^3$$

Integrated rate law

$$kt = \frac{1}{2}\left(\frac{1}{([A]_0 - x)^2} - \frac{1}{[A]_0^2}\right)$$

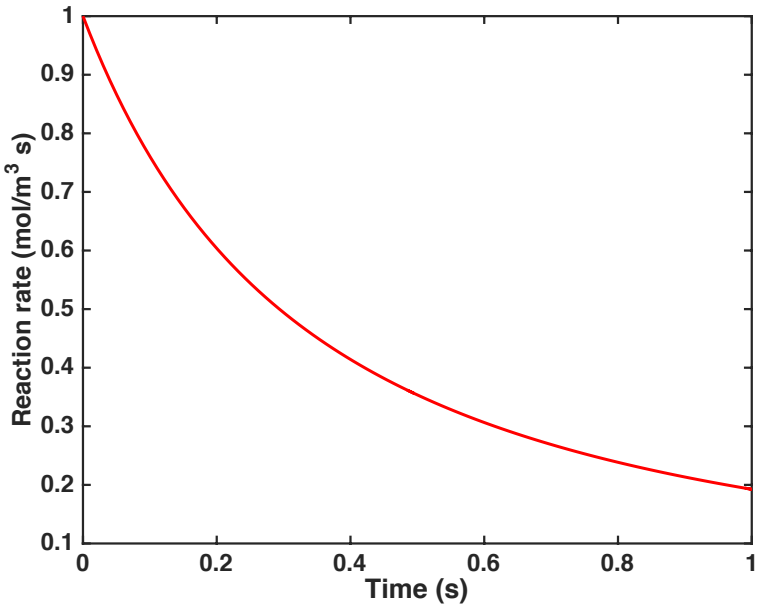
Mathematical description of reaction rate (15)

Third-order reaction with 1 reactant (2)

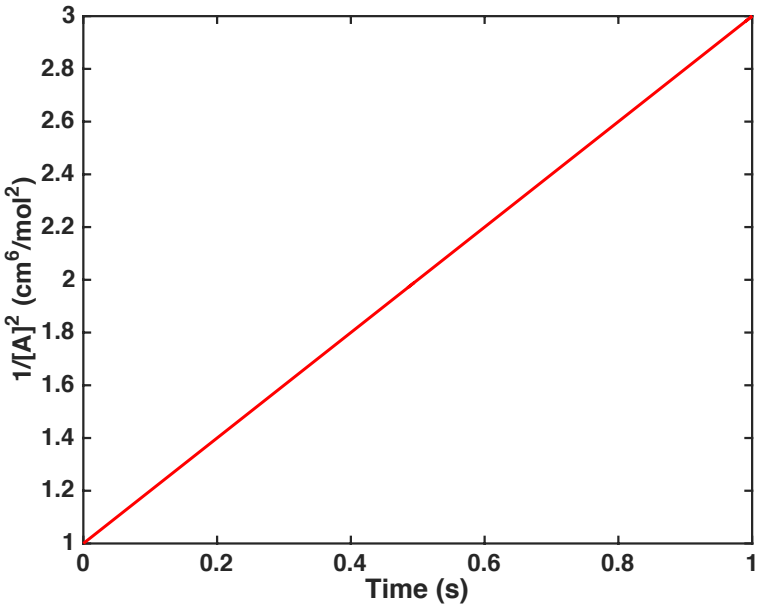
Half-reaction time

$$t_{1/2} = \frac{3}{2k[A]_0^2}$$

Evolution of reaction rate and reactant concentration



a) Reaction rate



b) Reactant concentration

Mathematical description of reaction rate (16)

Third-order reaction with 2 reactants (1)

Table of advancement of a third-order reaction with 2 reactants

	A	+	$2B$	=	P
t=0	a		b		0
t	a - x		b - 2x		x

Reaction rate

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k(a-x)(b-x)^2$$

Integrated rate law

$$kt = \frac{1}{(2[A]_0-[B]_0)^2}\left(\frac{2x(2[A]_0-[B]_0)}{[B]_0([B]_0-2x)} + \ln\left(\frac{[A]_0([B]_0-2x)}{[B]_0([A]_0-x)}\right)\right)$$

Mathematical description of reaction rate (17)

Reaction with fractional order with 1 reactant (1)

Reaction rate

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a-x)^n$$

Integrated rate law

$$kt = \frac{1}{n-1}\left(\frac{1}{([A]_0-x)^{n-1}} - \frac{1}{[A]_0^{n-1}}\right)$$

Half-reaction time

$$t_{1/2} = \frac{1}{k(n-1)[A]_0^{n-1}}\left(2^{n-1}-1\right)$$

Mathematical description of reaction rate (18)

Reaction with fractional order with 2 reactants (1)

Table of advancement

	A	+	sB	=	P
t=0	a		b		0
t	a - x		b - sx		x

Reaction rate

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\left(a-x\right)^{n_A}\left(sa-sx\right)^{n_B}$$

Setting k'=ks^{n_B} and n=$n_A + n_B$

$$\frac{dx}{dt} = ks^{n_B}\left(a-x\right)^{n_A+n_B} = k'\left(a-x\right)^{n}$$

Integrated rate law

$$k't = \frac{1}{n-1}\left(\frac{1}{([A]_0-x)^{n-1}} - \frac{1}{[A]_0^{n-1}}\right)$$

Mathematical description of reaction rate (19)

Reaction rate for variable volume reactor

Reaction

$$A + B = P$$

Reaction rate

$$v = -\frac{1}{V}\frac{dN_A}{dt} = k\left(\frac{N_A}{V}\right)^{n_A}\left(\frac{N_B}{V}\right)^{n_B}$$

The variation of [A]

$$-\frac{d[A]}{dt} = -\frac{d}{dt}\left(\frac{N_A}{V}\right) = -\frac{1}{V}\frac{dN_A}{dt} + \frac{N_A}{V^2}\frac{dV}{dt}$$

Which leads to

$$-\frac{d[A]}{dt} = k[A]^{n_A}[B]^{n_B} + \frac{[A]}{V}\frac{dV}{dt}$$

Mathematical description of reaction rate (20)

Correlation between concentration and time (1)

If only one reactant

- **if** $[A]$**=f(**t**) is linear, n=0**
- **if** $\ln[A]$**=f(**t**) is linear, n=1**
- **if** $1/[A]$**=f(**t**) is linear, n=2**
- **if** $1/[A]^2$**=f(**t**) is linear, n=3**

Mathematical description of reaction rate (21)

Correlation between concentration and time (2)

If two reactants : method of the degenerated order

$$b \gg a$$

$$x < a$$

=> $b \approx cst$

Rate law

$$v = k(a-x)^{n_A}(b-x)^{n_B} \approx k(a-x)^{n_A} b^{n_B} = k_{app}(a-x)^{n_A}$$

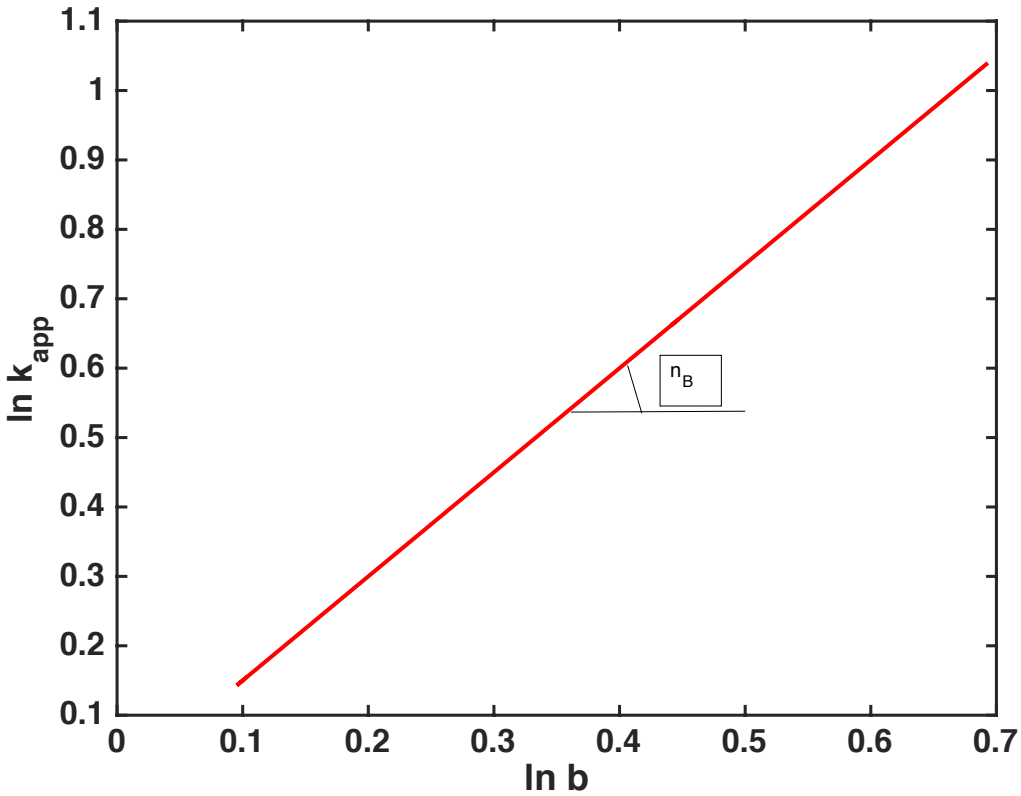
with $k_{app} = kb^{n_B}$

=> $\ln k_{app} = n_B \ln b + \ln k$

Mathematical description of reaction rate (22)

Correlation between concentration and time (3)

Plot $\ln K_{app}$**=f(**$\ln b$**)**



=> slope=n_B

Mathematical description of reaction rate (23)

The method of half-reaction time

If reaction order is 1

$$t_{1/2} = \ln 2 / k$$

=> the half-reaction time is independent [A]

If reaction order is different from 1

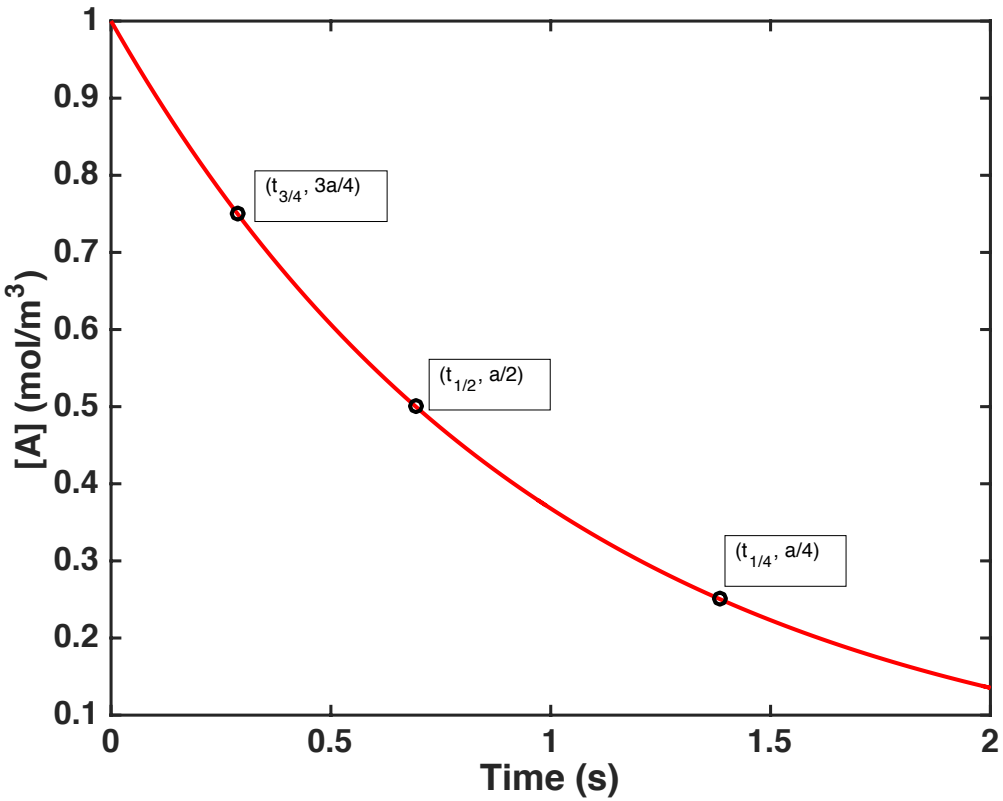
$$\ln t_{1/2} = (1-n)\ln a + \ln\left(\frac{2^{n-1}-1}{(n-1)k}\right)$$

=> Slope of $\ln t_{1/2}$=f($\ln a$) corresponds to 1-n

Mathematical description of reaction rate (24)

The method of successive characteristic times (1)

Several characteristic times can be defined



Mathematical description of reaction rate (25)

The method of successive characteristic times (2)

If reaction order is 1

$$\frac{t_{1/p}}{t_{1/q}} = \frac{\ln(p/(p-1))}{\ln(q/(q-1))}$$

If reaction order is different from 1

$$\frac{t_{1/p}}{t_{1/q}} = \frac{\left(\frac{p}{p-1}\right)^{n-1} - 1}{\left(\frac{q}{q-1}\right)^{n-1} - 1}$$

Characteristic values of time ratios

n	1	2	3
$t_{1/2}/t_{1/4}$	2.41	3	3.86
$t_{1/2}/t_{1/3}$	1.71	2	2.40
$t_{3/4}/t_{1/2}$	2	3	5

Mathematical description of reaction rate (26)

The Van't Hoff method (1)

Reaction

$$A + sB = P$$

Reaction rate

$$v = k\,(a-x)^{n_A}\,(b-sx)^{n_B}$$

If [B]=cst and [A]$\neq$cst

$$v' = k\,(a'-x)^{n_A}\,(b-sx)^{n_B}$$

Rate ratio

$$\frac{v}{v'} = \frac{(a-x)^{n_A}}{(a'-x)^{n_A}}$$

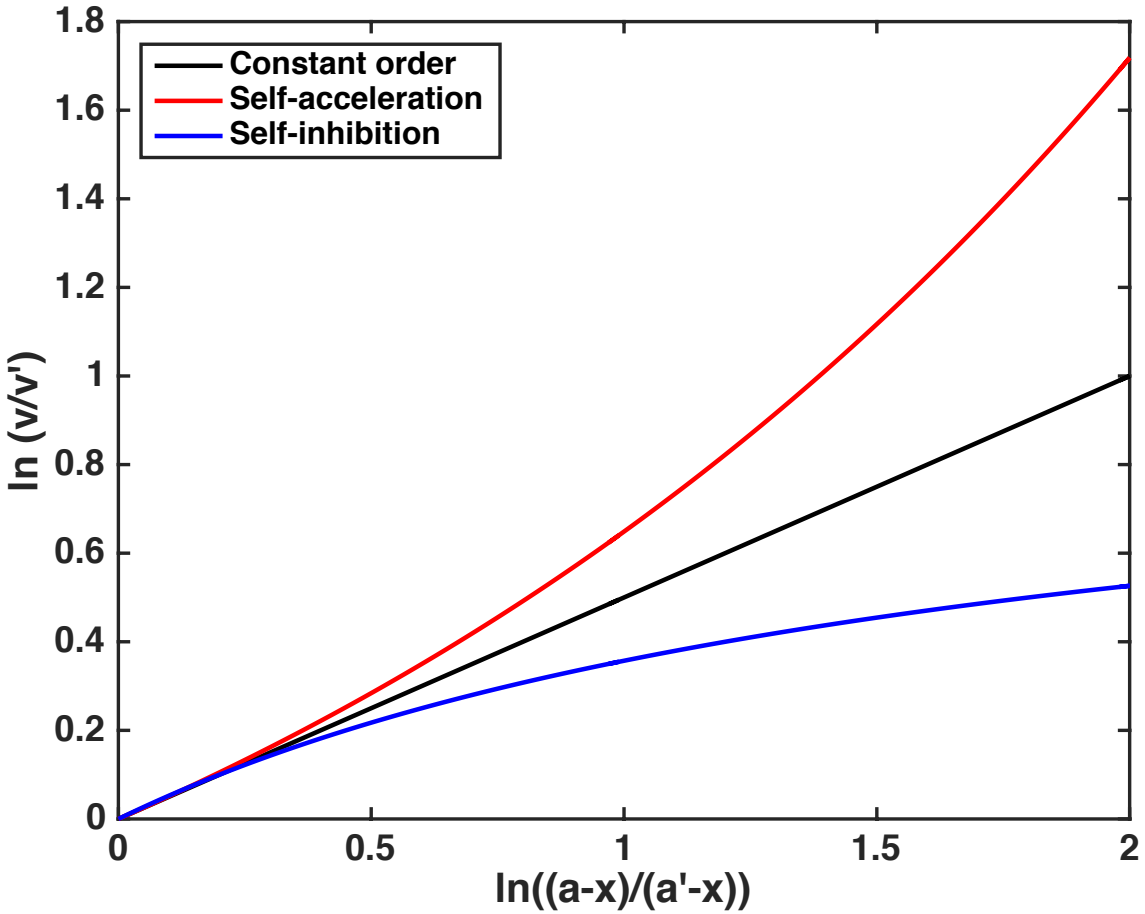
Reaction order with respect to A

$$\text{=> } n_A = \frac{\ln\left(\frac{v}{v'}\right)}{\ln\left(\frac{a-x}{a'-x}\right)}$$

Mathematical description of reaction rate (27)

The Van't Hoff method (2)

Plot $\ln(v/v')$**=f**$(\ln((a-x)/(a'-x)))$



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4. The kinetics of complex reactions
5. Kinetics modeling in combustion
6. Chemical kinetics studies in combustion
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The kinetics of complex reactions (1)

Elementary reaction

Chemical processes can be divided into a number of elementary chemical steps which are called “elementary reactions”

elementary reaction is defined as a chemical reaction in which the reactants form the products within a single reaction step and with a single transition state

Example of an important elementary reaction in combustion

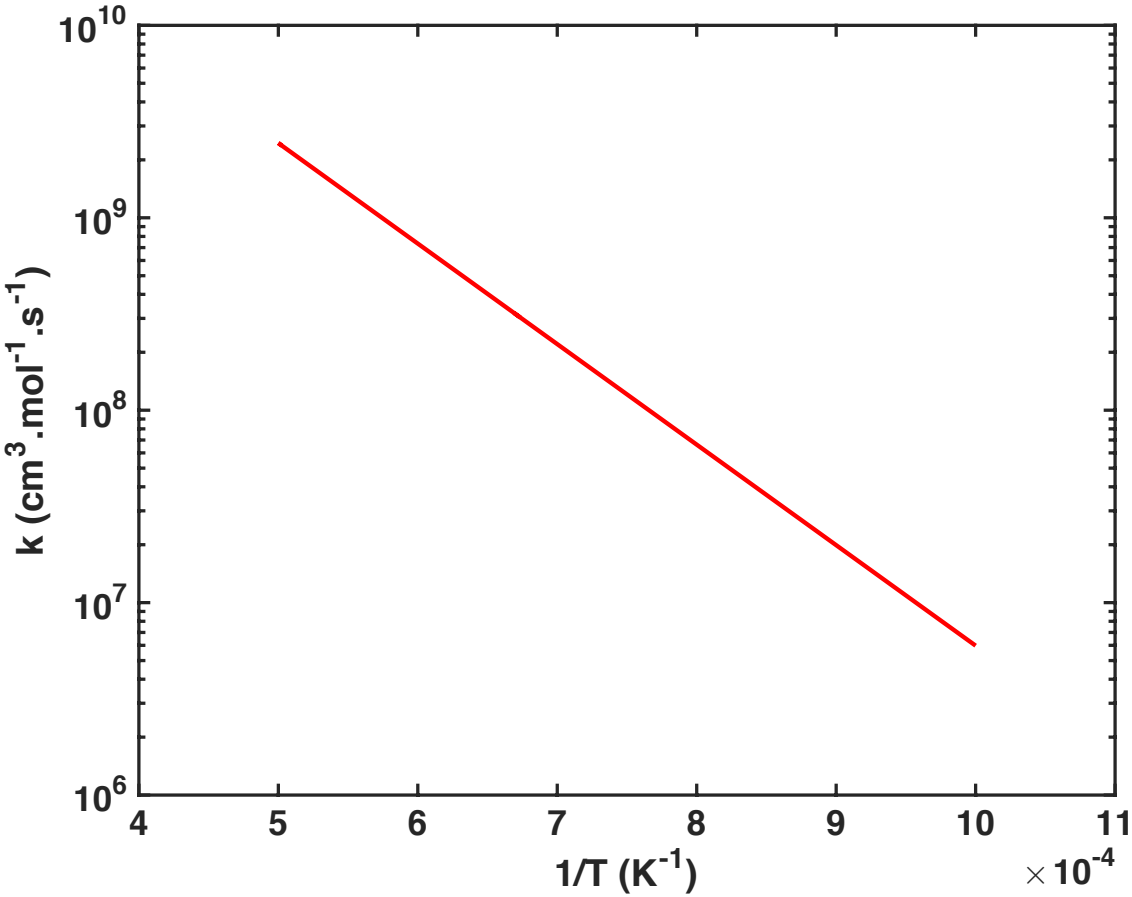
$$H + O_2 = OH + O$$

A series of elementary reactions which describe the kinetics of a chemical system is referred to as “reaction mechanism” or “reaction model”

The kinetics of complex reactions (2)

Temperature-dependence of reaction rate (1)

Reaction rate (k) usually increases with T



The kinetics of complex reactions (3)

Temperature-dependence of reaction rate (2)

Arrhenius law

$$k = A\exp\left(\frac{-E_a}{RT}\right)$$

- A **is the pre-exponential factor or collision frequency factor**
- E_a **is the activation energy**

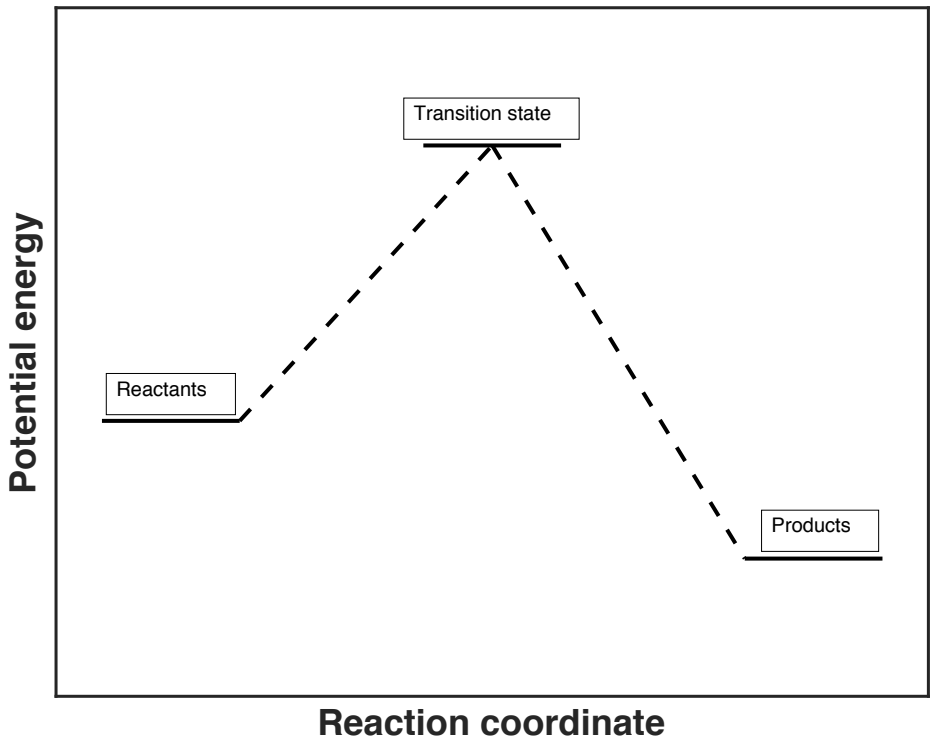
Physical interpretation

The pre-exponential factor corresponds to the total rate of collision between the molecules

The kinetics of complex reactions (4)

Temperature-dependence of reaction rate (3)

Activation energy



Activation energy corresponds to the energy to provide to transform the reactants into products

The kinetics of complex reactions (5)

Temperature-dependence of reaction rate (4)

The product $A\exp\left(\frac{-E_a}{RT}\right)$ **represents**

The kinetics of complex reactions (5)

Temperature-dependence of reaction rate (4)

The product $A\exp\left(\frac{-E_a}{RT}\right)$ **represents**

the rate of collisions which lead to the formation of the product

The kinetics of complex reactions (6)

Successive elementary reactions (1)

Succession of elementary reactions

$$A \xrightarrow{k_a} I \xrightarrow{k_b} P$$

Variations of the chemical species concentration

$$\frac{d[A]}{dt} = -k_a[A]$$

$$\frac{d[I]}{dt} = k_a[A] - k_b[I]$$

$$\frac{d[P]}{dt} = k_b[I]$$

The kinetics of complex reactions (7)

Successive elementary reactions (2)

Equations for species concentrations

$$[A] = [A]_0 \exp(-k_a t)$$

$$[I] = \frac{k_a}{k_b - k_a} (\exp(-k_a t) - \exp(-k_b t)) [A]_0$$

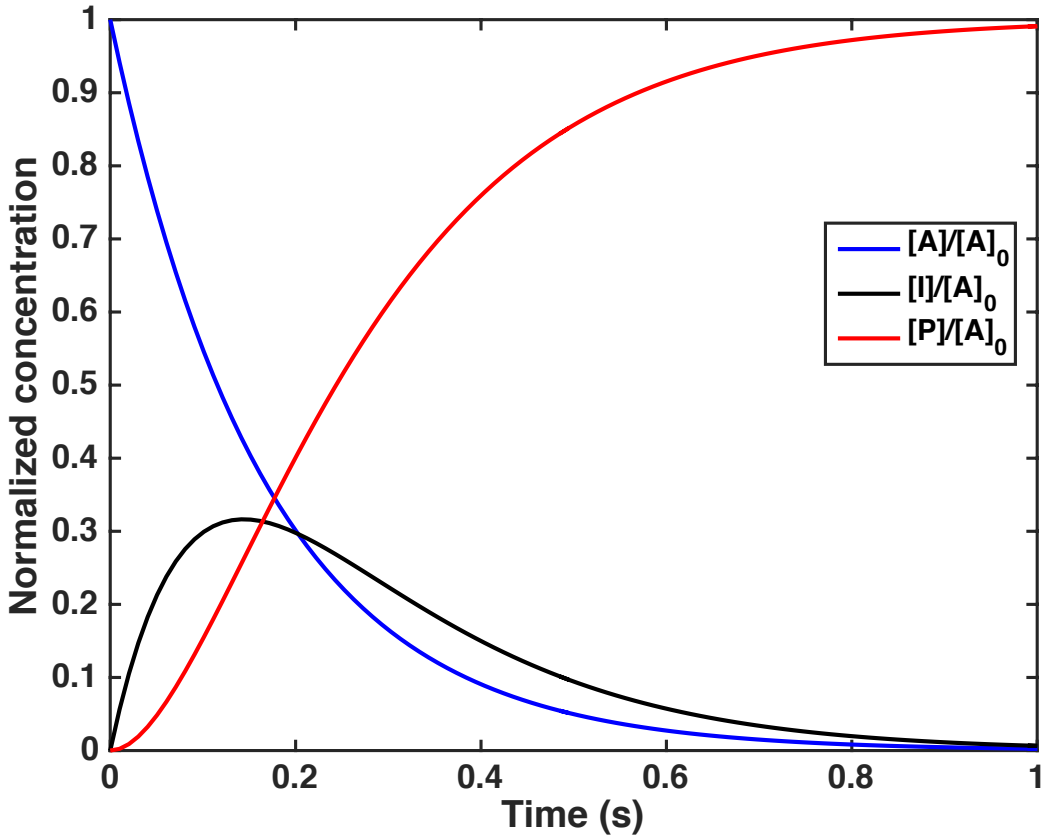
At all time : $[A] + [I] + [P] = [A]_0$

=> $[P] = \left(1 + \frac{k_a \exp(-k_b t) - k_b \exp(-k_a t)}{k_b - k_a}\right) [A]_0$

The kinetics of complex reactions (8)

Successive elementary reactions (3)

Species concentrations can be calculated



with $[A]_0$=1 ; k_a=6 ; and k_b=8

The kinetics of complex reactions (9)

Successive elementary reactions (4)

Determine which is the rate-limiting step

Reaction rate

$$v = \frac{d[P]}{dt} = k_b[I] = \frac{k_a k_b}{k_b - k_a}\left(\exp(-k_a t) - \exp(-k_b t)\right)[A]_0$$

If $k_a \gg k_b$

$$v = [A]_0 k_b \exp(-k_b t)$$

If $k_a \ll k_b$

$$v = [A]_0 k_a \exp(-k_a t)$$

The kinetics of complex reactions (9)

Successive elementary reactions (4)

Determine which is the rate-limiting step

Reaction rate

$$v = \frac{d[P]}{dt} = k_b[I] = \frac{k_a k_b}{k_b - k_a}\left(\exp(-k_a t) - \exp(-k_b t)\right)[A]_0$$

If $k_a \gg k_b$

$$v = [A]_0 k_b \exp(-k_b t)$$

If $k_a \ll k_b$

$$v = [A]_0 k_a \exp(-k_a t)$$

=> The overall reaction rate of a chemical process described by successive elementary reactions is controlled by the rate of the slowest step

The kinetics of complex reactions (10)

Competitive elementary reactions

$$A \xrightarrow{k_a} P_1$$

$$A \xrightarrow{k_b} P_2$$

Reaction rate

$$v = -\frac{d[A]}{dt} = k_a[A] + k_b[A] = (k_a + k_b)[A]$$

If $k_a \gg k_b$

$$v = k_a[A]$$

If $k_a \ll k_b$

$$v = k_b[A]$$

The kinetics of complex reactions (10)

Competitive elementary reactions

$$A \xrightarrow{k_a} P_1$$

$$A \xrightarrow{k_b} P_2$$

Reaction rate

$$v = -\frac{d[A]}{dt} = k_a[A] + k_b[A] = (k_a + k_b)[A]$$

If $k_a \gg k_b$

$$v = k_a[A]$$

If $k_a \ll k_b$

$$v = k_b[A]$$

=> The overall reaction rate of a chemical process described by competitive elementary reactions is controlled by the rate of the fastest step

The kinetics of complex reactions (11)

Quasi-steady-state approximation (1)

Reaction sequence

$$A \xrightarrow{k_a} I \xrightarrow{k_b} P$$

Variations of the chemical species concentration

$$\frac{d[A]}{dt} = -k_a[A]$$

$$\frac{d[I]}{dt} = k_a[A] - k_b[I]$$

$$\frac{d[P]}{dt} = k_b[I]$$

The kinetics of complex reactions (12)

Quasi-steady-state approximation (2)

Equations for species concentrations

For A : $[A] = [A]_0 \exp(-k_a t)$

Applying QSSA to I : $\frac{d[I]}{dt} \simeq 0$

=> $[I] = \frac{k_a}{k_b}[A] = \frac{k_a}{k_b}[A]_0 \exp(-k_a t)$

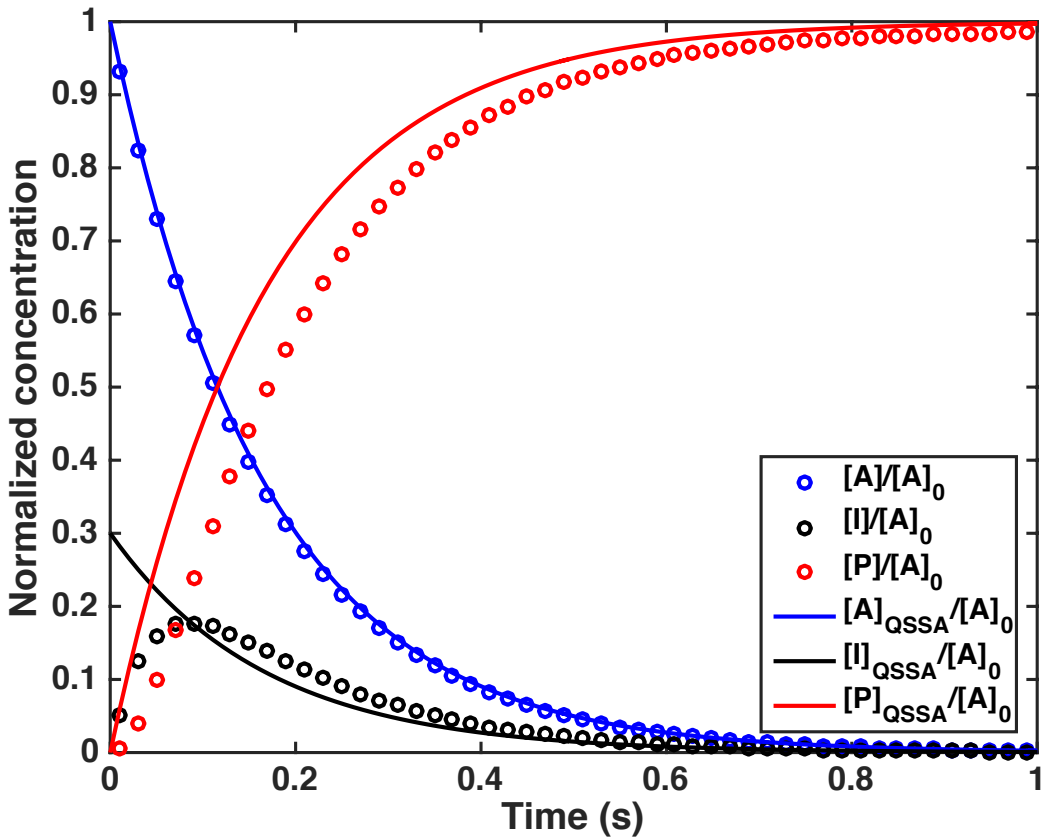
For P : $\frac{d[P]}{dt} = k_b[I] \simeq k_b \frac{k_a}{k_b}[A] = k_a[A]$

=> $[P] = k_a[A] \int_0^t \exp(-k_a t)\, dt = (1 - \exp(-k_a t))\,[A]_0$

The kinetics of complex reactions (13)

Quasi-steady-state approximation (3)

Species concentrations can be calculated



with $[A]_0$=1 ; k_a=6 ; and k_b=20

The kinetics of complex reactions (14)

Quasi-steady-state approximation (4)

Kinetics implication of QSSA

Assumption : $[I]$ is small with respect to $[A]_0$ at all time

$$\textbf{=>}\ \frac{[I]}{[A]_0} = \frac{k_a}{k_b}\exp(-k_a t) \ll 1$$

$$\textbf{=>}\ k_b \gg k_a$$

The kinetics of complex reactions (15)

Quasi-steady-state approximation (5)

Application of QSSA to more complex reaction mechanism

- $\mathbf{R_1}$: $H + O + M = OH* + M$
- $\mathbf{R_2}$: $OH* + M => OH + M$
- $\mathbf{R_3}$: $OH* => OH* + h\nu$
- $\mathbf{R_4}$: $N_2O + H => N_2 + OH*$

Applying QSSA to OH*

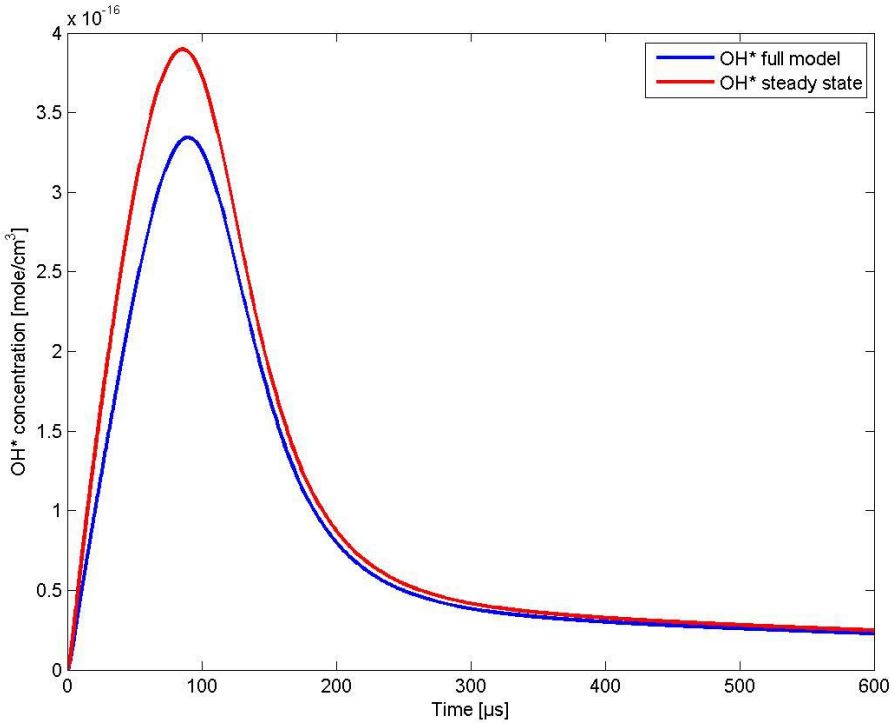
$$\frac{d[OH*]}{[dt]} = k_1[H][O][M] - k_{-1}[OH*][M] - k_2[OH*][M] - k_3[OH*] + k_4[N_2O][H] = 0$$

$$\mathbf{=>}[OH*]_{QSSA} = \frac{k_1[H][O][M] + k_4[N_2O][H]}{(k_2 + k_{-1})[M] + k_3}$$

The kinetics of complex reactions (16)

Quasi-steady-state approximation (6)

Comparison with full model



with [O] and [H] calculated with the detailed model

The kinetics of complex reactions (17)

Basic definitions (1)

A chain reaction mechanism corresponds to an ensemble of elementary reactions which are repeated cyclically

In a chain reaction, a reaction intermediate produced in one step generates an intermediate in a subsequent step, then that intermediate generates another intermediate, and so on

The intermediates are referred to as “chain carriers”

The kinetics of complex reactions (18)

Basic definitions (2)

Different types of reactions encountered in a chain mechanism

- **Initiation reaction : the initial step of formation of the chain carriers**
- **Propagation reaction : a chain carrier reacts with the reactant to form a new chain carrier**
- **Branching reaction : produces more than one chain carrier**
- **Inhibition reaction : a chain carrier reacts with the reaction product, which slows down the overall reaction rate**
- **Termination reaction : chain carriers react together to form a stable species**

Two kind of chain reactions

- **Linear chain mechanism : no chain branching reaction are taking place**
- **Chain branching mechanism : a number of chain branching reactions are taking place**

The kinetics of complex reactions (19)

Linear chain mechanism (1)

Example : gas phase synthesis of hydrogen bromide (HBr) from hydrogen and bromine

$$H_2(g) + Br_2(g) = 2HBr(g)$$

Elementary reactions

- **Initiation reaction : (1)** $Br_2 + M => 2Br + M$
- **Propagation reaction : (2)** $Br + H_2 => HBr + H$
- **Propagation reaction : (3)** $H + Br_2 => HBr + Br$
- **Inhibition reaction : (4)** $HBr + H => Br + H_2$
- **Termination reaction : (5)** $2Br + M => Br_2 + M$

The kinetics of complex reactions (20)

Linear chain mechanism (2)

Rate law : cannot be used directly because it includes the concentrations of intermediate species

$$v = \frac{d[HBr]}{dt} = k_2[Br][H_2] + k_3[H][Br_2] - k_4[HBr][H]$$

Applying QSSA to H and Br

$$\frac{d[H]}{dt} = k_2[Br][H_2] - k_3[H][Br_2] - k_4[HBr][H] = 0$$

and

$$\frac{d[Br]}{dt} = 2k_1[Br_2][M] - k_2[Br][H_2] + k_3[H][Br_2] + k_4[HBr][H] - 2k_5[Br]^2[M] = 0$$

The kinetics of complex reactions (21)

Linear chain mechanism (3)

We obtain

$$[H] = \frac{k_2[Br][H_2]}{k_3[Br_2]+k_4[HBr]}$$

and

$$[Br] = \left(\frac{k_1}{k_5}[Br_2]\right)^{1/2}$$

The rate law becomes

$$v = \frac{d[HBr]}{dt} = \frac{2k_2\left(\frac{k_1}{k_5}\right)^{1/2}[H_2][Br_2]^{3/2}}{[Br_2]+\frac{k_4}{k_3}[HBr]}$$

The kinetics of complex reactions (22)

Linear chain mechanism (4)

The rate law can be re-written as

$$v = \frac{d[HBr]}{dt} = \frac{k_a[H_2][Br_2]^{3/2}}{[Br_2]+k_b[HBr]}$$

where

$$k_a = 2k_2\left(\frac{k_1}{k_5}\right)^{1/2}$$

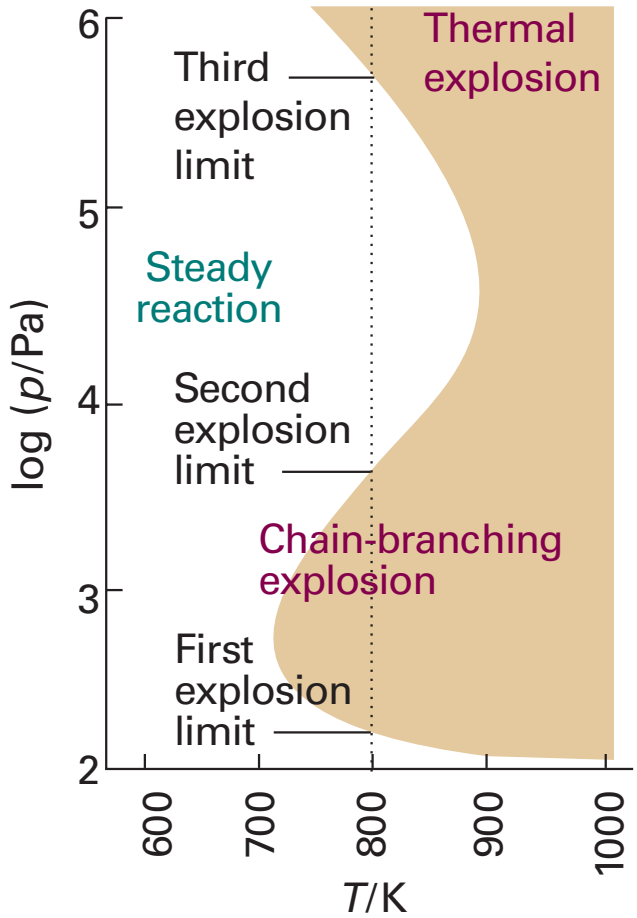
and

$$k_b = \frac{k_4}{k_3}$$

The kinetics of complex reactions (23)

Chain branching mechanism (1)

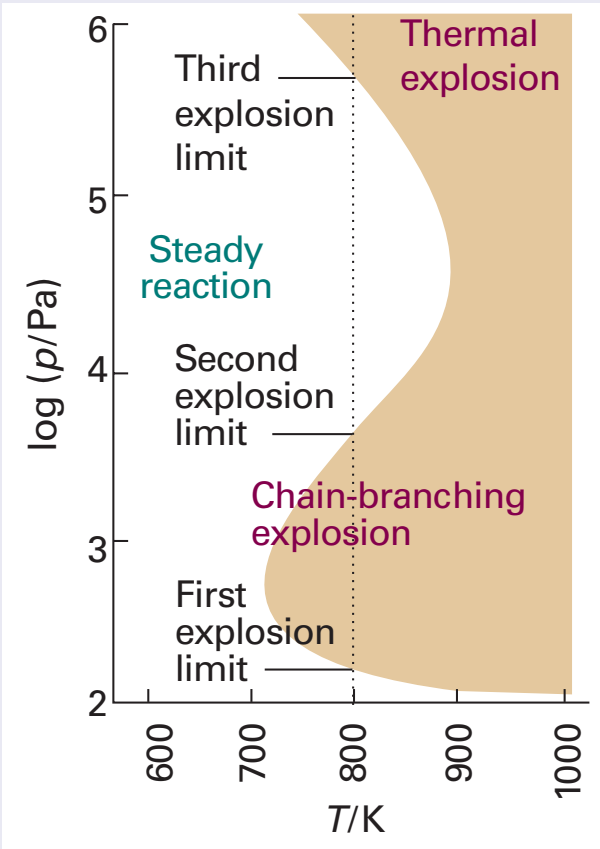
Example : the explosion limits of H_2-O_2 mixtures



The kinetics of complex reactions (24)

Chain branching mechanism (2)

Explosion limits



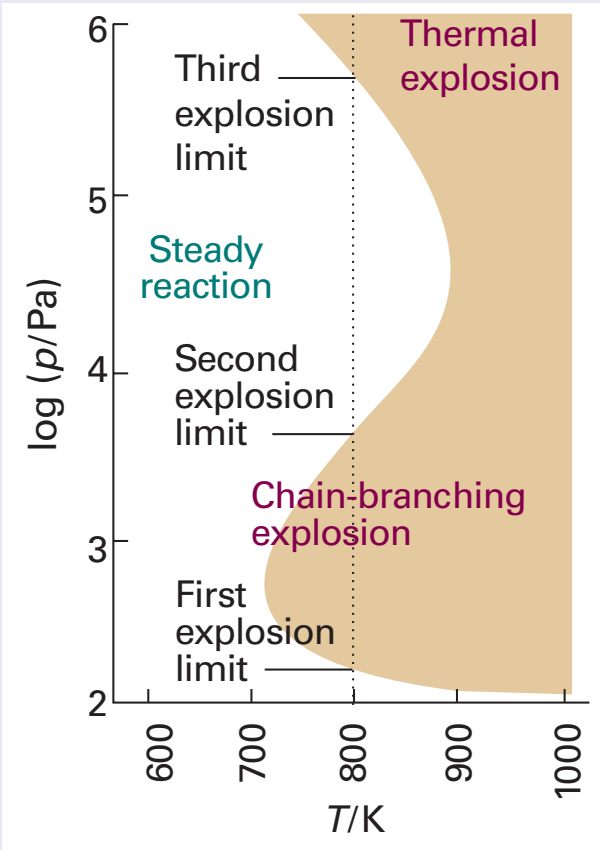
Reaction mechanism

- Initiation reaction : (1)
  $H_2 + O_2 => H + HO_2$
- Propagation reaction : (2)
  $H_2 + OH => H_2O + H$
- Propagation reaction : (3)
  $HO_2 + H => 2OH$
- Branching reaction : (4)
  $H + O_2 => OH + O$
- Branching reaction : (5)
  $O + H_2 => OH + H$
- Termination reaction : (6)
  $H + wall => \frac{1}{2} H_2$
- Termination reaction : (7)
  $H + O_2 + M => HO_2 + M$

The kinetics of complex reactions (25)

Chain branching mechanism (3)

Explosion limits



Between 1st and 2nd explosion limit

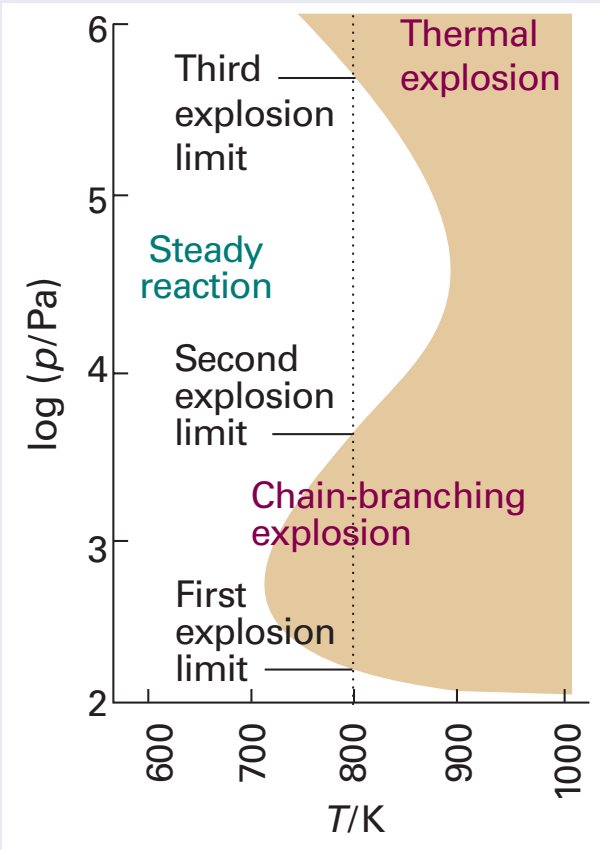
Reaction mechanism

- Initiation reaction : (1)
  $H_2 + O_2 => H + HO_2$
- Propagation reaction : (2)
  $H_2 + OH => H_2O + H$
- Propagation reaction : (3)
  $HO_2 + H => 2OH$
- Branching reaction : (4)
  $H + O_2 => OH + O$
- Branching reaction : (5)
  $O + H_2 => OH + H$
- Termination reaction : (6)
  $H + wall => \frac{1}{2} H_2$
- Termination reaction : (7)
  $H + O_2 + M => HO_2 + M$

The kinetics of complex reactions (26)

Chain branching mechanism (4)

Explosion limits



Between 2nd and 3rd explosion limit

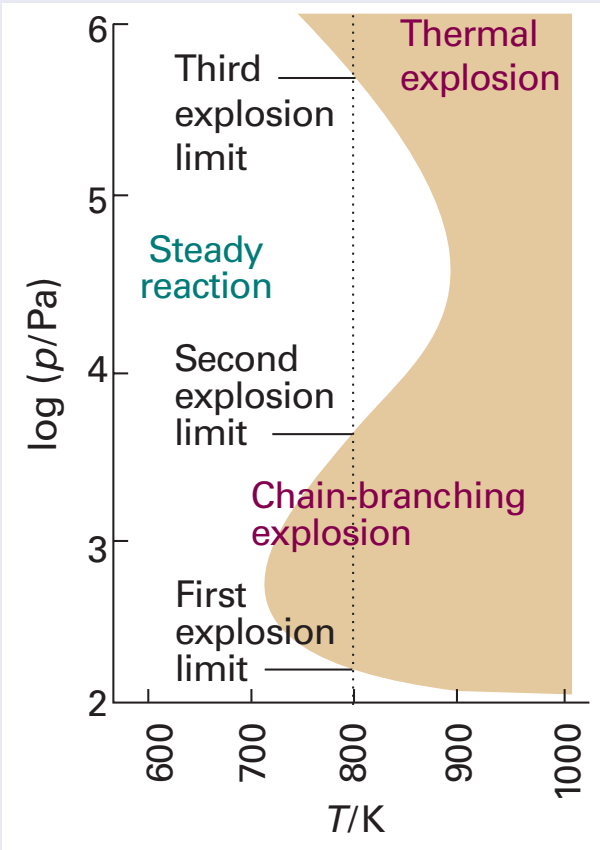
Reaction mechanism

- Initiation reaction : (1)
  $H_2 + O_2 => H + HO_2$
- Propagation reaction : (2)
  $H_2 + OH => H_2O + H$
- Propagation reaction : (3)
  $HO_2 + H => 2OH$
- Branching reaction : (4)
  $H + O_2 => OH + O$
- Branching reaction : (5)
  $O + H_2 => OH + H$
- Termination reaction : (6)
  $H + wall => \frac{1}{2} H_2$
- Termination reaction : (7)
  $H + O_2 + M => HO_2 + M$

The kinetics of complex reactions (27)

Chain branching mechanism (5)

Explosion limits



Above the 3rd explosion limit

Reaction mechanism

- Initiation reaction : (1)
  $H_2 + O_2 => H + HO_2$
- Propagation reaction : (2)
  $H_2 + OH => H_2O + H$
- Propagation reaction : (3)
  $HO_2 + H => 2OH$
- Branching reaction : (4)
  $H + O_2 => OH + O$
- Branching reaction : (5)
  $O + H_2 => OH + H$
- Termination reaction : (6)
  $H + wall => \frac{1}{2} H_2$
- Termination reaction : (7)
  $H + O_2 + M => HO_2 + M$

The kinetics of complex reactions (28)

Chain branching mechanism (6)

Kinetic analysis : conditions for chain branching explosion

Rate of chain carrier formation

$$v_{rad} = \frac{d[H]}{dt} = k_1[H_2][O_2] + k_2[H_2][OH] - k_3[HO_2][H] - k_4[H][O_2] + k_5[O][H_2] - k_6[H] - k_7[H][O_2][M]$$

Applying the QSSA to OH and O gives

$$\frac{d[OH]}{dt} = -k_2[H_2][OH] + 2k_3[HO_2][H] + k_4[H][O_2] + k_5[O][H_2] = 0$$

and

$$\frac{d[O]}{dt} = k_4[H][O_2] - k_5[O][H_2] = 0$$

The kinetics of complex reactions (29)

Chain branching mechanism (7)

which leads to

$$[OH] = \frac{2k_3[HO_2][H]+k_4[H][O_2]+k_5[O][H_2]}{k_2[H_2]}$$

and

$$[O] = \frac{k_4[H][O_2]}{k_5[H_2]}$$

Rate of H production becomes

$$\frac{d[H]}{dt} = v_{ini} + (k_{lin} + k_{br} - k_{ter})[H]$$

with

$$v_{ini}=k_1[H_2][O_2] \quad k_{lin}=k_3[HO_2] \quad k_{br}=2k_4[O_2] \quad k_{ter}=k_6+k_7[O_2][M]$$

The kinetics of complex reactions (30)

Chain branching mechanism (8)

We set : k_{lin}=0

$$[H] = \frac{v_{ini}}{k_{br} - k_{ter}} \left(\exp\left(\left(k_{br} - k_{ter}\right) t\right) - 1\right)$$

If $k_{ter} \gg k_{br}$

$$[H] = \frac{v_{ini}}{k_{ter}}$$

$=>$ **No explosion since [H] is essentially constant**

If $k_{br} \gg k_{ter}$

$$[H] = \frac{v_{ini}}{k_{br}} \left(\exp\left(k_{br} t\right)\right)$$

$=>$ **Explosion since [H] increases exponentially**

The kinetics of complex reactions (31)

Chain branching mechanism (9)

Critical conditions if $k_{br} = k_{ter}$

Achieved for $\frac{2k_4}{k_7[M]} = 1$

$=>$ Second explosion limit criterion of Lewis and von Elbe

The kinetics of complex reactions (32)

Basic definitions (1)

Catalytic reactions are chemical reactions which are influence by the presence of a chemical compound, the catalyst, which is not included in the overall reaction equation

A catalyst is a chemical compound which accelerates, orientates, or inhibits a chemical reaction without being consumed. The catalyst is regenerated in the course of the reaction

The kinetics of complex reactions (33)

Basic definitions (2)

Three type of catalytic reactions

- **Homogeneous catalysis for which all the chemical species are in the same fluid phase (gas or liquid)**
- **Heterogeneous catalysis for which some of the chemical species are in the solid phase and others are in a fluid phase**
- **Enzymatic catalysis which is a biological process and is close to homogeneous catalysis**

The kinetics of complex reactions (34)

Thermodynamic conditions

Catalysts do not modify the state functions between the initial and the final states. Thus, the yield of a chemical reaction is not modified

Non-catalyzed reaction written as

$$A = P$$

Catalyzed reaction written as

$$A + C = P + C$$

Change of Gibbs energy

$$(\Delta_r G_1)_{P,T} = (\Delta_r G_2)_{P,T}$$

which leads to

$$K_P^1 = \frac{P_P}{P_A} = K_P^2$$

The kinetics of complex reactions (35)

Mode of action (1)

Non-catalyzed reaction written as

$$R_1 : A + B = P$$

Catalyzed reaction written as

$$R_2 : A + C = X$$

$$R_3 : X + B = P + C$$

Applying the QSSA to X

$$\frac{d[X]}{dt} = k_2[A][C] - k_3[X][B] = 0$$

which leads to

$$[X] = \frac{k_2[A][C]}{k_3[B]}$$

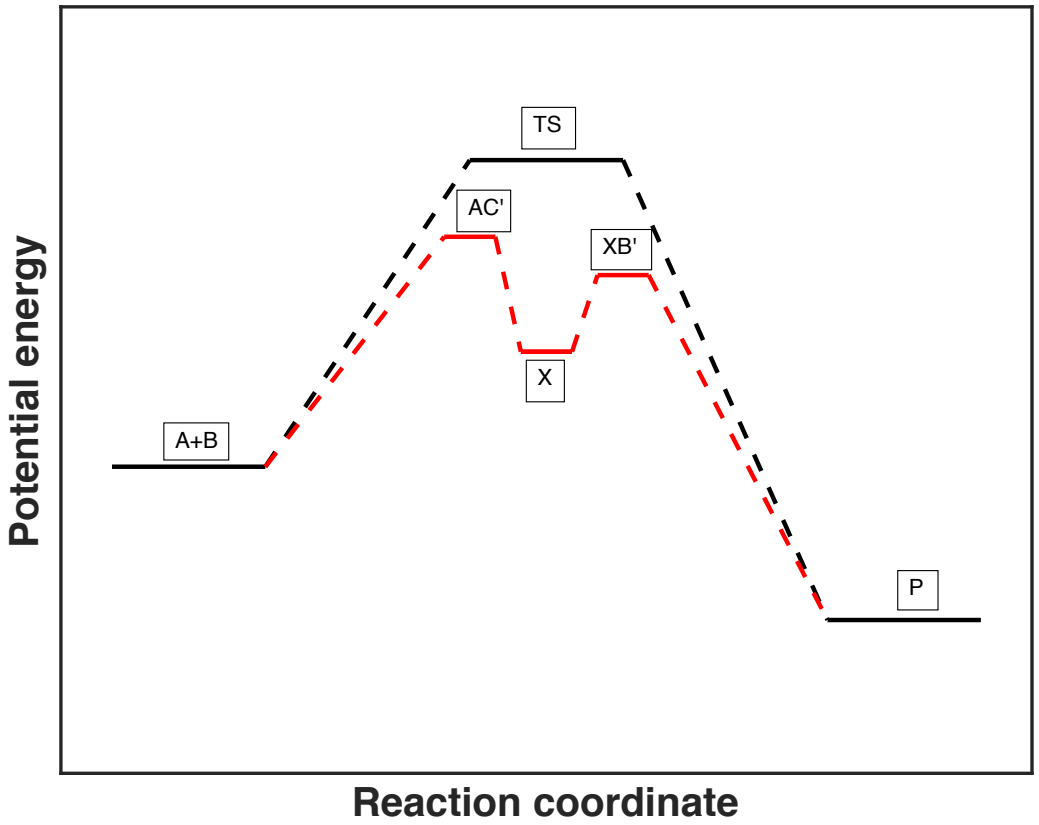
=> The reaction rate depends on [C]

$$v = \frac{d[P]}{dt} = k_3[X][B] = k_2[A][C]$$

The kinetics of complex reactions (36)

Mode of action (2)

Catalyst induces a decrease of the activation energy



The kinetics of complex reactions (37)

Atmospheric examples

Destruction of stratospheric ozone

$$NO + O_3 = NO_2 + O_2$$

$$NO_2 + O = NO + O_2$$

and

$$Cl + O_3 = ClO + O_2$$

$$ClO + O = Cl + O_2$$

Overall

$$O_3 + O = 2O_2$$

$=>$ **increase of UV radiation on Earth's surface**

The kinetics of complex reactions (38)

Effect of catalyst on the reaction mechanism (1)

Pyrolysis of acetaldehyde without catalyst

- **Initiation reaction : (1)** $CH_3CHO => CH_3 + CHO$
- **Propagation reaction : (2)** $CH_3CHO + CH_3 => CH_3CO + CH_4$
- **Propagation reaction : (3)** $CH_3CO => CH_3 + CO$
- **Termination reaction : (4)** $2CH_3 => C_2H_6$

Applying the QSSA to CH_3CO and CH_3

$$\frac{d[CH_3CO]}{dt} = k_2[CH_3CHO][CH_3] - k_3[CH_3CO] = 0$$

and

$$\frac{d[CH_3]}{dt} = k_1[CH_3CHO] - k_2[CH_3CHO][CH_3] + k_3[CH_3CO] - 2k_4[CH_3]^2 = 0$$

The kinetics of complex reactions (39)

Effect of catalyst on the reaction mechanism (2)

Overall reaction rate

$$v = \frac{d[CH_4]}{dt} = k_2[CH_3CHO][CH_3]$$

which becomes

$$v = \frac{d[CH_4]}{dt} = k_2\left(\frac{k_1}{2k_4}\right)^{1/2}[CH_3CHO]^{3/2}$$

The kinetics of complex reactions (40)

Effect of catalyst on the reaction mechanism (3)

Pyrolysis of acetaldehyde with catalyst (I_2)

- **Initiation reaction : (1)** $I_2 + M <=> 2I + M$
- **Propagation reaction : (2)** $I + CH_3CHO => HI + CH_3CO$
- **Propagation reaction : (3)** $CH_3CO => CH_3 + CO$
- **Catalyst regeneration reaction : (4)** $CH_3 + HI => CH_4 + I$

Applying the QSSA to I and HI

$$\frac{d[I]}{dt} = 2k_1[I_2][M] - 2k_{-1}[I]^2[M] - k_2[I][CH_3CHO] + k_4[CH_3][HI] = 0$$

and

$$\frac{d[HI]}{dt} = k_2[I][CH_3CHO] - k_4[CH_3][HI] = 0$$

The kinetics of complex reactions (41)

Effect of catalyst on the reaction mechanism (4)

Overall reaction rate

$$v = \frac{-d[CH_3CHO]}{dt} = k_2[CH_3CHO][I]$$

which becomes

$$v = \frac{-d[CH_3CHO]}{dt} = k_2 \left(\frac{k_1}{k_{-1}}\right)^{1/2} [I_2]^{1/2}[CH_3CHO]$$

=> depends on $[I_2]$

The kinetics of complex reactions (42)

Auto-catalysis (1)

Auto-catalyzed reactions are characterized by an induction period, during which the reaction rate is low, followed by a sudden increase of the reaction rate

$$A + B => 2B + P$$

Reaction rate

$$v = -\frac{d[A]}{dt} = \frac{dx}{dt} = k(a-x)(b+x)$$

After integration

$$t = \frac{1}{k(a+b)} \ln \frac{a}{b} \frac{(b+x)}{(a-x)}$$

The kinetics of complex reactions (43)

Auto-catalysis (2)

Maximum reaction rate

$$v_{max} = \frac{d^2x}{dt^2} = 0 = \frac{d(k(a-x)(b+x))}{dt} = -b - x + a - x$$

which leads to

$$x_{max} = \frac{a-b}{2}$$

We finally obtain

$$v_{max} = \frac{k}{4}(a+b)^2$$

Time at which v=v_{max}

$$t_{max} = \frac{1}{k(a+b)} \ln \frac{a}{b}$$

The kinetics of complex reactions (44)

Auto-catalysis (3)

At half-reaction time, $[A]$=a/2

$$t_{1/2} = \frac{1}{k(a+b)} \ln \frac{2b+a}{b}$$

If $a \gg b$:

$$t_{max} \simeq t_{1/2} \simeq \frac{1}{ka} \ln \frac{a}{b}$$

=> the maximum reaction rate is not reached at initial time but at a time close to $t_{1/2}$

The kinetics of complex reactions (45)

Auto-catalyzed oscillating reaction (1)

Lokta-Volterra mechanism described as

- $\mathbf{R_1}$: $A + X => 2X$
- $\mathbf{R_2}$: $X + Y => 2Y$
- $\mathbf{R_3}$: $Y => B$

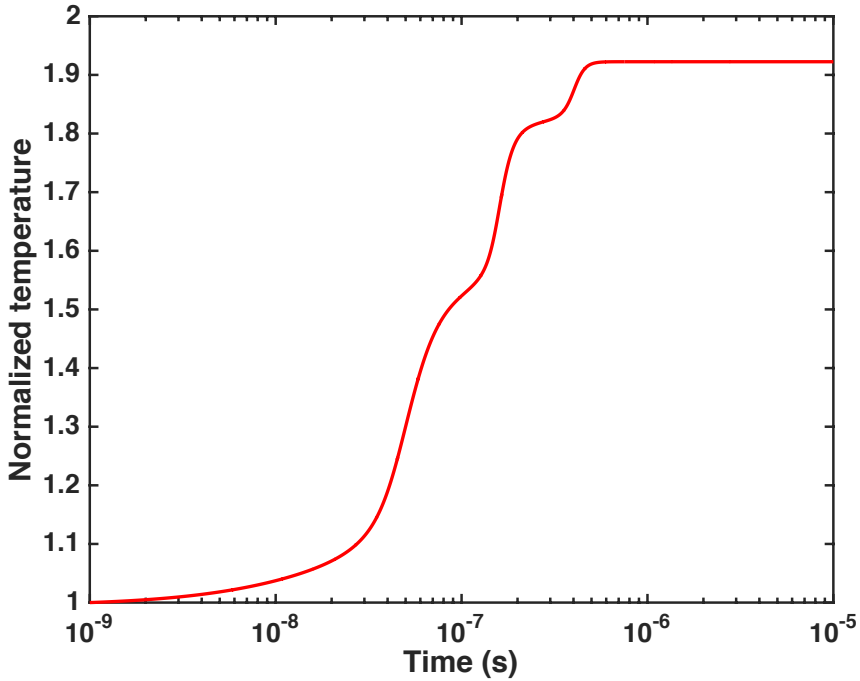
Assumptions

- $\mathbf{R_1}$ and $\mathbf{R_2}$ are non-thermal
- $\mathbf{R_3}$ is exothermic
- $\mathbf{R_2}$ is much faster than $\mathbf{R_1}$ and $\mathbf{R_3}$

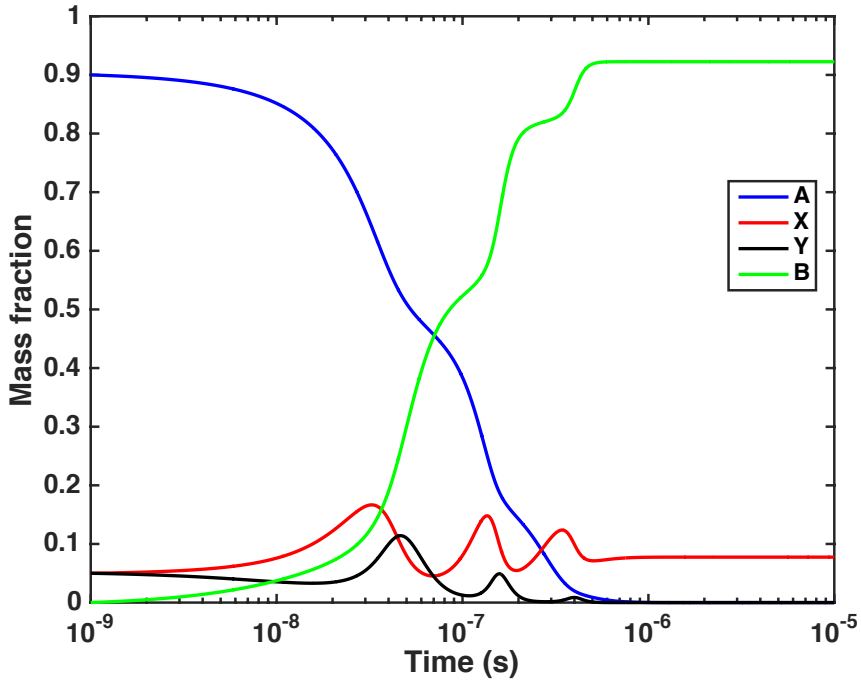
The kinetics of complex reactions (46)

Auto-catalyzed oscillating reaction (2)

Temperature and species profiles



a) Temperature profile



b) Species profiles

=> Oscillations of the intermediate species mole fractions + non-monotonous temperature profile

The kinetics of complex reactions (47)

Photochemistry (1)

Photochemical processes correspond to chemical sequences which are initiated by photons. In general, high energy photons, in the UV range, are used to break the chemical bonds of the reactants

Example : formation of tropospheric ozone

- $\mathbf{R_1}$: $OH + CO => HOCO$
- $\mathbf{R_2}$: $HOCO + O_2 => HO_2 + CO_2$
- $\mathbf{R_3}$: $HO_2 + NO => OH + NO_2$
- $\mathbf{R_4}$: $NO_2 + h\nu => NO + O$
- $\mathbf{R_5}$: $O + O_2 => O_3$

Reaction R_4 is a photochemical reaction

The kinetics of complex reactions (48)

Photochemistry (2)

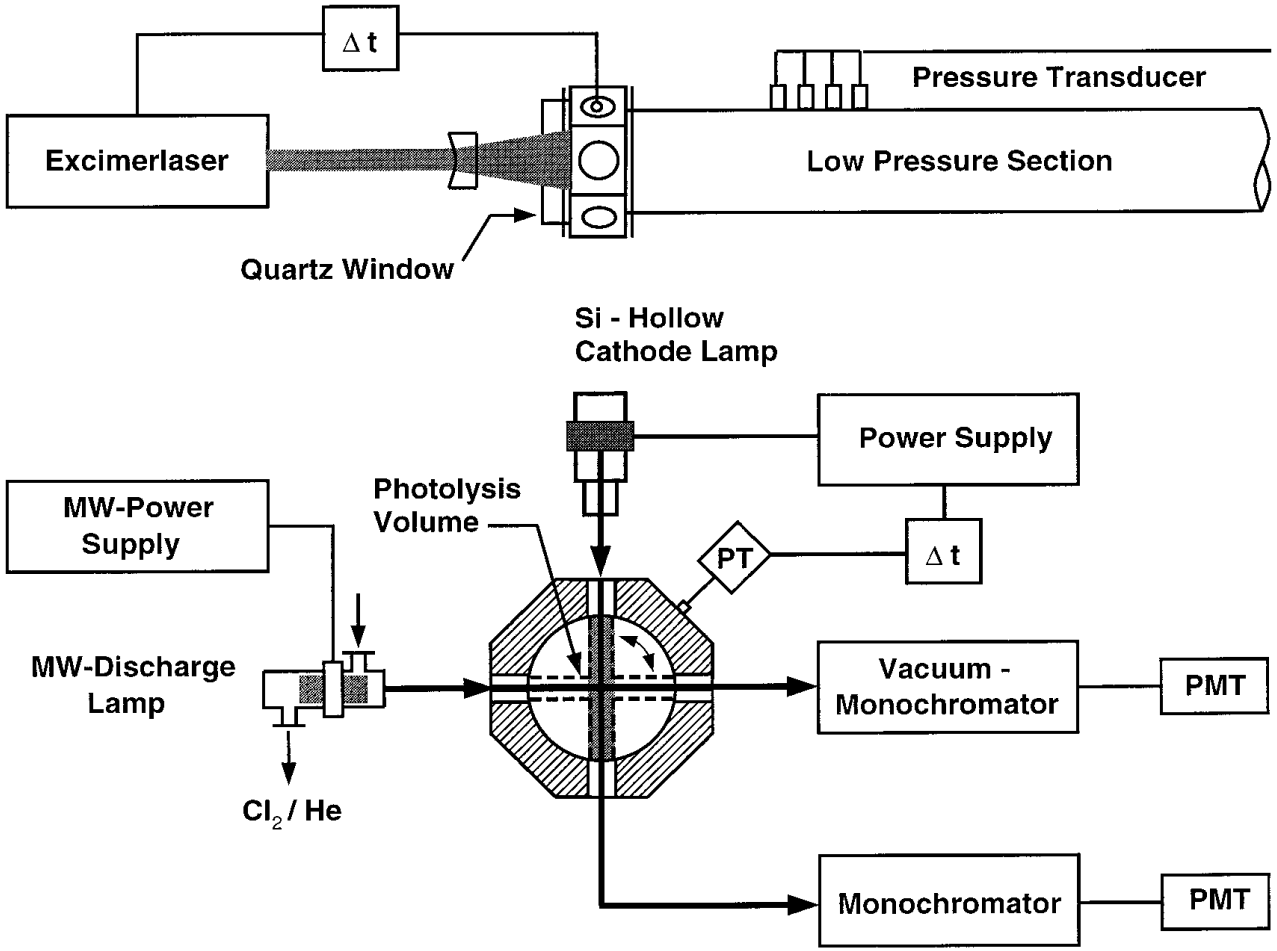
In combustion, photochemical processes can be used to measure elementary rate constants or to initiate ignition of combustible mixtures.

The laser flash photolysis consists in initiating the formation of chain carriers by a pulse of laser light

The kinetics of complex reactions (49)

Photochemistry (3)

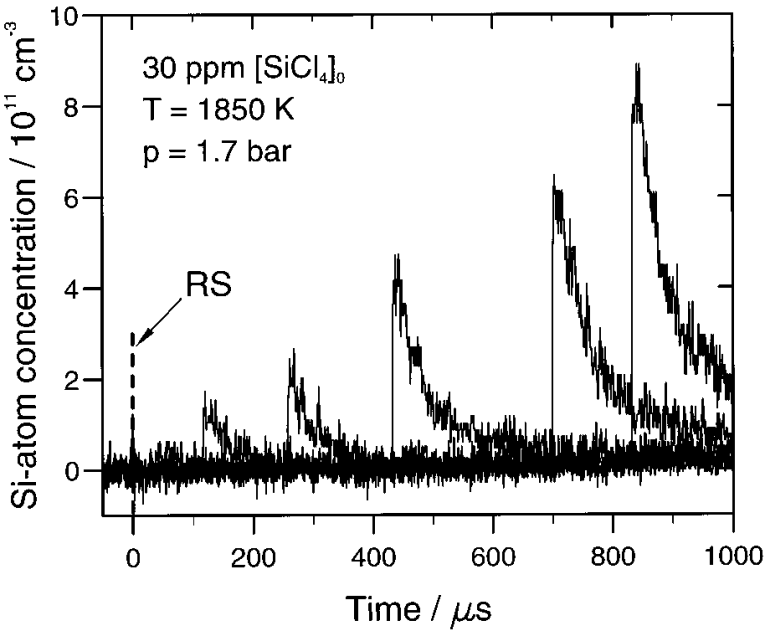
Example 1 : shock-tube/pulsed laser photolysis facility



The kinetics of complex reactions (50)

Photochemistry (4)

Typical experimental signal



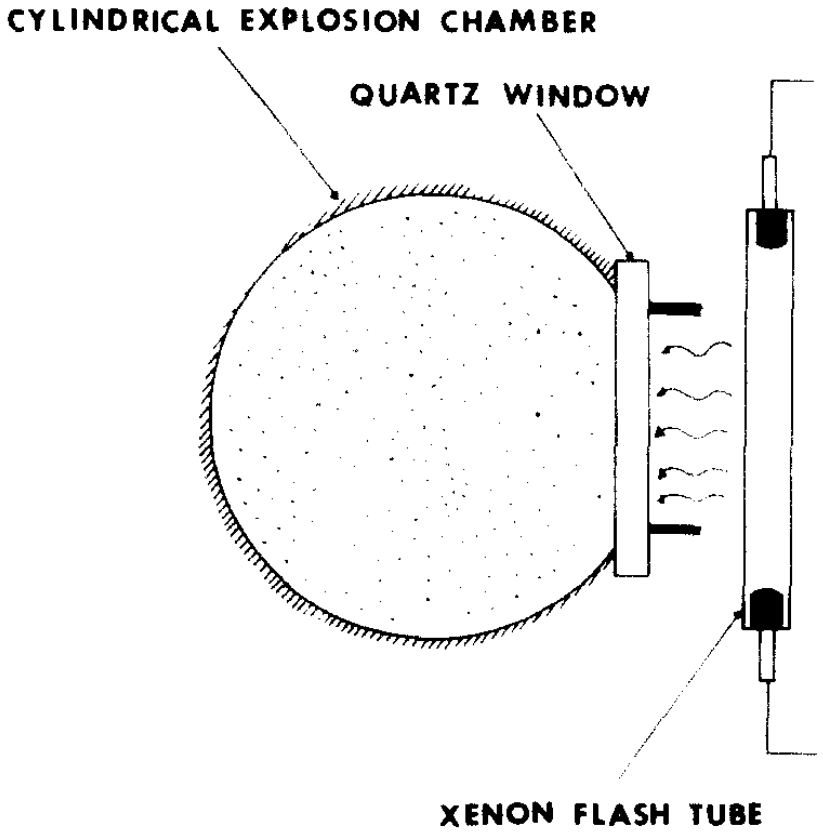
For each pulse of light, rapid formation of radical

$=>$ Reaction rate can be measured from the signal decrease

The kinetics of complex reactions (51)

Photochemistry (5)

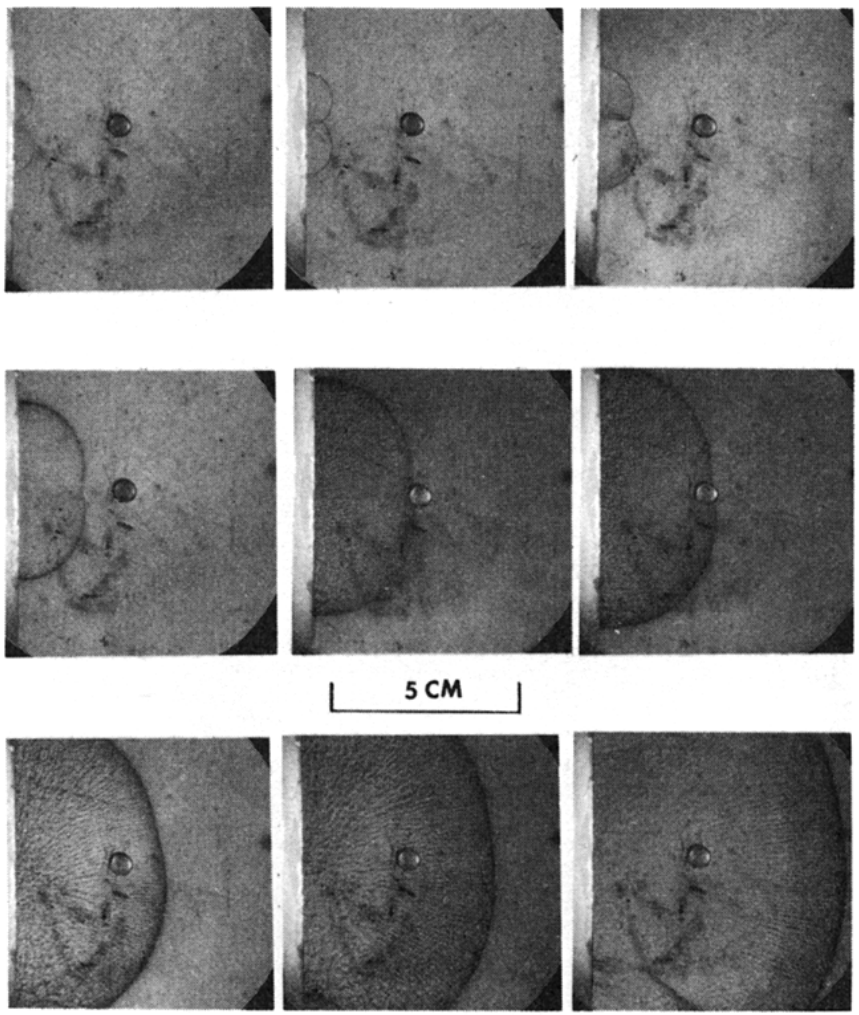
Example 2 : explosion initiation by flash photolysis



The kinetics of complex reactions (52)

Photochemistry (6)

Detonation initiation in a H_2-Cl_2 mixture



outlines

1 Introduction

2 Basic definitions in chemical kinetics

3 Mathematical description of reaction rate

4 The kinetics of complex reactions

5 Kinetics modeling in combustion

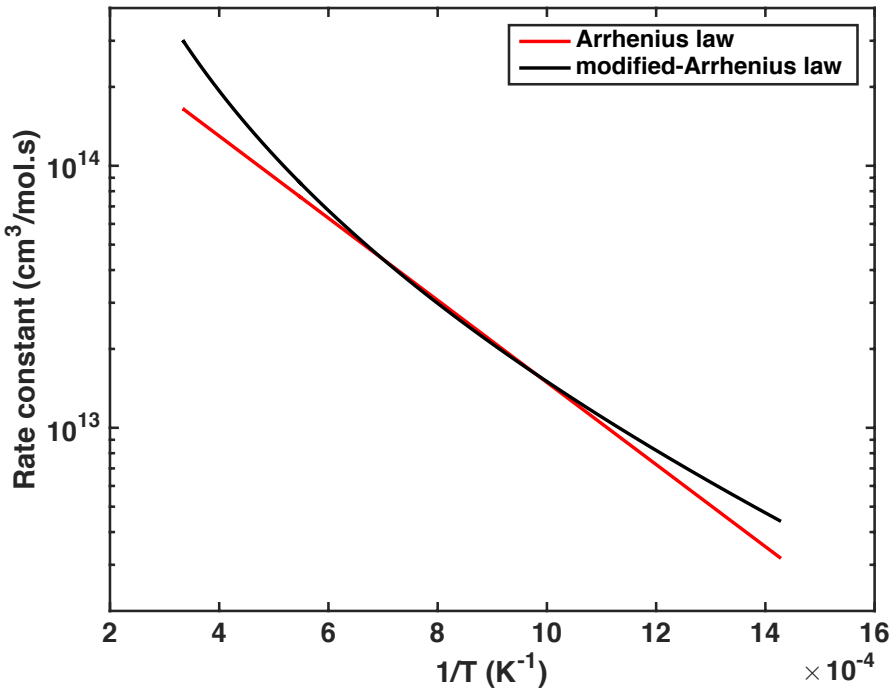
6 Chemical kinetics studies in combustion

7 Numerical analyses of chemical kinetics

Kinetics modeling in combustion (1)

Modified-Arrhenius law

$$k = AT^n \exp\left(\frac{-E_a}{RT}\right)$$



Declaration in software

$R = P$ A n E_a

Kinetics modeling in combustion (2)

Reversible and irreversible reactions (1)

Reversible reaction

$$R = P$$

Irreversible reaction

$$R => P$$

If available, the kinetics parameters of the reverse reaction may be provided to the software

$$R = P \qquad A_f \quad n_f \quad E_{a,f}$$

REV / $A_b \quad n_b \quad E_{a,b}$/

Kinetics modeling in combustion (3)

Reversible and irreversible reactions (2)

If no kinetics parameters are available for the reverse reaction, its rate constant is calculated using the equilibrium constant (k_{eq})

$$k_{eq} = \frac{k_f}{k_b} = k_p \left(\frac{P}{RT}\right)^{\sum \nu_j} = \left(\frac{P}{RT}\right)^{\sum \nu_j} \exp\left(\frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT}\right)$$

where

$$\frac{\Delta_r S^\circ}{R} = \sum_{j=1}^{J} \nu_j \frac{S_j^\circ}{R}$$

and

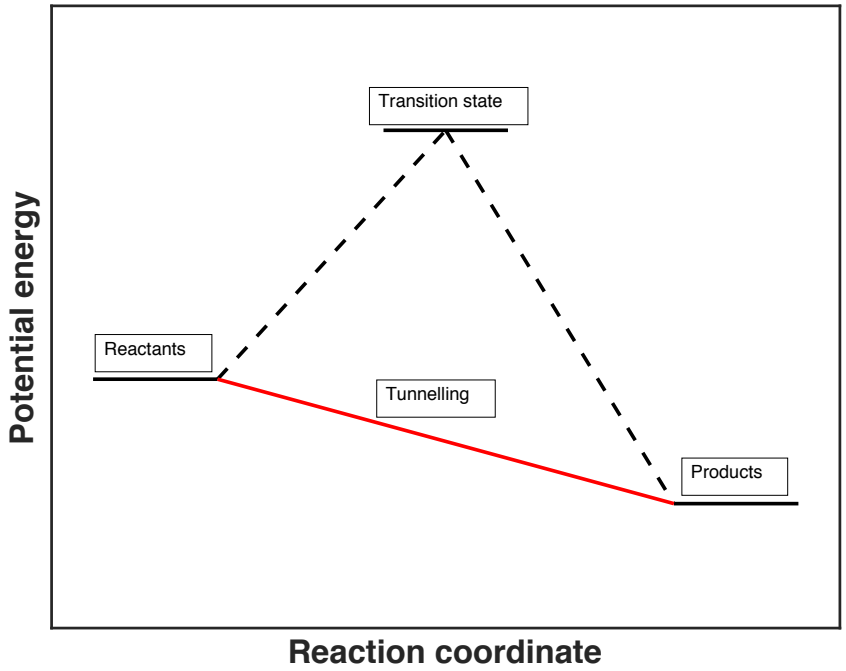
$$\frac{\Delta_r H^\circ}{RT} = \sum_{j=1}^{J} \nu_j \frac{H_j^\circ}{RT}$$

Kinetics modeling in combustion (4)

Second-order reactions (1)

In general, second-order reactions are simply described using the modified-Arrhenius law. However, for some reactions, the rate constant is much higher at low temperature that can be expected from extrapolating the high-temperature rate constant

$\Rightarrow$ **Quantum tunneling effect**



Kinetics modeling in combustion (5)

Second-order reactions (2)

To account for the quantum tunneling effect

$$k_{tot} = A_1 T^{n_1} \exp\left(\frac{-E_{a,1}}{RT}\right) + A_2 T^{n_2} \exp\left(\frac{-E_{a,2}}{RT}\right)$$

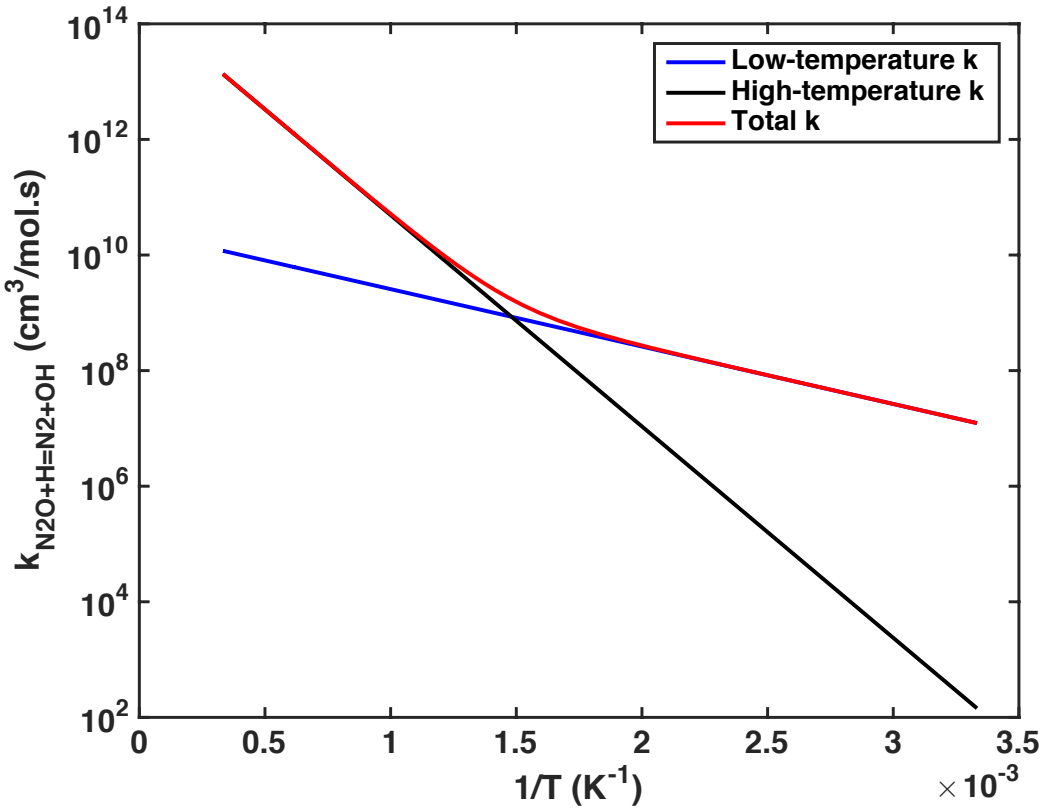
Declaration in combustion software

$R = P$	A_1	n_1	$E_{a,1}$
Dup			
$R = P$	A_2	n_2	$E_{a,2}$
Dup			

Kinetics modeling in combustion (6)

Second-order reactions (3)

Graphical representation



Kinetics modeling in combustion (7)

Unimolecular reactions (1)

Physical issue with unimolecular reactions : only one species involved $=>$ no collision ?

Lindemann-Hinshelwood mechanism : A energetically excited by collision with M

$$R_1 : A + M \rightarrow A^* + M$$

$$R_2 : A^* + M \rightarrow A + M$$

$$R_3 : A^* \rightarrow P$$

Kinetics modeling in combustion (8)

Unimolecular reactions (2)

Rate laws for these 3 reactions

$$\frac{d[A^*]}{dt} = k_1[A][M]$$

$$\frac{d[A^*]}{dt} = -k_2[A^*][M]$$

$$\frac{d[A^*]}{dt} = -k_3[A^*]$$

The QSSA can be applied to A^*

$$\frac{d[A^*]}{dt} = k_1[A][M] - k_2[A^*][M] - k_3[A^*] \simeq 0$$

which leads to

$$[A^*] = \frac{k_1[A][M]}{k_2[M]+k_3}$$

Kinetics modeling in combustion (9)

Unimolecular reactions (3)

The reaction rate is given

$$v = \frac{d[P]}{dt} = k_3[A^*] = \frac{k_1 k_3 [A][M]}{k_2[M]+k_3}$$

In the limit of low pressure, $k_2[M] \ll k_3$, so that

$$v = \frac{d[P]}{dt} = k_1[A][M]$$

$=>$ second-order rate law with a pressure-dependence

In the limit of high pressure, $k_2[M] \gg k_3$, so that

$$v = \frac{d[P]}{dt} = \frac{k_1 k_3}{k_2}[A] = k'[A]$$

$=>$ first-order rate law, with no pressure-dependence

Kinetics modeling in combustion (10)

Lindemann and Troe formalism (1)

Used to describe

Decomposition reactions : $CH_4 + M = CH_3 + H + M$

Recombination reactions : $H + OH + M = H_2O + M$

For the Lindemann model, the declaration in combustion software

$CH_4(+M) = CH_3 + H(+M)$ $\qquad A_\infty \quad n_\infty \quad E_{a,\infty}$

LOW / $A_0 \quad n_0 \quad E_{a,0}$ /

The rate constant is calculated using

$$k = k_\infty \left(\frac{P_r}{1+P_r} \right) F$$

Kinetics modeling in combustion (11)

Lindemann and Troe formalism (2)

with

$$P_r = \frac{k_0[M]}{k_\infty}$$

$$k_0 = A_0 T^{n_0} \exp\left(\frac{-E_{a,0}}{RT}\right)$$

$$k_\infty = A_\infty T^{n_\infty} \exp\left(\frac{-E_{a,\infty}}{RT}\right)$$

F is the "broadening factor"

For Lindemann : F=1

Kinetics modeling in combustion (12)

Lindemann and Troe formalism (3)

For the Troe model, the declaration in combustion software

$CH_4(+M) = CH_3 + H(+M)$ A_∞ n_∞ $E_{a,\infty}$

LOW / A_0 n_0 $E_{a,0}$ /

TROE / α T^{***} T^{*} T^{**} /

The rate constant is calculated using

$$k = k_\infty \left(\frac{P_r}{1+P_r} \right) F$$

with

$$\log F = \left(1 + \left(\frac{\log P_r + c}{n - d(\log P_r + c)} \right)^2 \right)^{-1} \log F_{cent}$$

Kinetics modeling in combustion (13)

Lindemann and Troe formalism (4)

$$c = -0.4 - 0.67 \log F_{cent}$$

$$n = 0.75 - 1.27 \log F_{cent}$$

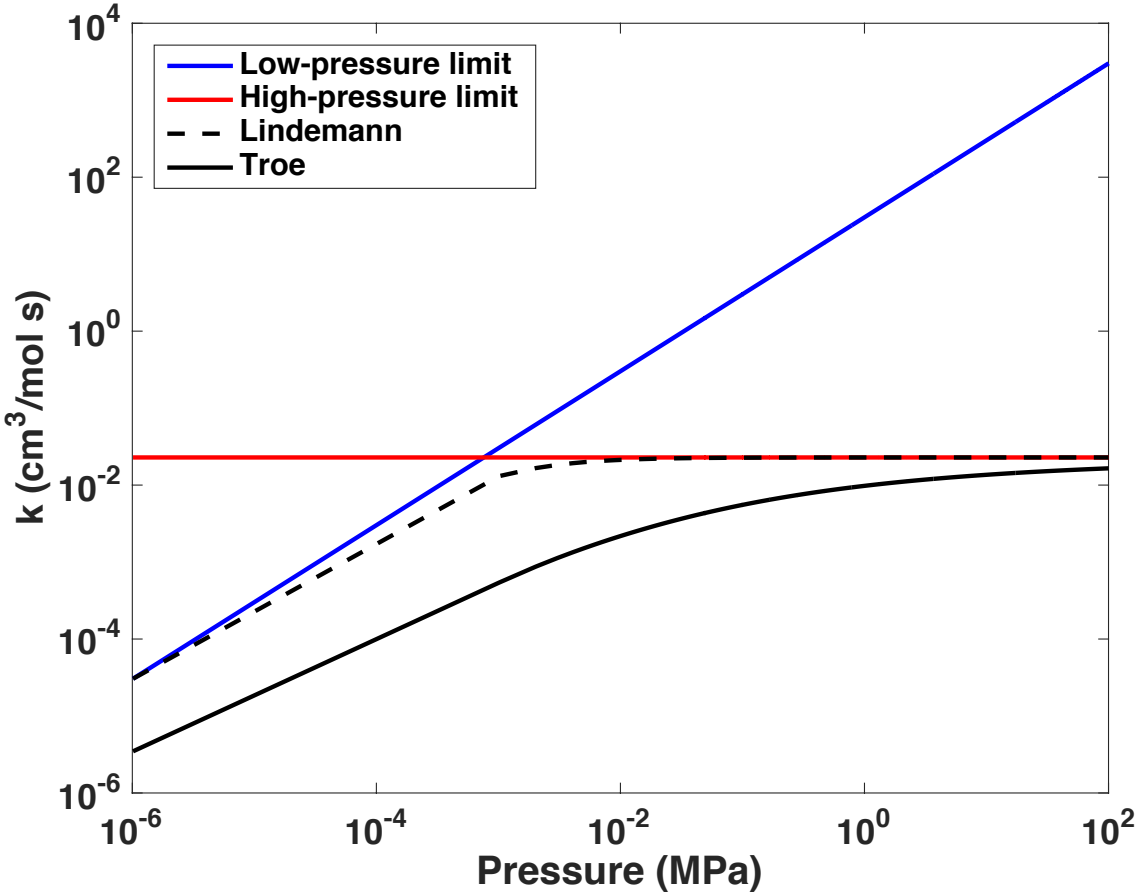
$$d = 0.14$$

$$F_{cent} = (1-\alpha) \exp\left(\frac{-T}{T^{***}}\right) + \alpha \exp\left(\frac{-T}{T^{*}}\right) + \exp\left(\frac{-T^{**}}{T}\right)$$

Kinetics modeling in combustion (14)

Lindemann and Troe formalism (5)

Example : decomposition of CH_3CHO



Kinetics modeling in combustion (15)

PLOG formalism (1)

New formalism, more precise than Troe model, rate given at several pressure

$CH_3 + OH = CH_2O + M_2 \quad 3.50 \times 10^{+05} \quad 1.441 \quad -3.24 \times 10^{3}$
PLOG / 0.01 $3.50 \times 10^{+05}$ 1.44 $-3.24 \times 10^{+03}$ /
PLOG / 0.1 $8.85 \times 10^{+05}$ 1.33 $-2.98 \times 10^{+003}$ /
PLOG / 1 $1.65 \times 10^{+07}$ 0.97 $-2.01 \times 10^{+003}$ /
PLOG / 10 $5.37 \times 10^{+09}$ 0.29 $2.80 \times 10^{+002}$ /
PLOG / 100 $9.49 \times 10^{+18}$ -2.20 $9.77 \times 10^{+003}$ /

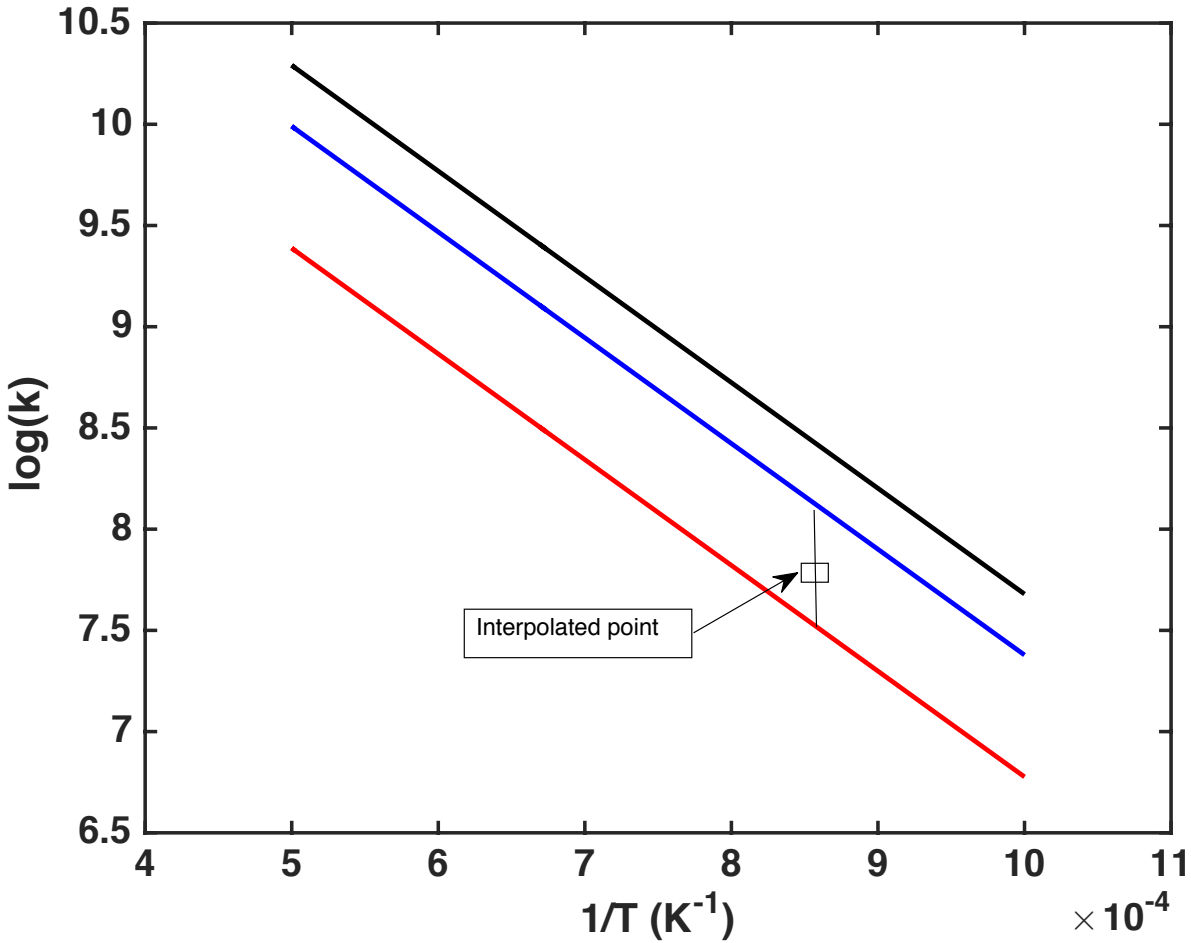
For intermediate pressures, the rate constant is interpolated using

$$\log k(T, P_i) = \log k_i + (\log P - \log P_i) \frac{\log k_{i+1} - \log k_i}{\log P_{i+1} - \log P_i}$$

Kinetics modeling in combustion (16)

PLOG formalism (2)

Graphical representation



outlines

1 Introduction

2 Basic definitions in chemical kinetics

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4 The kinetics of complex reactions

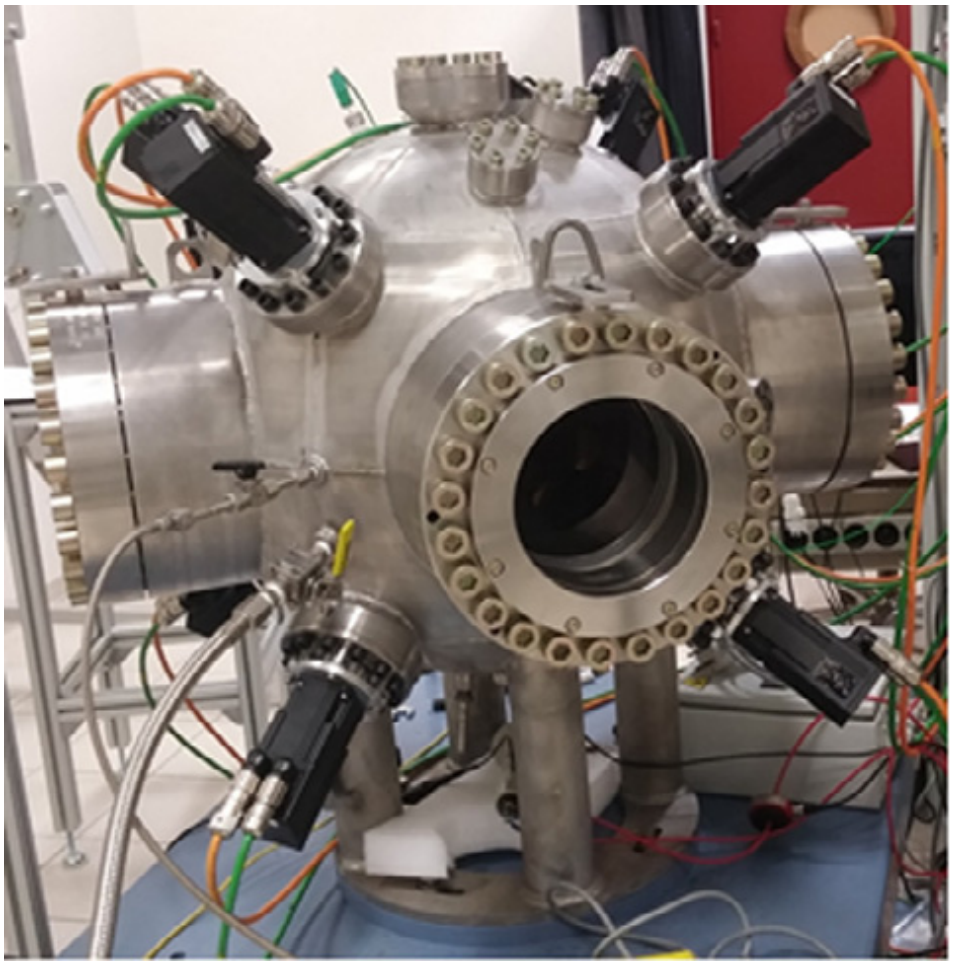
5 Kinetics modeling in combustion

6 Chemical kinetics studies in combustion

7 Numerical analyses of chemical kinetics

Chemical kinetics studies in combustion (1)

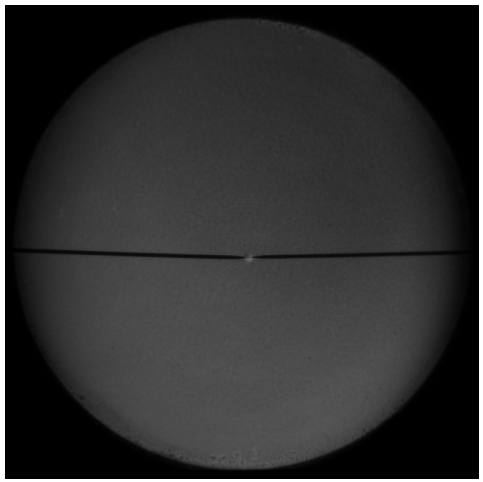
Constant volume combustion chamber (1)



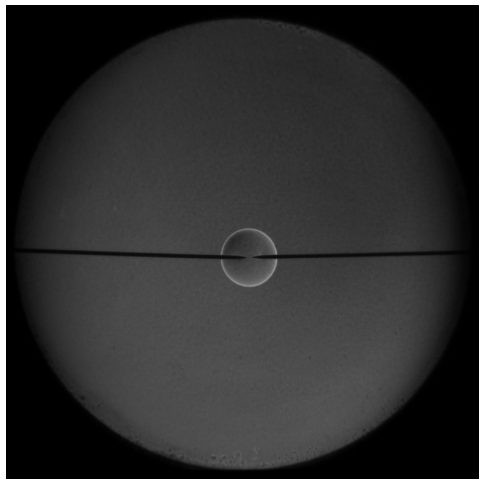
$=>$ **Mainly used to measure the laminar flame speed**

Chemical kinetics studies in combustion (2)

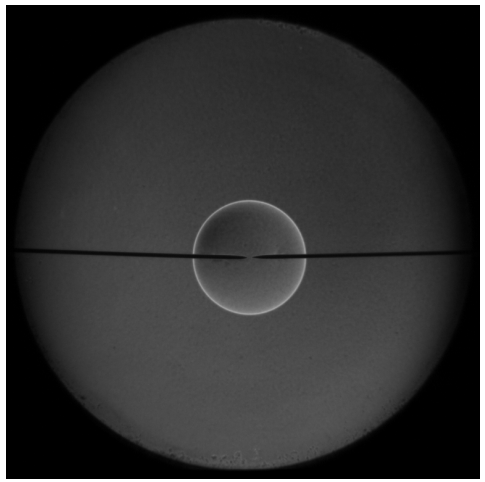
Constant volume combustion chamber (2)



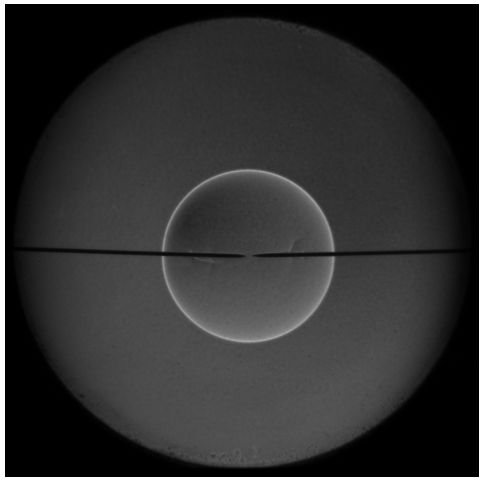
a) t=0 ms



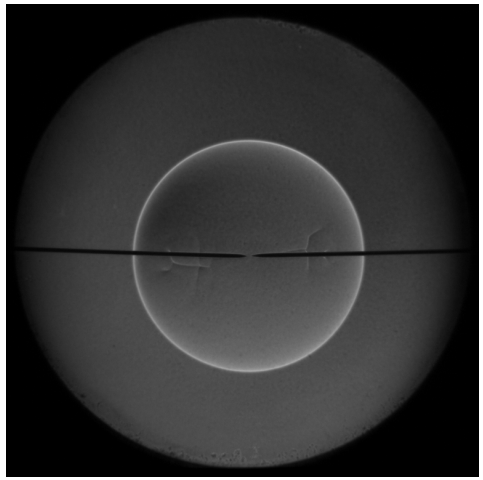
b) t=0.72 ms



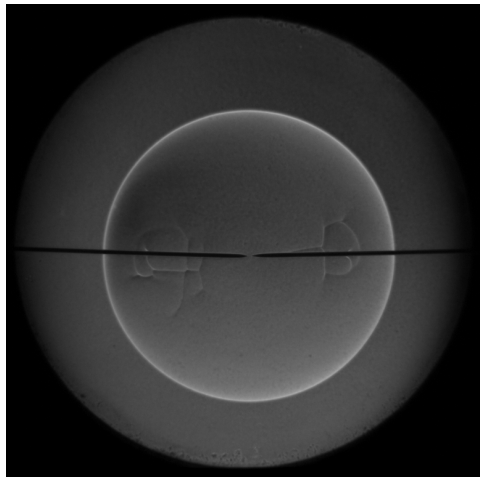
c) t=1.52 ms



d) t=2.32 ms



e) t=3.12 ms



f) t=3.92 ms

Chemical kinetics studies in combustion (3)

Constant volume combustion chamber (3)

$=>$ Laminar flame speed can be extracted from R_f=f(t), but not directly !

Relationship between the flame speed and stretch rate

$$S_L = S_L^0 - L \cdot K$$

Stretch rate given by

$$K = \frac{1}{A} \cdot \frac{dA}{dt}$$

For a spherical flame, the flame surface is given by

$$A = 4 \cdot \pi \cdot R_f^2$$

Stretch rate becomes

$$K = 2 \cdot \frac{V_S}{R_f}$$

with

$$V_S = \frac{dR_f}{dt}$$

Chemical kinetics studies in combustion (4)

Constant volume combustion chamber (4)

Relationship S_L and V_S

$$S_L = \frac{1}{\sigma} \cdot \left(V_S + \frac{R_b}{3 \cdot P_b} \cdot \frac{dP_b}{dt} \right)$$

with

$$\sigma = \frac{\rho_u}{\rho_b}$$

For P$\simeq$cst

$$S_L = \frac{V_S}{\sigma}$$

We finally obtain

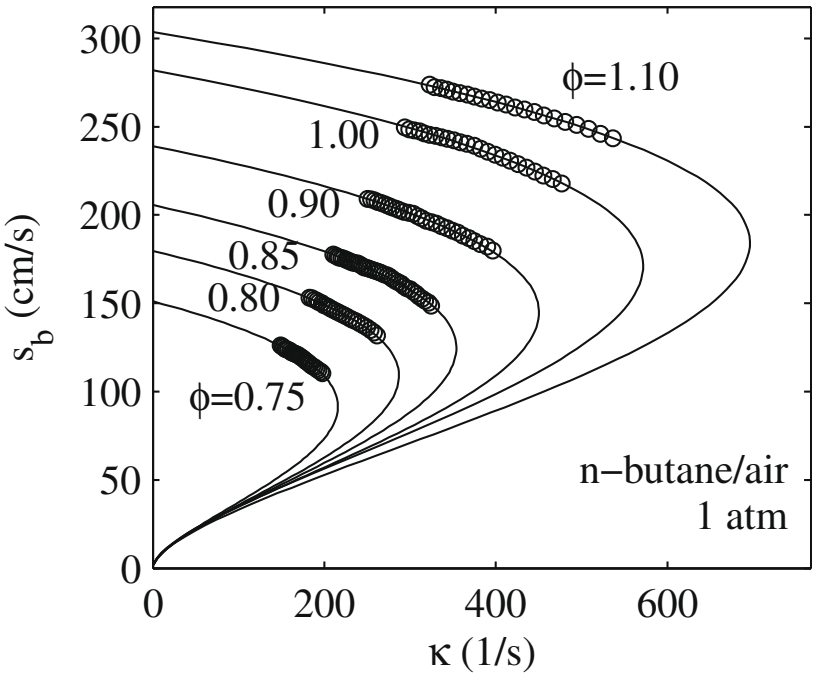
$$V_S^0 \cdot t = R_f + 2\sigma L \cdot ln(R_f) + C_{st}$$

$=>$ **Linear equation. There are other linear and non-linear equations !**

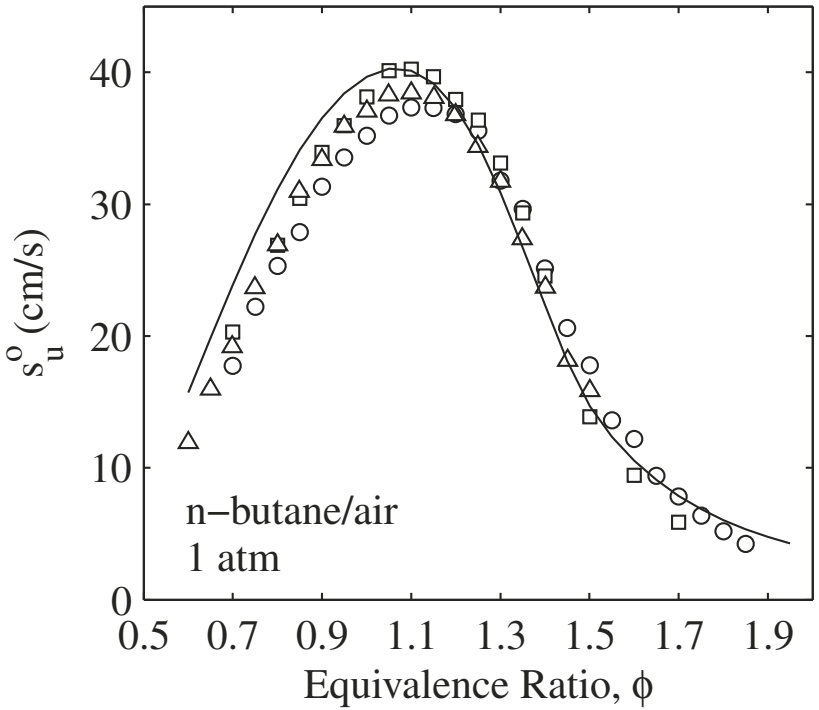
Chemical kinetics studies in combustion (5)

Constant volume combustion chamber (5)

Example of results



a) Non-linear extrapolation

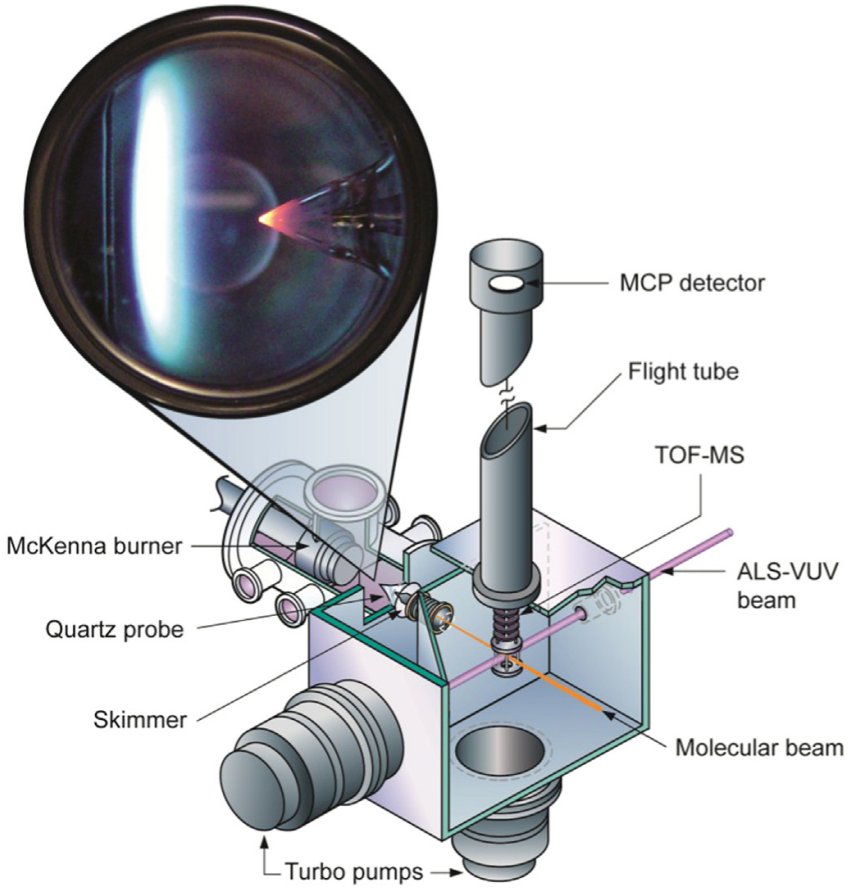


e) $S_L^0 = f(\Phi)$

$\Rightarrow$ **Chemical model predictions compared to experimental results**

Chemical kinetics studies in combustion (6)

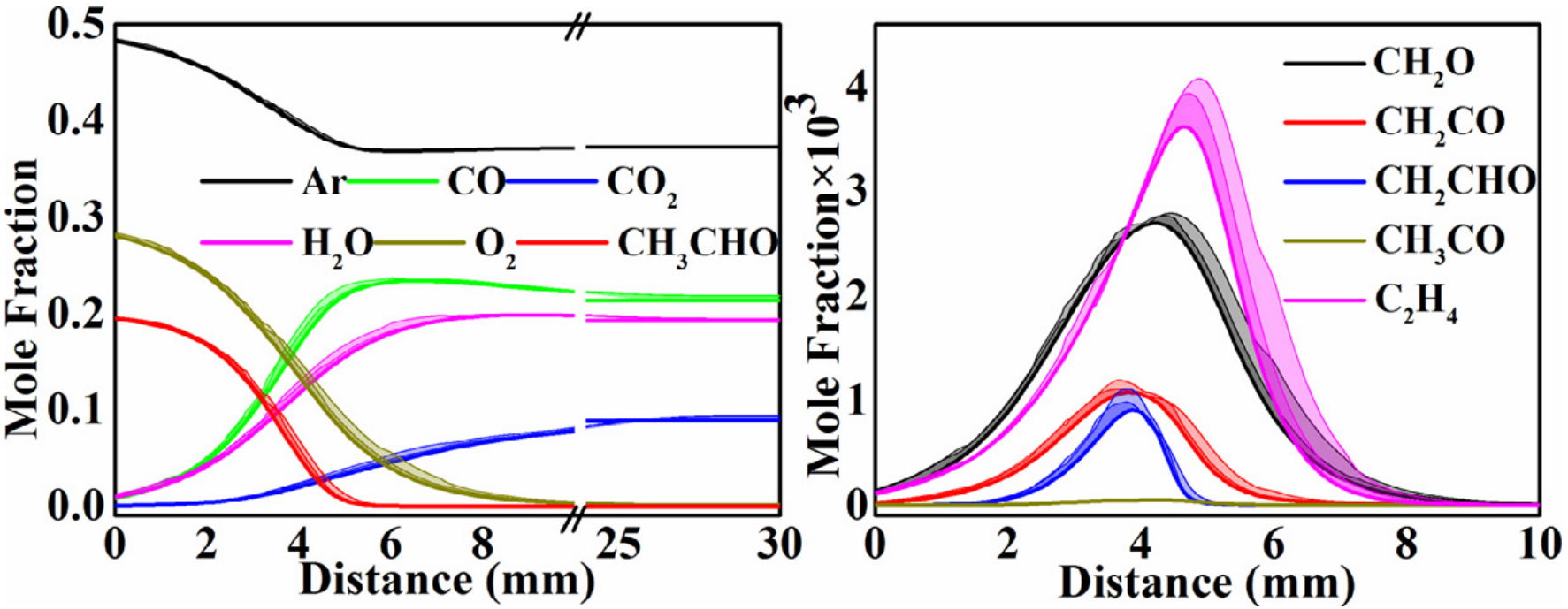
Flat flame burner (1)



$=>$ **Species profiles as a function of distance above the burner**

Chemical kinetics studies in combustion (7)

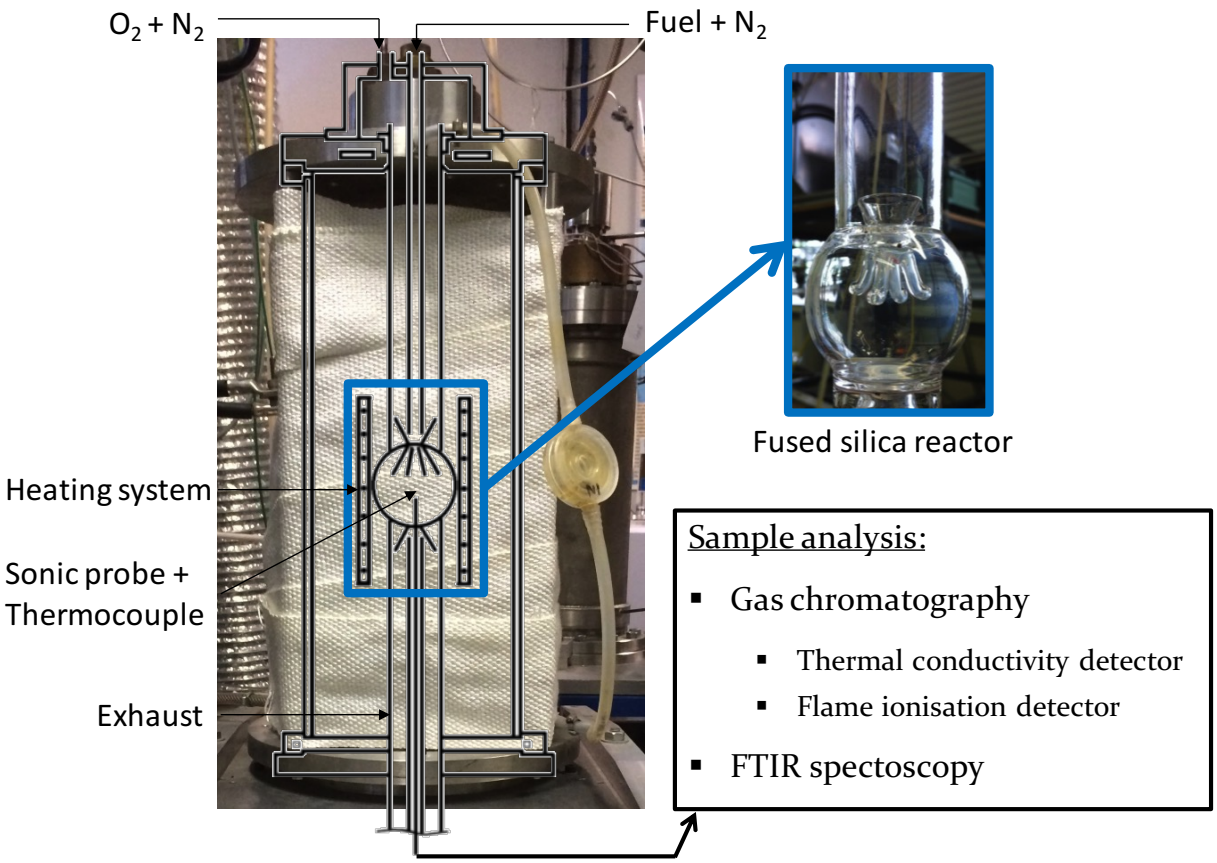
Flat flame burner (2)



$=>$ **Comparison with the predictions of a chemical model**

Chemical kinetics studies in combustion (8)

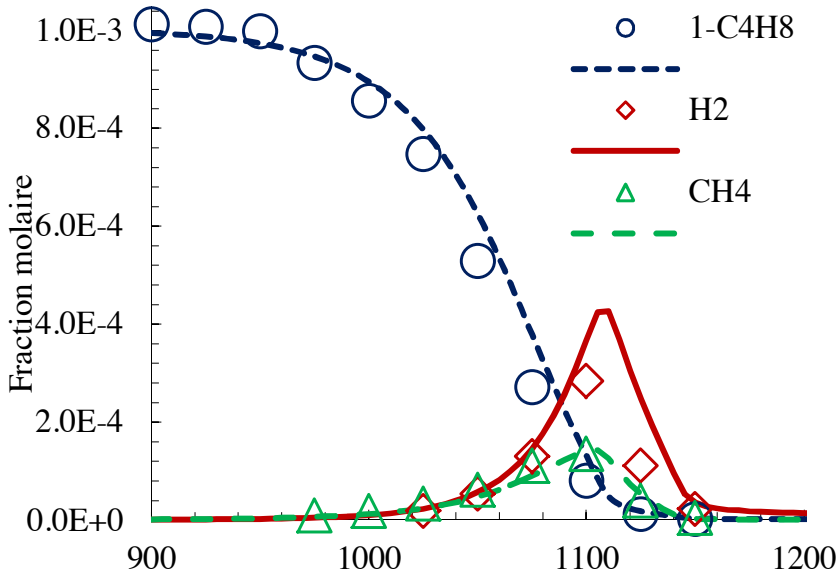
Jet-stirred reactor (1)

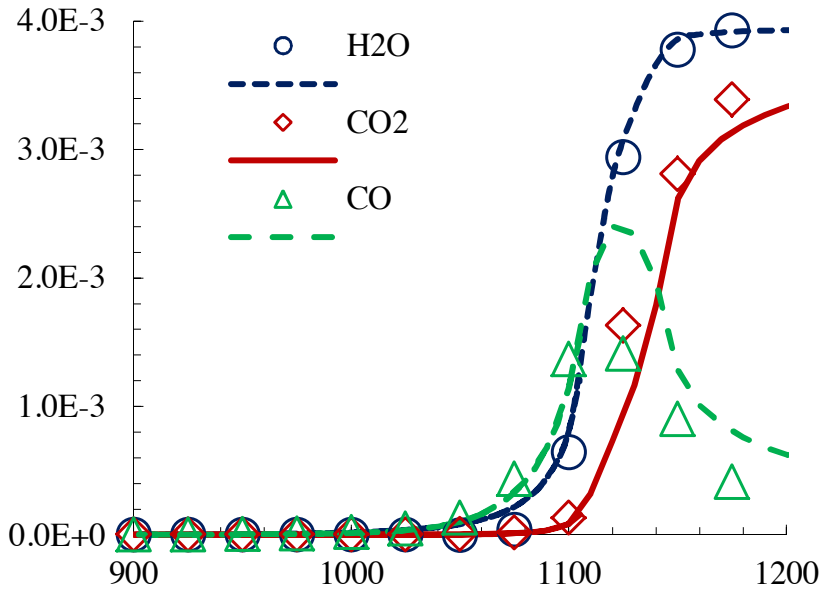


$=>$ **Species profiles as a function of temperature or residence time**

Chemical kinetics studies in combustion (9)

Jet-stirred reactor (2)

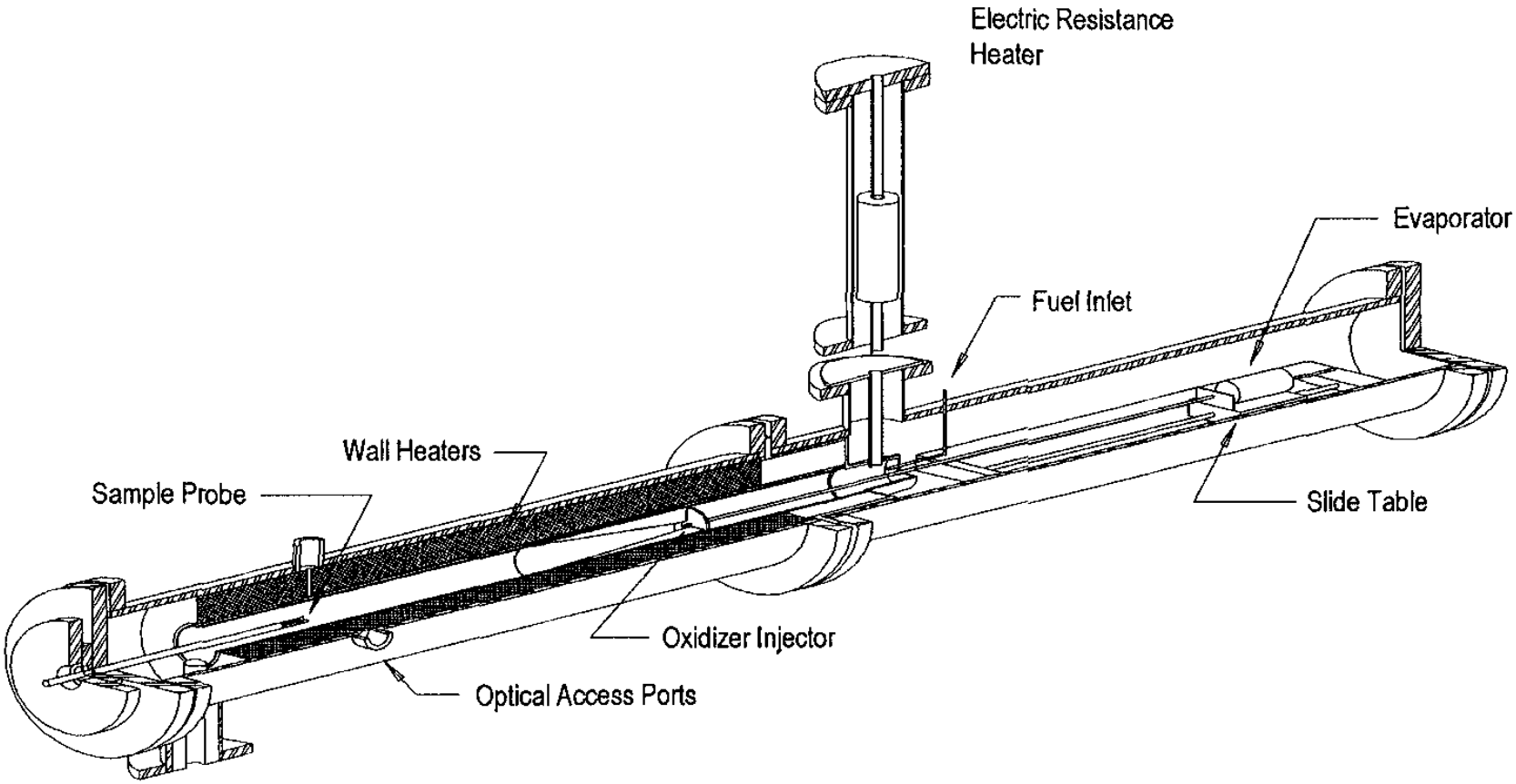




$=>$ **Comparison with the predictions of a chemical model**

Chemical kinetics studies in combustion (10)

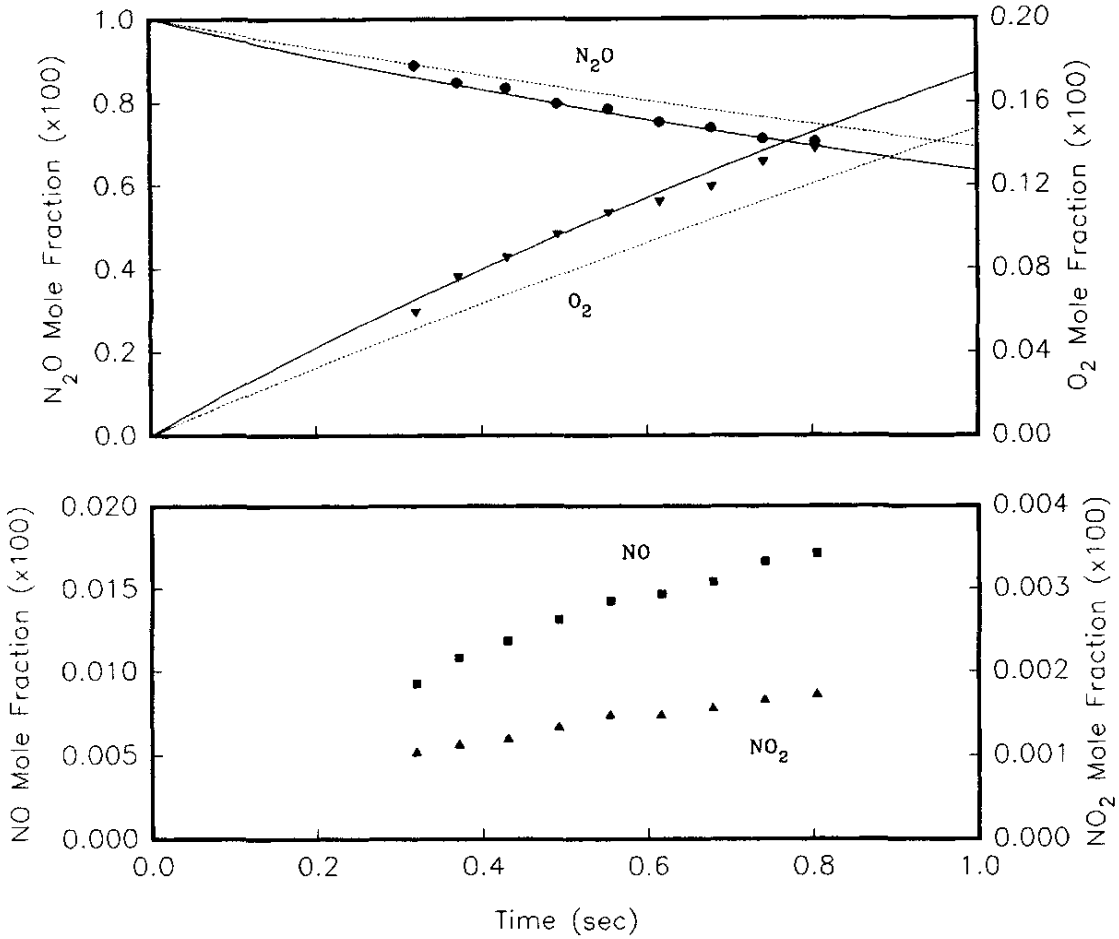
Flow reactor (1)



$=>$ **Species profiles as a function of temperature or residence time**

Chemical kinetics studies in combustion (11)

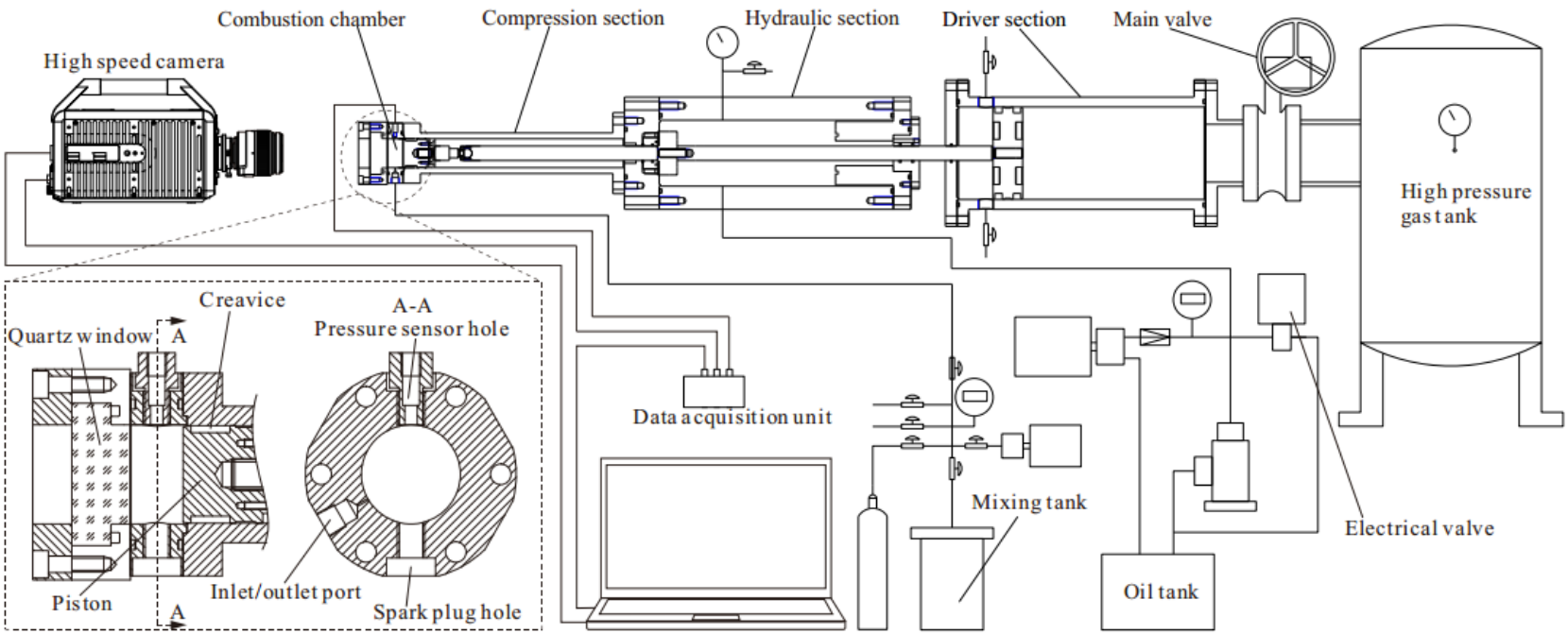
Flow reactor (2)



$\Rightarrow$ **Comparison with the predictions of a chemical model**

Chemical kinetics studies in combustion (12)

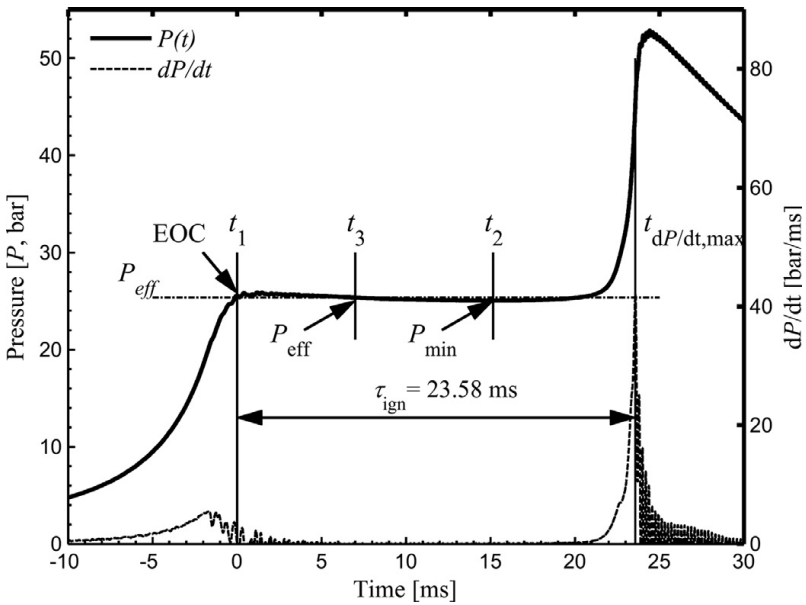
Rapid compression machine (1)

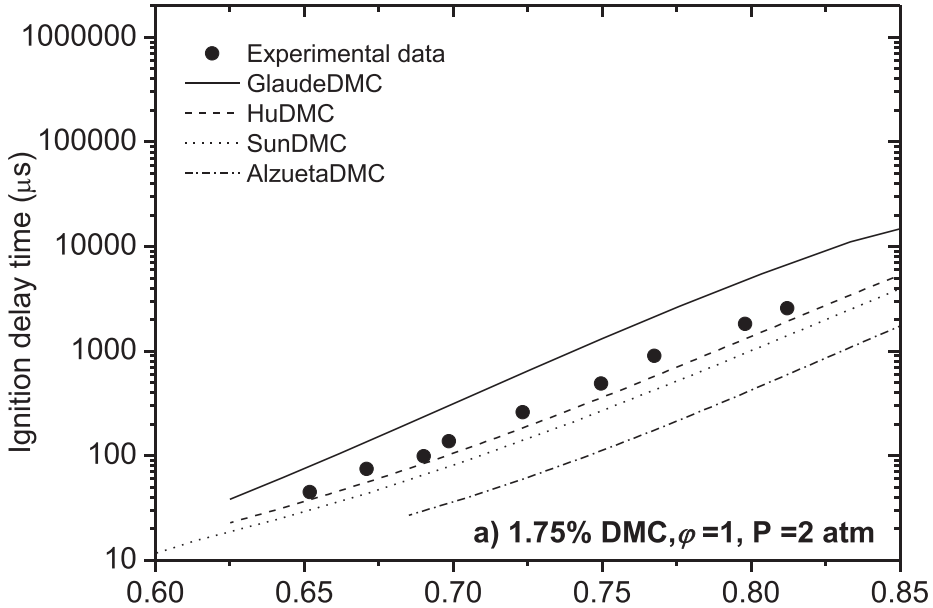


$=>$ **Ignition delay-time or species profiles as a function of temperature or residence time**

Chemical kinetics studies in combustion (13)

Rapid compression machine (2)

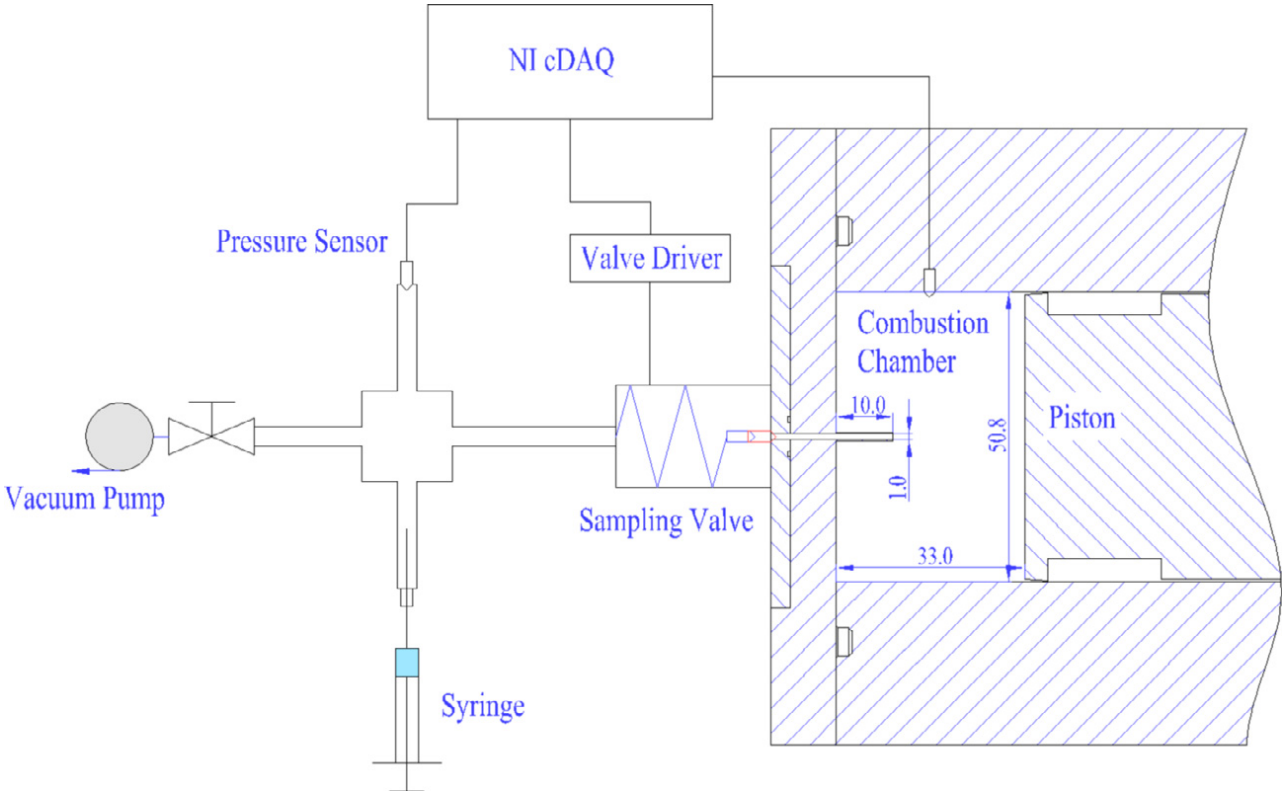




=> **Comparison with the predictions of a chemical model**

Chemical kinetics studies in combustion (14)

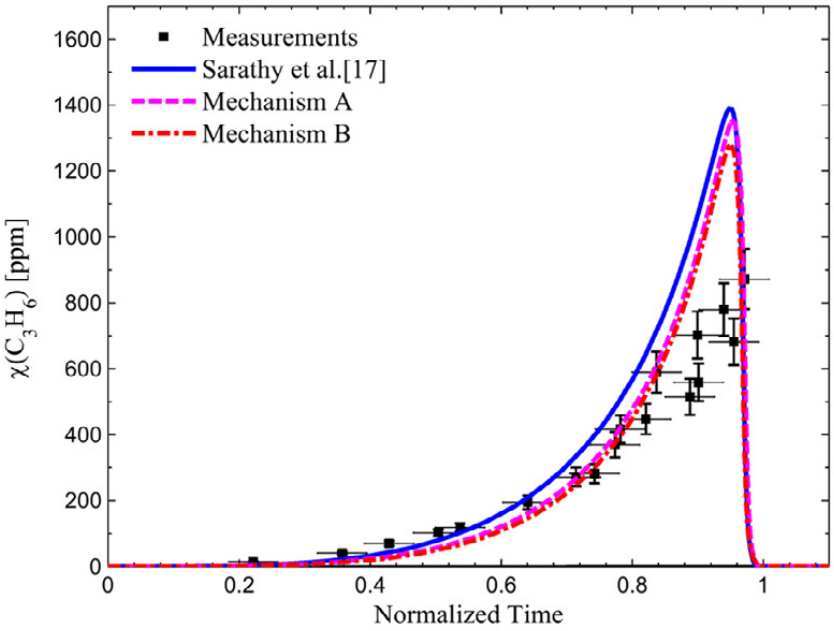
Rapid compression machine (3)

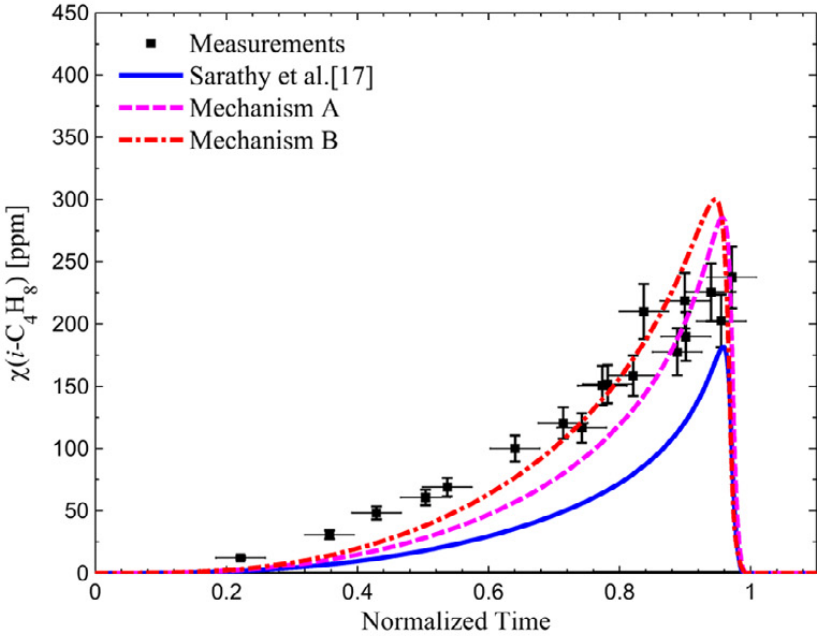


$=>$ **Sampling system**

Chemical kinetics studies in combustion (15)

Rapid compression machine (4)

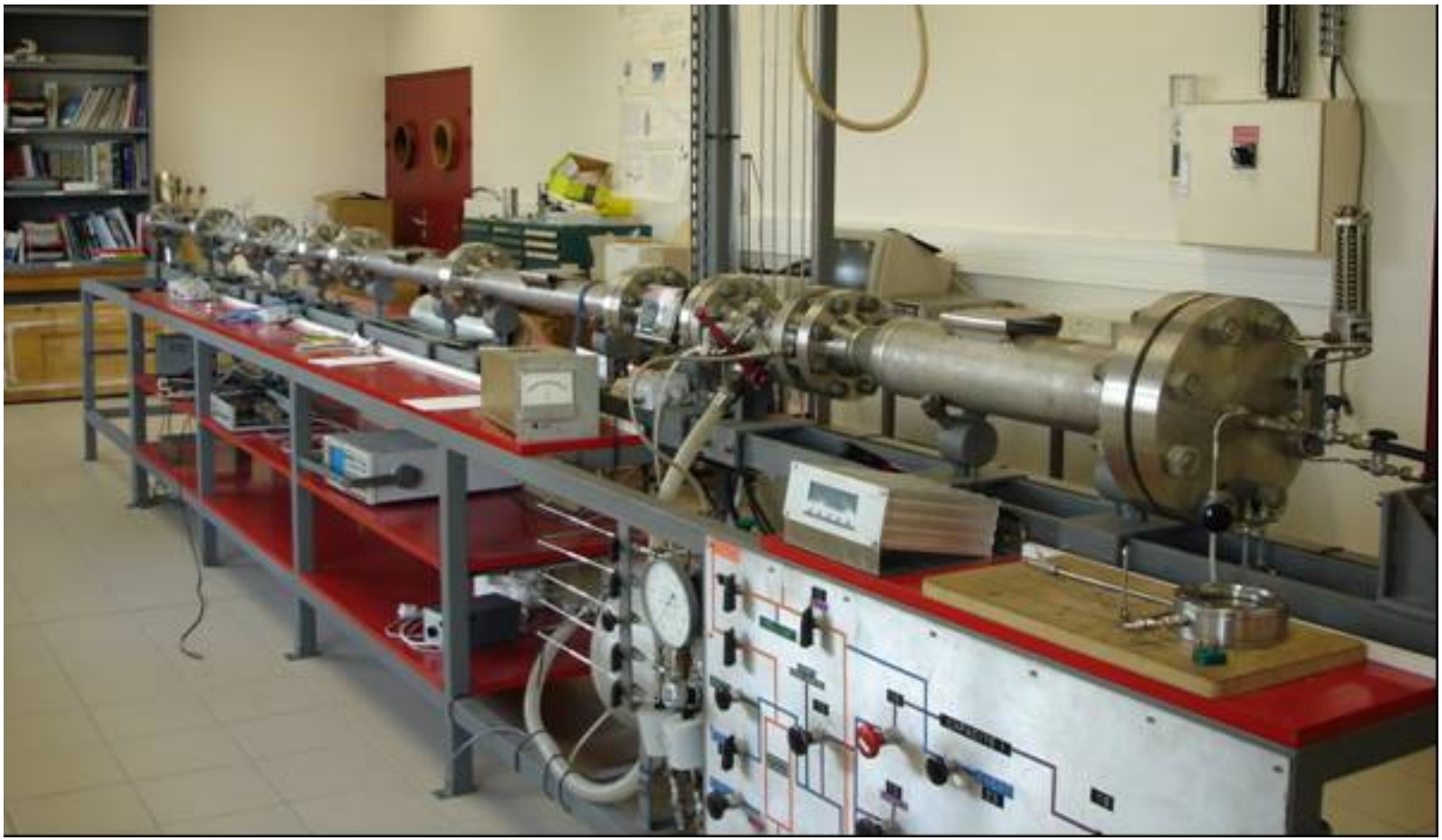




$\Rightarrow$ **Comparison with the predictions of a chemical model**

Chemical kinetics studies in combustion (16)

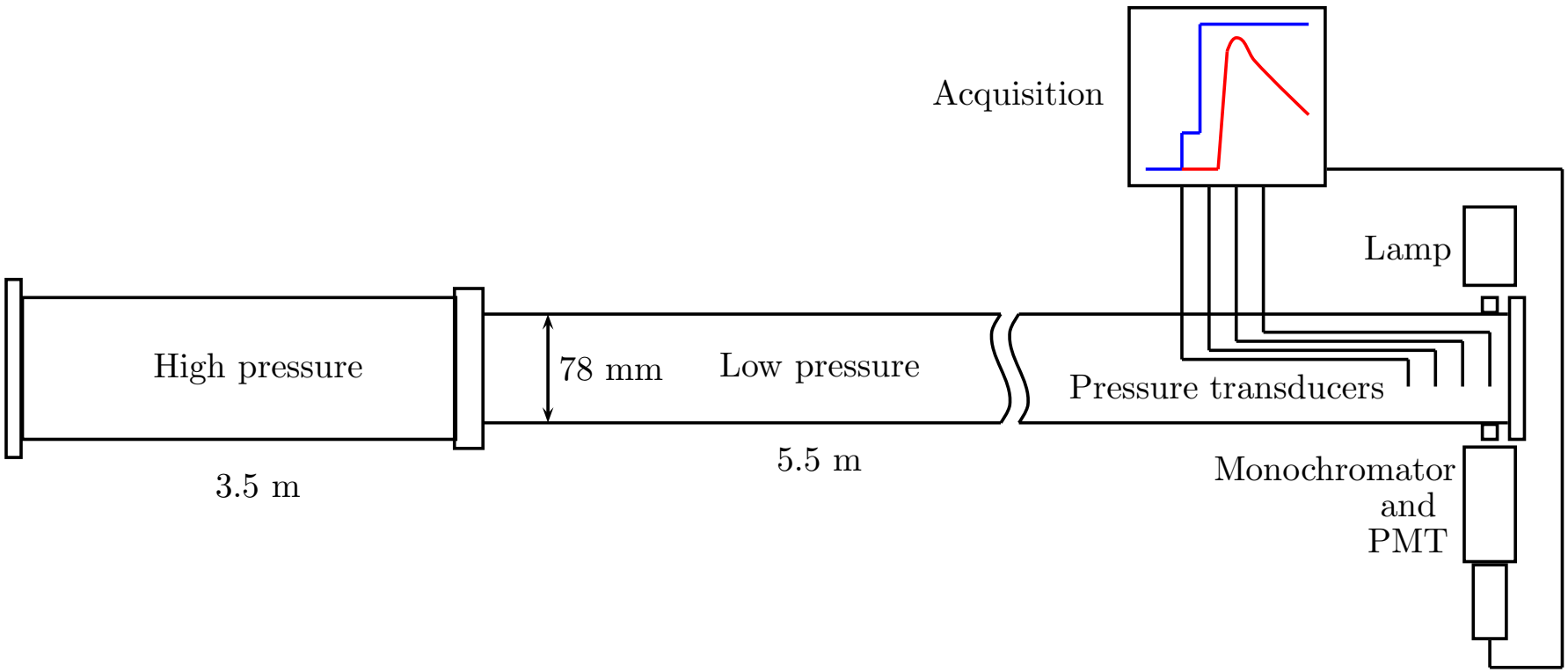
Shock tube (1)



$\Rightarrow$ **Enables to study reactive systems at high-temperature ($>$1000 K)**

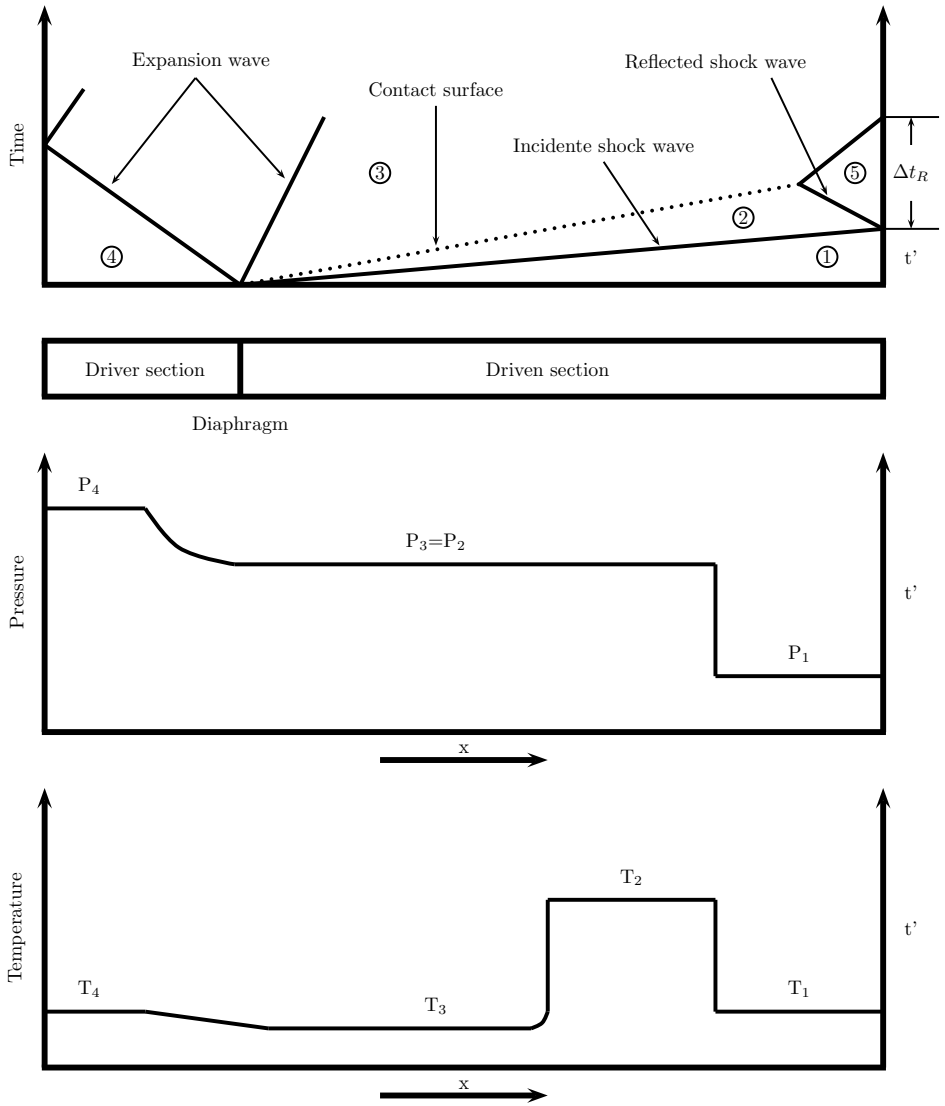
Chemical kinetics studies in combustion (17)

Shock tube (2)



Chemical kinetics studies in combustion (18)

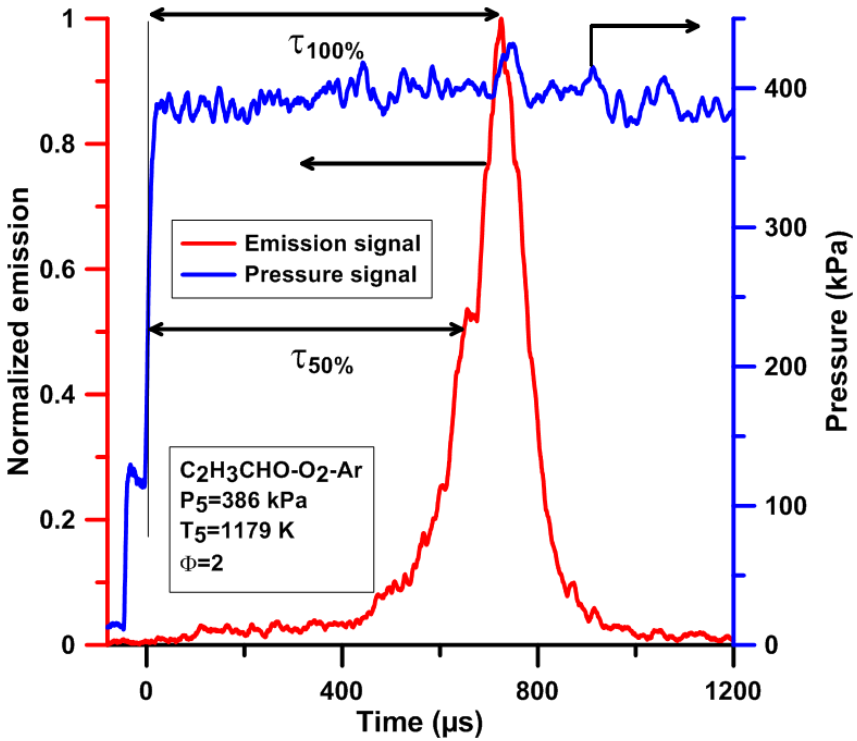
Shock tube (3)

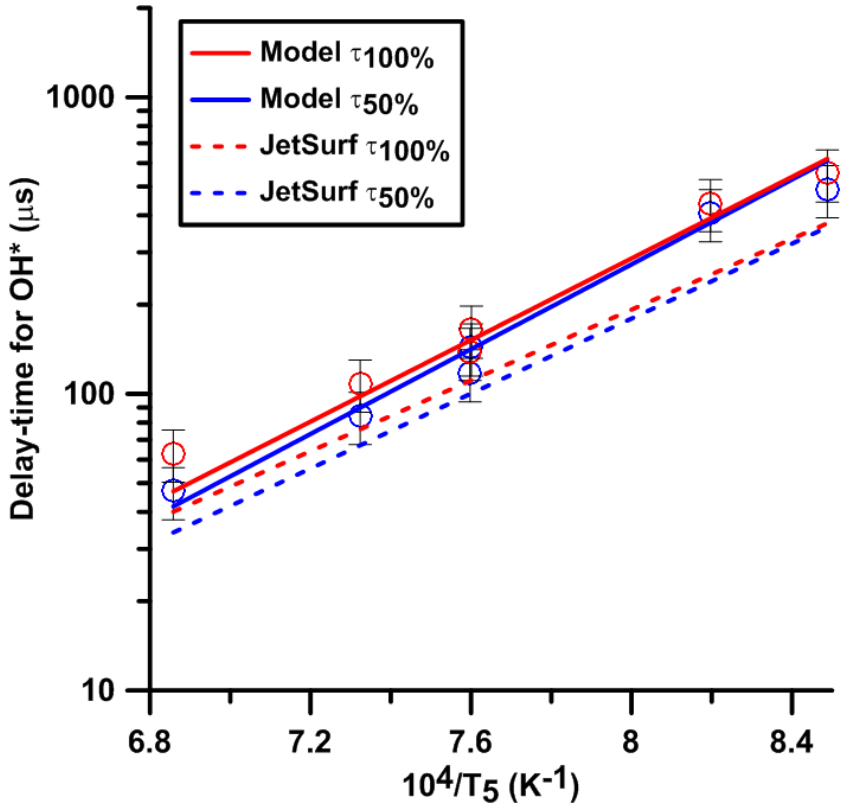


Chemical kinetics studies in combustion (19)

Shock tube (4)

Typical data : ignition delay-time (or characteristic time of reaction)



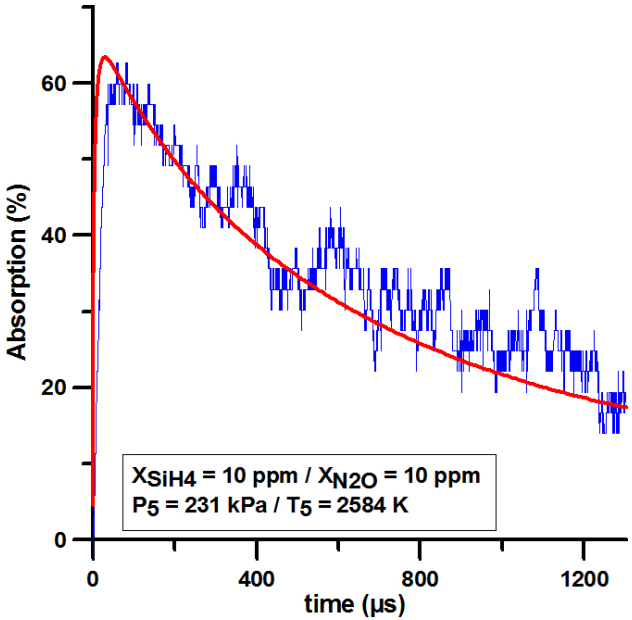


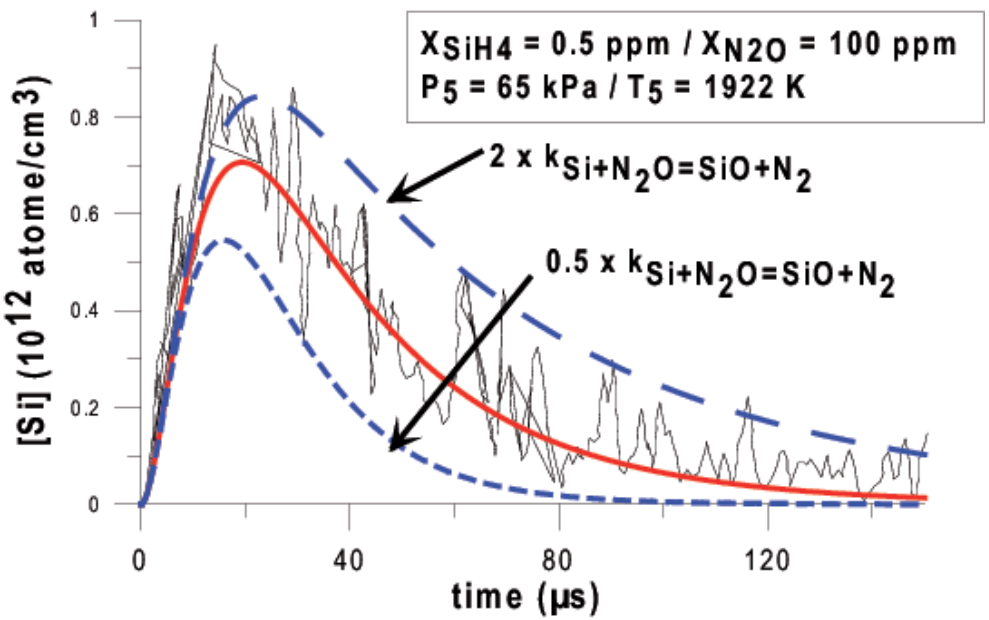
=> **Comparison with the predictions of a chemical model**

Chemical kinetics studies in combustion (20)

Shock tube (5)

Typical data : absorption or species concentration profiles



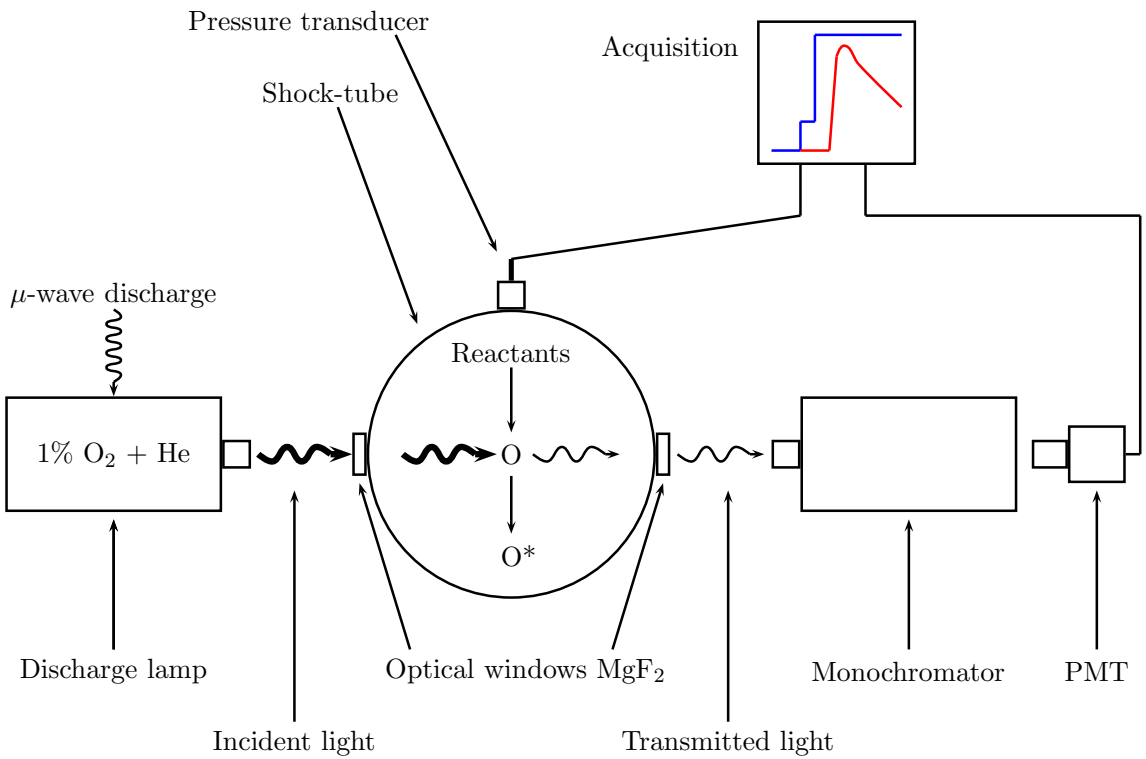


=> **Comparison with the predictions of a chemical model**

Chemical kinetics studies in combustion (21)

Rate constant measurement (1)

Spectroscopic diagnostics : atomic resonant absorption spectroscopy

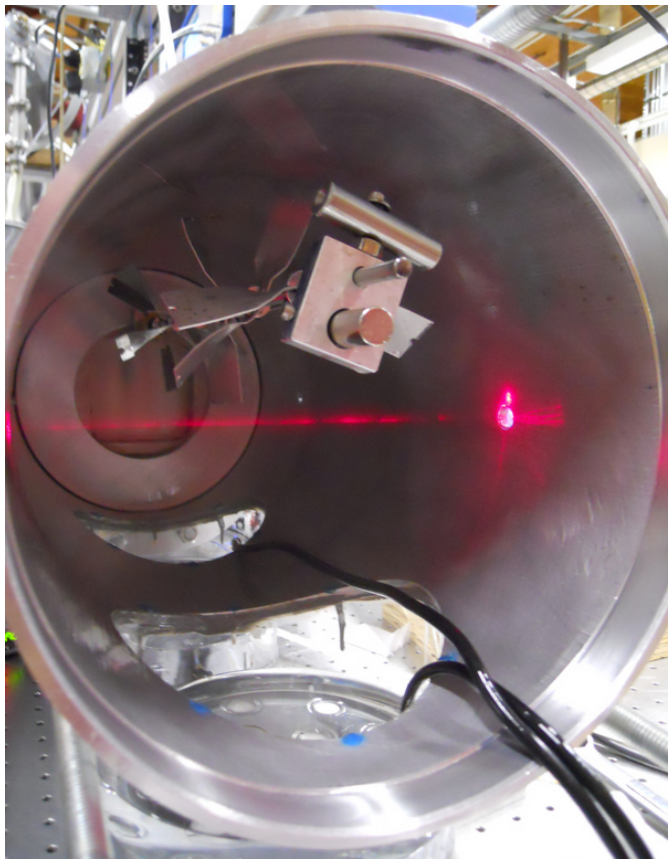


$\Rightarrow$ **Enables quantitative measurement of atom concentration**

Chemical kinetics studies in combustion (22)

Rate constant measurement (2)

Spectroscopic diagnostics : laser absorption

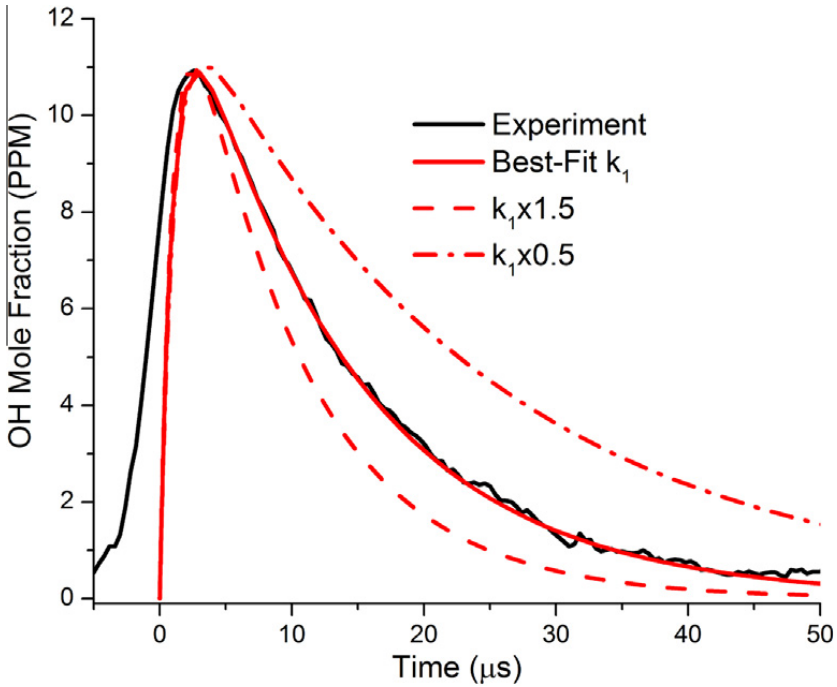


$=>$ **Enables quantitative measurement of molecule concentration**

Chemical kinetics studies in combustion (23)

Rate constant measurement (3)

Species concentration profiles



$\Rightarrow$ **Fitted with a computed profile**

Chemical kinetics studies in combustion (24)

Rate constant measurement (4)

In general, pseudo-first order approximation

$$v = k[A][B]$$

If [A]$\gg$[B] and [A]$\simeq$cst

$$v = k[A][B] = k_{ap}[B]$$

Rate constant deduced from

$$k = \frac{k_{ap}}{[A]}$$

Chemical kinetics studies in combustion (25)

Rate constant measurement (5)

Sources of uncertainty

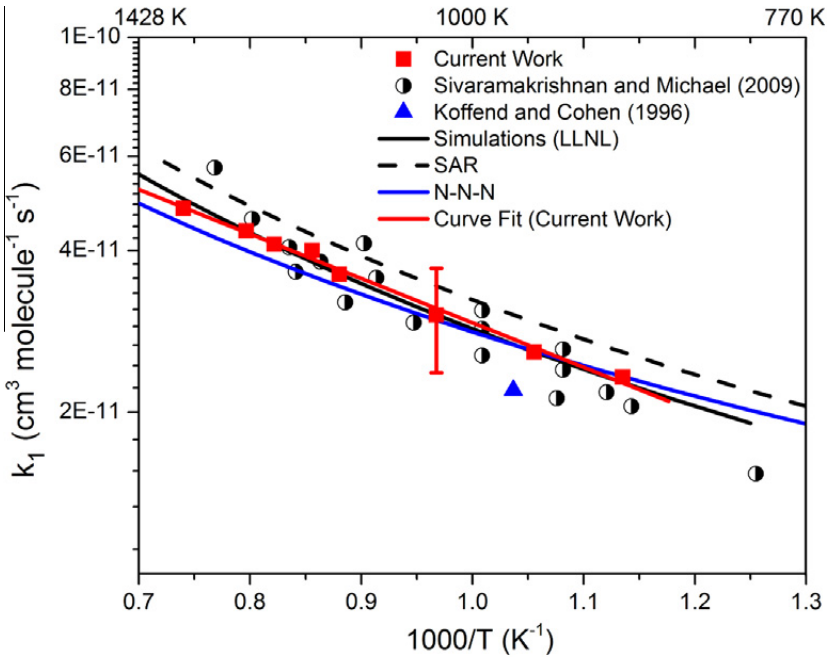
- the temperature
- the mixture composition
- the time zero
- the absorption coefficient
- the laser wavelength
- the fitting procedure
- the rate constant of secondary reactions

$=>$ **In general, between 10 and 30%**

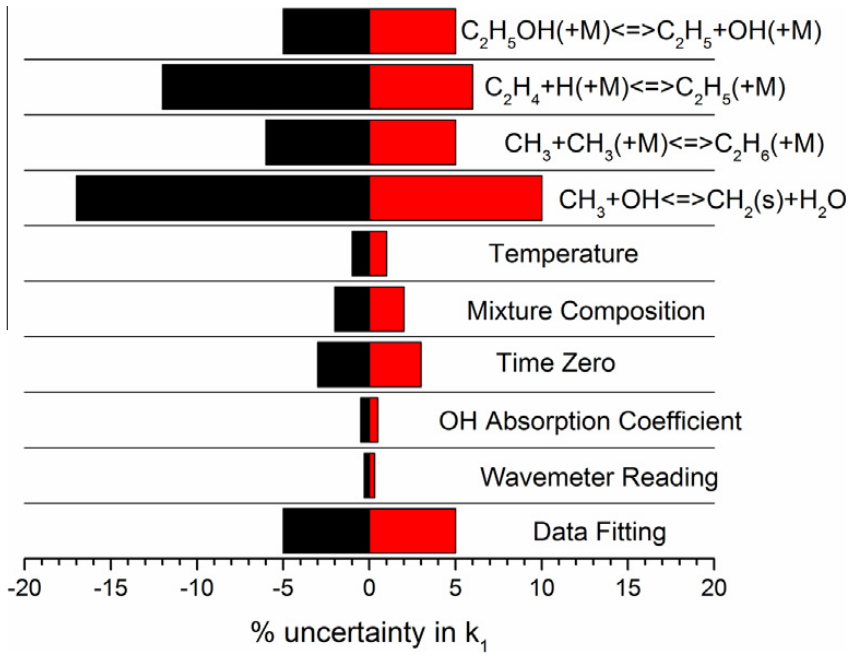
Chemical kinetics studies in combustion (26)

Rate constant measurement (6)

Arrhenius and uncertainty plots



a) Arrhenius plot



b) Uncertainty plot

outlines

1 Introduction

2 Basic definitions in chemical kinetics

3 Mathematical description of reaction rate

4 The kinetics of complex reactions

5 Kinetics modeling in combustion

6 Chemical kinetics studies in combustion

7 Numerical analyses of chemical kinetics

Numerical analyses of chemical kinetics (1)

Rate of energy release per reaction

Given by

$$q_i = \nu_i \Delta_r H_i$$

Corresponds to the rate of energy production/consumption per unit volume for each reaction

$=>$ useful to determine the important chemical pathways responsible for temperature increase

Numerical analyses of chemical kinetics (2)

Rate of production (1)

Given by

$$ROP_j = \sum_{i=1}^{N} \nu_j \nu_{i,j}$$

$=>$ Enables to identify the most important chemical pathways producing or consuming a given chemical species

Numerical analyses of chemical kinetics (3)

Rate of production (2)

Example for a chain reaction mechanism

- **R_1 : R+M=>B+M**
- **R_2 : R+B=>B+B**
- **R_3 : R+B+M=>C+M**
- **R_4 : C+M=>B+B+M**
- **R_5 : B+B+M=>P+M**

Rate of production for B

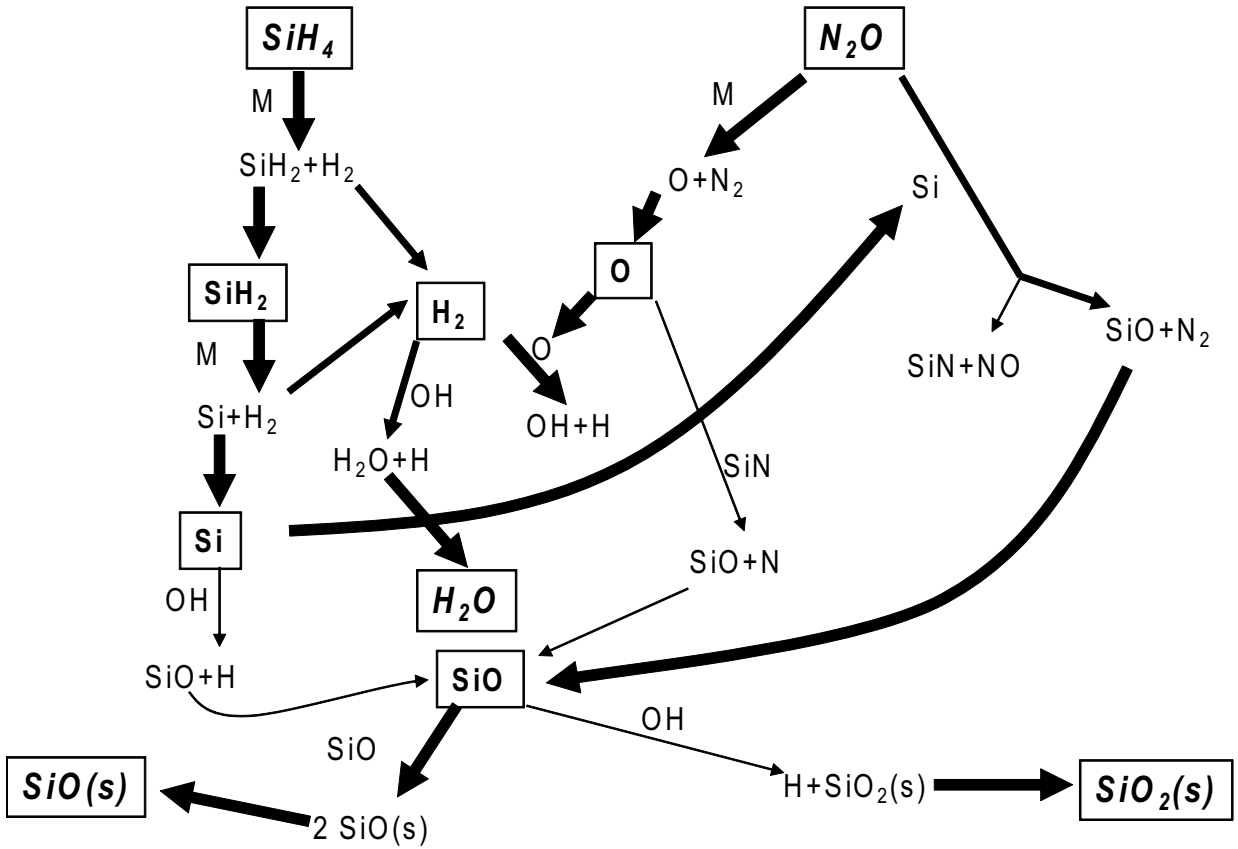
$$ROP_B = \frac{d[B]}{dt} = \sum_{i=1}^{N} \nu_B \nu_{i,B} =$$
$$k_1[R][M] + k_2[R][B] - k_3[R][B][M] + 2k_4[C][M] - 2k_5[B][B][M]$$

Numerical analyses of chemical kinetics (4)

Rate of production (3)

If ROP are calculated for each important species, it is possible to establish chemical pathway diagrams

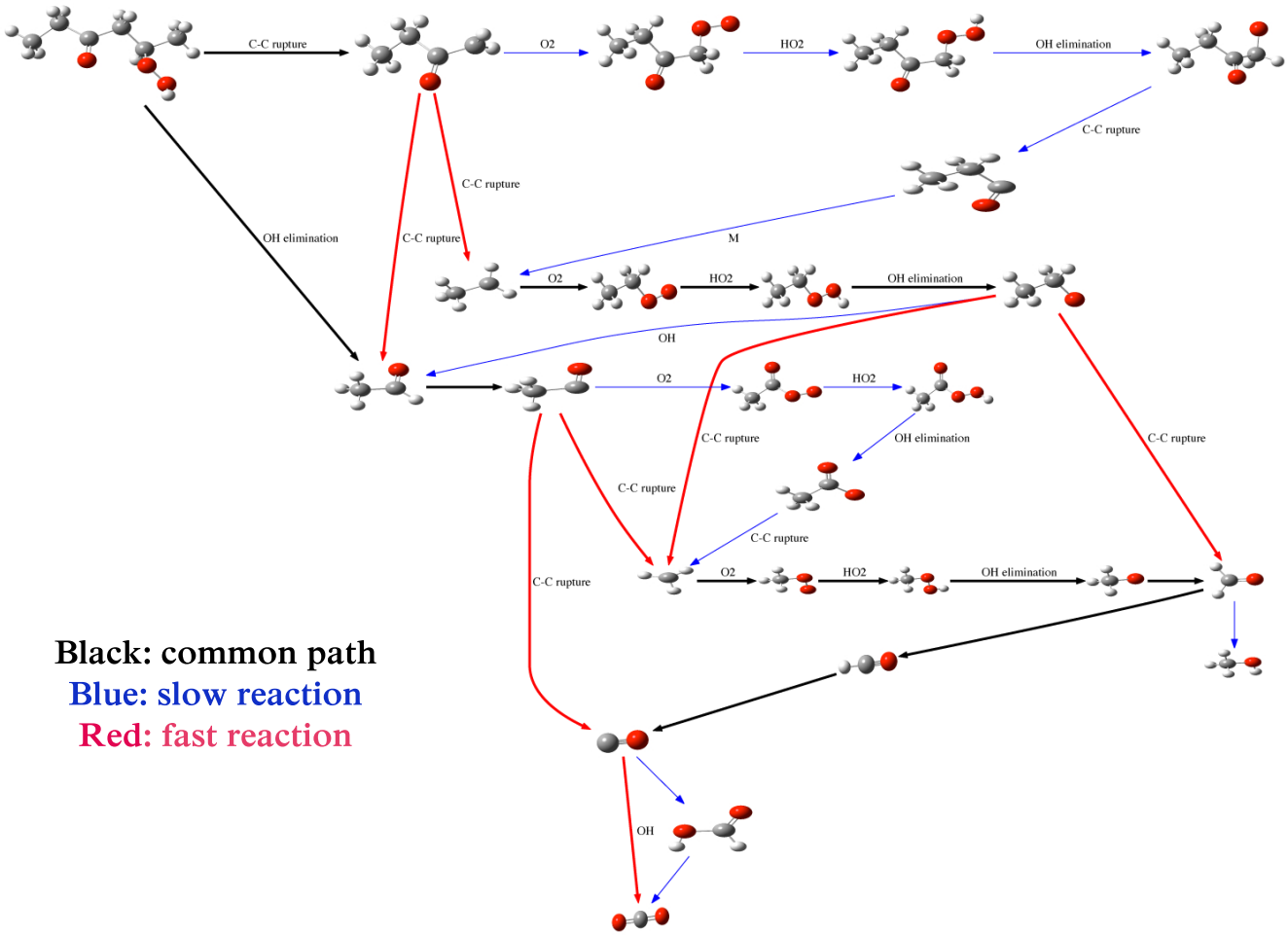
Silane oxidation by nitrous oxide



Numerical analyses of chemical kinetics (5)

Rate of production (4)

Low-temperature oxidation of *n*-hexane by oxygen



Numerical analyses of chemical kinetics (6)

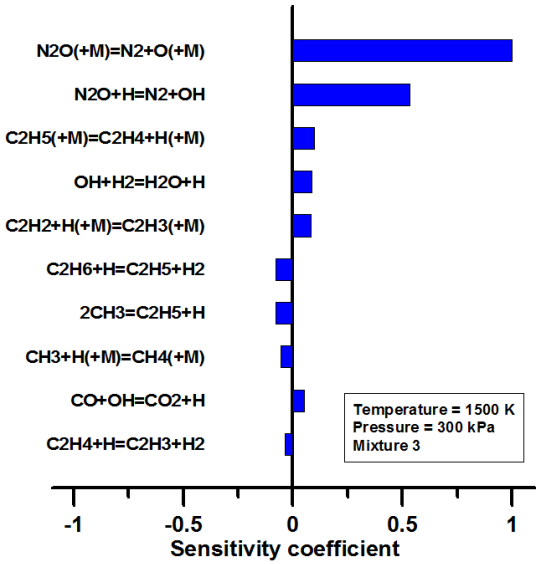
Sensitivity analysis

Defined as

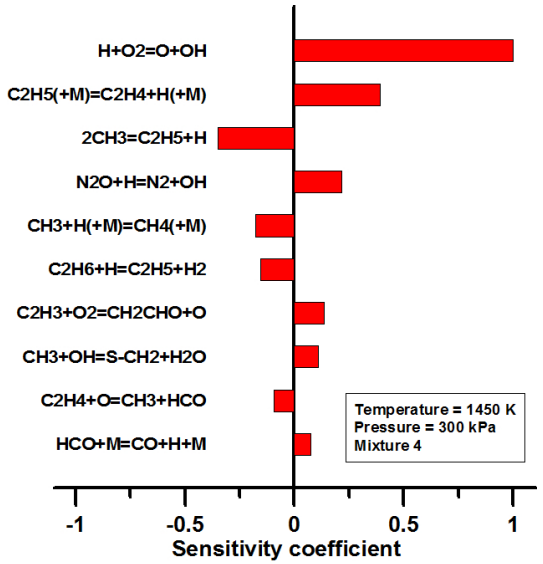
$$C_i = \frac{\partial \ln[j]}{\partial \ln k_i} = \frac{k_i}{[j]} \frac{\partial [j]}{\partial k_i}$$

$\Rightarrow$ Enables to identify the most important reactions with respect to a given parameter of the chemical system

Example : oxidation of ethane



a) Mixture with N_2O

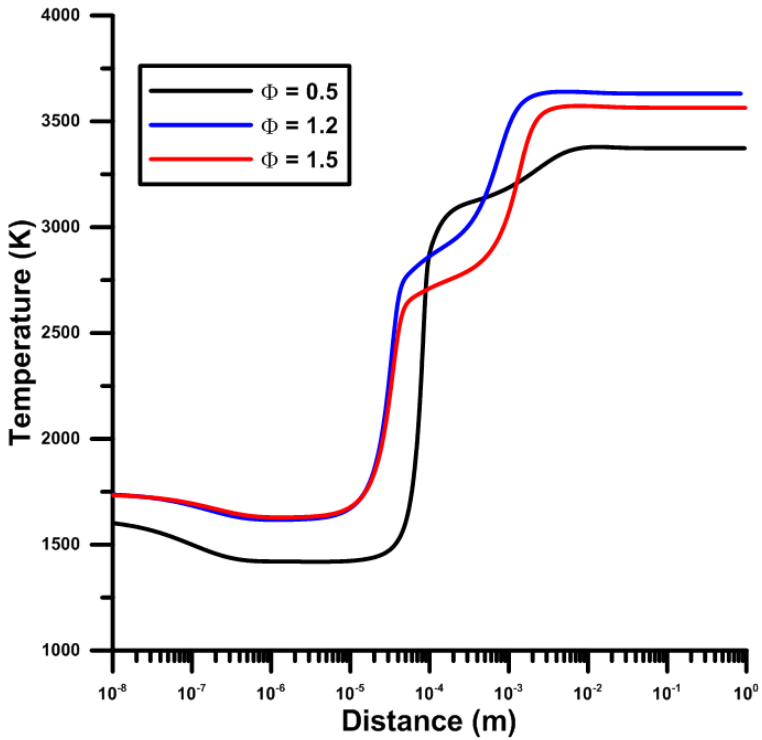


b) Mixture with N_2O and O_2

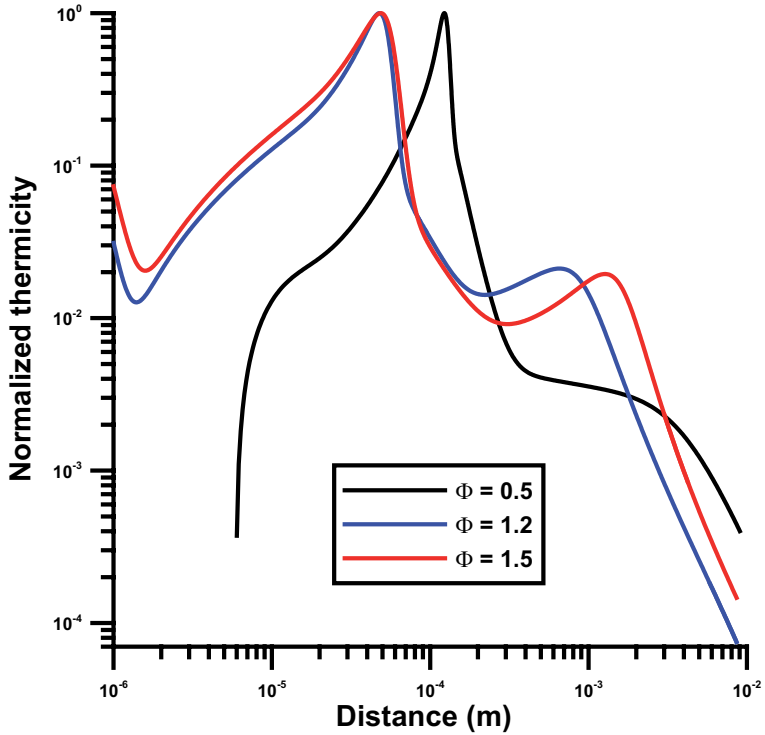
Numerical analyses of chemical kinetics (7)

Example of application (1)

Two-step energy release profile in H_2-NO_2/N_2O_4 mixtures



a) Temperature profiles



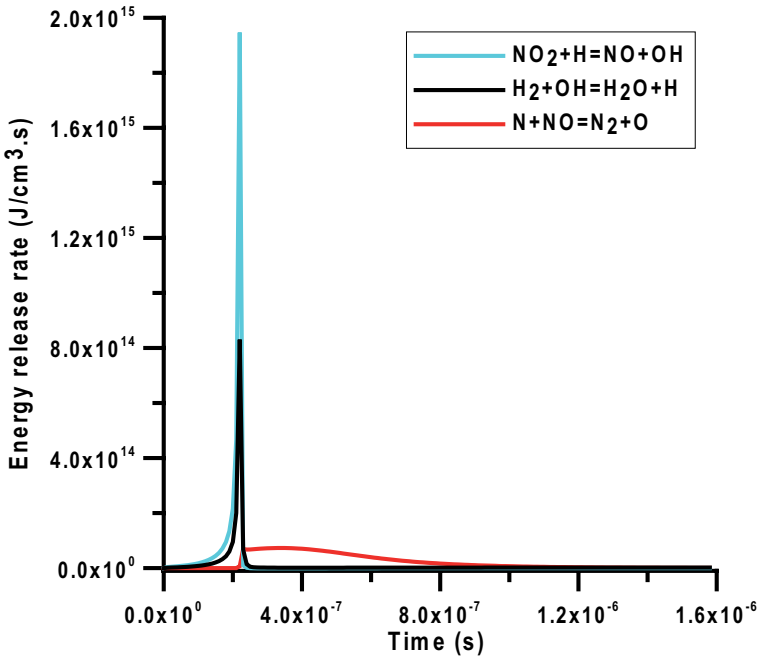
b) Thermicity profiles

$\Rightarrow$ **Where is it coming from ?**

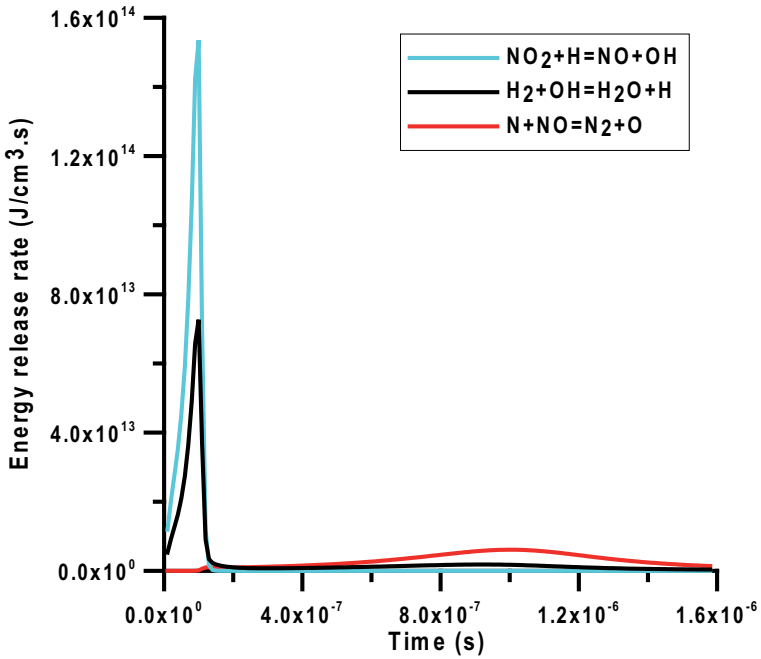
Numerical analyses of chemical kinetics (7)

Example of application (2)

First step : analysis of the energy release rate per reaction



a) Lean mixture



b) Rich mixture

1st step : linear chain $NO_2+H=NO+OH$; $H_2+OH=H_2O+H$

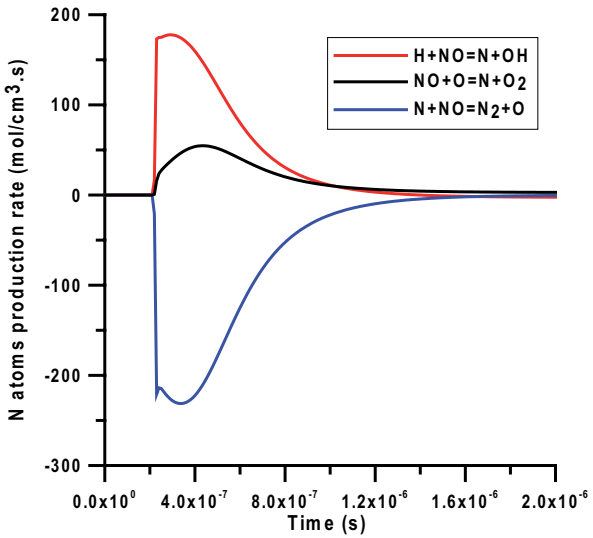
2nd step : formation of N_2 : $N+NO=N_2+O$

Numerical analyses of chemical kinetics (8)

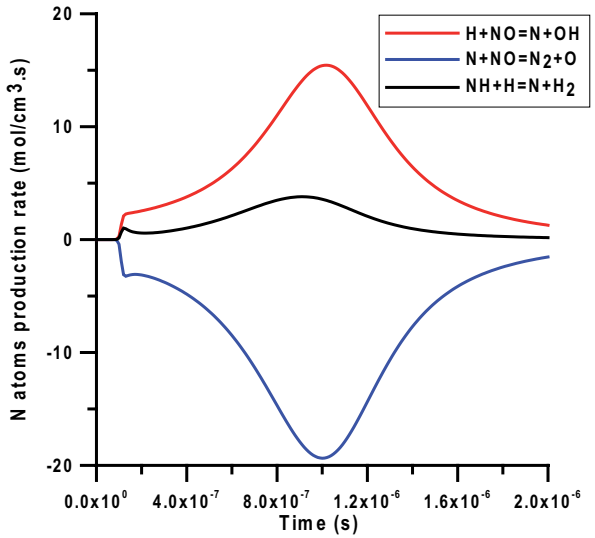
Example of application (3)

Second step : ROP for N

Lean mixture



Rich mixture



After 1st step : T = 3200 K
Fast H+NO=N+OH

After 1st step : T = 2600 K
Slow H+NO=N+OH

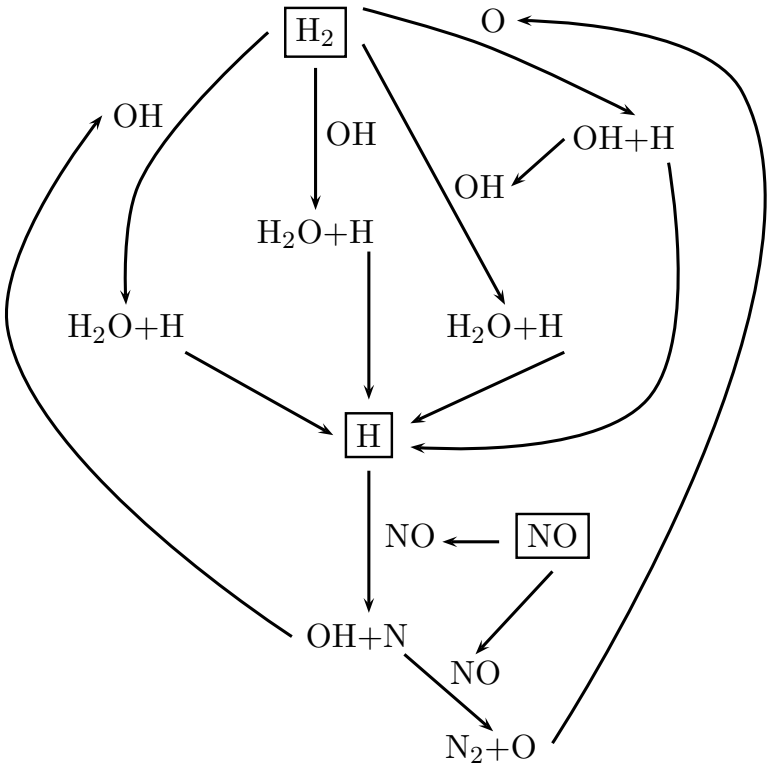
Production by H+NO=N+OH (high Ea)
Consumption by N+NO=N_2+O (Ea = 0)

Numerical analyses of chemical kinetics (9)

Example of application (4)

Third step : identifying reaction mechanism

Chain branching process for N atoms production



Overall : 1 OH $\Rightarrow$ 4 H $\Rightarrow$ 4 N